Real Analysis

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Contents

1 Introduction

The following is intended for anyone who stumbles over these notes. This is intended to be my personal notes in real analysis. I will hopefully be attending MAT2400 Real Analysis at the University of Oslo in the spring of 2026. However, as I am not a program student at this institution, but just someone who takes individual courses there of my own volition, I do not and cannot attend lectures and therefore I have to learn the material on my own. As far as I understand, while this course is called Real Analysis, it is a bit different than a first course from what I understand. The earlier exams give a hint of functional analysis and also include topics such as fourier analysis, meassure- and integration theory among other things. Thus the different supplementary coursematerial I will use to aid myself in learning the content of this course will most likely be a bit scattered and all over the place, which these notes will undoubtedly reflect. I will do my best to keep things organized for my own sake, but keep this in mind if you are someone who intends to use these notes to learn Real Analysis.

2 Basic Banch Space theory

The following section of notes is derived from the first video in the lecture series MIT 18.102 Introduction to Functional Analysis, Spring 2021 (found on youtube).

Definition 2.1 (Vector Space). *A vector space V over a field $\mathbb{F}$ is a nonempty set of elements called "vectors" together with a binary operation $+$ on V and a binary function $\cdot$ which maps elements of $V, \mathbb{F}$ to V satisfying:*

1. *Associativity of vector addition:*
$$u + (v + w) = (u + v) + w, \forall u, v, w \in V$$
2. *Commutativity of vector addition:*
$$u + v = v + w, \forall u, v \in V$$
3. *Identity element:*
$$\exists 0 \in V : v + 0 = 0 + v = v, \forall v \in V$$
4. *Each $v \in V$ has an inverse $-v$ under the vector-addition operation.*
5. *Scalar multiplication is compatible with field multiplication:*
$$a(bv) = (ab)v$$
where $a, b \in \mathbb{F}$ and $v \in V$.
6. *The multiplicative identity $1 \in \mathbb{F}$ satisfies:*
$$1v = v, \forall v \in V$$
7. *Distributivity of scalar multiplication with respect to vector addition:*
$$a(u + v) = au + av$$
where $a \in \mathbb{F}$ and $u, v \in V$.
8. *Distributivity of scalar multiplication with respect to field addition:*
$$(a + b)v = av + bv$$
where $a, b \in \mathbb{F}$ and $v \in V$.

When proving that something is a vector space, most of these follow naturally from showing closure under addition and scalar multiplication and those two properties, are generally enough to show that it is indeed a vector space.
A subspace U of V is a set $U \subseteq V$ which is also a vector space. It is enough to show that $U \subseteq V$ and that it is closed under the two operations.

Some typical examples of vector spaces are $\mathbb{F}^n$ where $\mathbb{F}$ is the reals or the complex numbers. We also have spaces like the space of real polynomials of degree $\leq n$, i.e. $\mathcal{P}_n = \{\sum_{i=0}^n \alpha_i x^i : \alpha_i \in \mathbb{R}\}$, which is itself a subspace of the space of continuous real-valued functions $C(\mathbb{R})$.

So $\mathbb{R}^2$ and $C(\mathbb{R})$ are both vector spaces over $\mathbb{R}$, but they have one really big difference, that being the dimension.

Definition 2.2. *Let V be a vector space. A set $\{v_1, \dots, v_n\} \subseteq V$ is linearly independent if*

$$\sum_{i=1}^{n} \alpha_i v_i = 0 \Leftrightarrow \alpha_1 = \cdots = \alpha_n = 0 \in \mathbb{F}$$

Note: the right-to-left direction of this implication is always true.

The two spaces discussed above are different in dimension, $\mathbb{R}^2$ being 2-dimensional and the other being infinite-dimensional. One definition of finite-dimensional is that every linearly independent set in the space is finite. I however like the definition using bases more. Both of these definitions are equivalent.

We won't give a rigorous definition of a basis, but in short a basis of V is a linearly independent set of vectors which spans V, i.e. every vector in V can be expressed as a linear combination of basis-vectors. If the basis is finite then V is finite dimensional. Moreover, if the basis is finite then the dimension of V is the number of basis-vectors. Note that if a finite dimensional space V has a basis with n elements then every basis of V has n elements. A space is infinite-dimensional if no finite set of linearly independent vectors spans the space.

2.1 Norms and Metrics

Definition 2.3 (Norm). *Let V be a vector space. A norm $||\cdot||$ is a function from $V \to [0, \infty)$ satisfying:*

1. $||v|| = 0$ *if and only if* $v = 0$.
2. $||\alpha v|| = |\alpha| \cdot ||v||$ *where α is an element of the ground field.*
3. $||v + w|| \leq ||v|| + ||w||$.

The tuple $(V, ||\cdot||)$ is called a normed space.

Example 2.1. $||x||_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$ *defines a norm on $\mathbb{R}^n$. In fact it constitutes a norm on $\mathbb{C}^2$ as well. Formally, if we take $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$ the p-norm of a vector $v \in \mathbb{F}^n$ ($p \in [1, \infty]$) is*

$$||v||_p := \begin{cases} \left(\sum_{i=1}^{n} |v_i|^p \right)^{1/p} & p < \infty \\ \max_{i=1,\ldots,n} |v_i| & p = \infty \end{cases}$$

Norms give us a notion of the "length" of a vector. Now all we need to do analysis on spaces is a notion of distance. Intuitively, norms already give us a notion of distance from 0.

Definition 2.4. *Let X be a set.*
A metric is a function $d : X^2 \to [0, \infty)$ satisfying:

1. $d(x, y) = 0$ *if and only if* $x = y$.
2. $d(x, y) = d(y, x)$.
3. $d(x, z) \leq d(x, y) + d(y, z)$.

The metric gives us a notion of distance. In a typical first course in analysis where we work on the reals, $d(a, b) = |a - b|$ is the metric we deal with.

Proposition 2.1. *Let V be a normed space with norm $||\cdot||$. Then we can define the distance (a metric) between two vectors by*

$$d(x,y) := ||x-y||$$

In other words you can define a metric in terms of the norm in any normed space. This metric is usually refered to as the metric induced by the norm.

We won't provide a proof of this as it's fairly intuitive. Now we can get a sense of convergence and continuity in vector spaces by saying that a sequence $\{a_n\}_{n\in\mathbb{N}}$ converges to a value a if

$$\forall\varepsilon>0, \exists N\in\mathbb{N} : n\geq N \Rightarrow ||a_n - a|| < \varepsilon$$

and a linear transformation (look up definition if necessary) $T \in \mathcal{L}(U,V)$ is (uniformly) continuous if

$$\forall\varepsilon>0, \exists\delta>0 : ||x-y||_U < \delta \Rightarrow ||Tx - Ty||_V < \varepsilon$$

for every $x, y \in U$. Notice that if you replace $||x-y||$ with $d(x,y)$ it looks like the standard definitions in terms of metric spaces.

2.2 Banach Spaces

Definition 2.5 (Banach Space). *A normed space V is a Banach Space if it is complete with respect to the metric induced by the norm, meaning that every cauchy sequence converges to a value in the space.*

Example 2.2. *$\mathbb{R}^n$ or $\mathbb{C}^n$ form Banach Spaces with respect to the ℓ^p norms (see Example 2.1).*

Theorem 2.1. *If X is a complete metric space, then $C_\infty(X)$ is a Banach Space.*

Recall that $C_\infty(X)$ is the space of bounded continuous functions on X.

Proof. We show that every Cauchy sequence in $C_\infty(X)$ converges to an element of $C_\infty(X)$.

Let $\{u_n\}_{n=1}^\infty \subseteq C_\infty(X)$ be a Cauchy sequence with respect to the supremum norm. Then for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$\|u_n - u_m\|_\infty < \varepsilon \quad \text{for all } n, m \geq N.$$

Equivalently,

$$|u_n(x) - u_m(x)| < \varepsilon \quad \text{for all } x \in X \text{ and all } n, m \geq N.$$

Fix $x \in X$. Then $\{u_n(x)\}_{n=1}^\infty$ is a Cauchy sequence in $\mathbb{R}$ (or $\mathbb{C}$), since

$$|u_n(x) - u_m(x)| \leq \|u_n - u_m\|_\infty.$$

Because $\mathbb{R}$ (or $\mathbb{C}$) is complete, the limit

$$u(x) := \lim_{n\to\infty} u_n(x)$$

exists. This defines a function $u : X \to \mathbb{R}$.

We now show that $u_n \to u$ uniformly on X. Let $\varepsilon > 0$ and choose N such that $\|u_n - u_m\|_\infty < \varepsilon$ for all $n, m \geq N$. Fix $n \geq N$ and $x \in X$. Taking the limit $m \to \infty$ gives

$$|u_n(x) - u(x)| = \lim_{m\to\infty} |u_n(x) - u_m(x)| \leq \varepsilon.$$

Since $x \in X$ was arbitrary, it follows that

$$\|u_n - u\|_\infty \leq \varepsilon \quad \text{for all } n \geq N.$$

Thus $u_n \to u$ uniformly on X.

Since each u_n is bounded and the convergence is uniform, the limit function u is bounded. Moreover, since each u_n is continuous and uniform limits of continuous functions are continuous, u is continuous on X.

Therefore $u \in C_\infty(X)$ and $\{u_n\}$ converges to u in the supremum norm. Hence $C_\infty(X)$ is complete, and thus a Banach space. $\square$

3 A step back into Calculus Theory

The following section is largely in accordance with chapter 2 of Lindstrøm's Spaces: An Introduction To Real Analysis.

Definition 3.1 (Limit of a sequence). *Let (x_n) be a sequence of real numbers and $x \in \mathbb{R}$.*
We say

$$x_n \to x$$

if for every $\varepsilon > 0$, there exists a natural number N such that

$$|x_n - x| < \varepsilon, \ \forall n \geq N$$

This is among the most important definitions in a first course in Real-Analysis. Intuitively what it captures is the sense of being able to get arbitrarily close to a value. We will all be familiar with examples from calculus like $\lim_{n\to\infty} \frac{1}{n} = 0$.

Theorem 3.1. *If a sequence converges, then its limit is unique.*

Proof. Suppose, for a contradiction, that we have some sequence (x_n) with

$$x_n \to x \text{ and } x_n \to y$$

of course with $x \neq y$.

x_n converges so we can choose any $\varepsilon > 0$ for the condition in definition 3.1 to hold.

Consider the case of $\varepsilon = \frac{|x-y|}{2}$. $\varepsilon > 0$ since $x \neq y$ and is therefore a value for which the condition must hold. Namely,

$$|x_n - x| < \varepsilon \text{ and } |x_n - y| < \varepsilon$$

leading to

$$|x - x_n| + |x_n - y| < 2\varepsilon = |x - y|$$

which, via triangle inequality, yields

$$|x - y| = |x - x_n + x_n - y| \leq |x - x_n| + |x_n - y| < |x - y|$$

in particular

$$|x - y| < |x - y|$$

which is not possible, a contradiction.

□

Theorem 3.2 (Algebra of Limits). *If $x_n \to x$ and $y_n \to y$, then:*

1. $x_n + y_n \to x + y$
2. $x_n y_n \to xy$
3. *If $y_n \neq 0$ and $y \neq 0$, then* $\frac{x_n}{y_n} \to \frac{x}{y}$.

Proof of (1): Sum of Limits. Let (x_n) and (y_n) be convergent sequences with $x_n \to x$ and $y_n \to y$.

Let $\varepsilon > 0$. Since both sequences converge to their respective limits we have some $N_1, N_2 \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon/2, \ \forall n \geq N_1$$
$$|y_n - y| < \varepsilon/2, \ \forall n \geq N_2$$

Let $N = \max\{N_1, N_2\}$, then $n \geq N$ satisfies $n \geq N_1$ and $N \geq N_2$ so

$$|x_n - x| + |y_n - y| < \varepsilon$$

Notice that via the triangle inequality we get

$$|(x_n + y_n) - (x + y)| = |(x_n - x) + (y_n - y)| \leq |x_n - x| + |y_n - y| < \varepsilon$$

completing the proof. □

Theorem 3.3. *A sequence x_n converges to x if and only if*

$$\limsup x_n = \liminf x_n = x$$

We will take this theorem without proof as it requires some background about monotone sequences and completeness which are covered later.

Definition 3.2 (Limit of a function). *Let $f : D \subset \mathbb{R} \to \mathbb{R}$, and let a be a limit point of D.*
We say

$$\lim_{x \to a} f(x) = L$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$$

Later, when we get to metric spaces again, this condition becomes

$$d_X(x, a) < \delta \Rightarrow d_Y(f(x), f(a)) < \varepsilon$$

for continuity.

Theorem 3.4 (Sequential criterion).

$$\lim_{x \to a} f(x) = L \Leftrightarrow \textit{for every sequence } x_n \to a \textit{ with } x_n \neq a, f(x_n) \to L$$

Proof sketch. We only prove $\Rightarrow$, for now.

Assume $\lim_{x \to a} f(x) = L$.

Let $x_n \to a$ with $x_n \neq a$.

Given $\varepsilon > 0$, choose $\delta > 0$ from the definition of the limit.

Since $x_n \to a$ there exists $N \in \mathbb{N}$ such that with $n \geq N$ we have

$$|x_n - a| < \delta$$

Hence for $n \geq N$,

$$|f(x_n) - L| < \varepsilon$$

Thus $f(x_n) \to L$. $\square$

4 Completeness

Definition 4.1 (Upper Bound). *A set $A \subset \mathbb{R}$ is bounded above if there exists $M \in \mathbb{R}$ such that*

$$a \leq M, \forall a \in A$$

Such an M is called an upper bound.

Definition 4.2 (Supremum). *Let $A \subset \mathbb{R}$.*
A number $s \in \mathbb{R}$ is the supremum of A if it is an upper bound of A and for any upper bound u of A we have

$$s \leq u$$

and we use the notation

$$s = \sup A$$

The completeness axiom says that every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$, this being the distinguishing quality separating $\mathbb{R}$ from $\mathbb{Q}$.

Sidenote: You can also characterize the reals as the *unique* ordered field containing $\mathbb{Q}$ which has the least upper bound property. More precisely, every $\mathbb{F}$ with this property satisfies $\mathbb{F} \cong \mathbb{R}$.

Proposition 4.1. *If A has a supremum then for every $\varepsilon > 0$, there exists $a \in A$ such that*

$$s - \varepsilon < a \leq s$$

Proof. Let $\varepsilon > 0$. If no $a \in A$ existed with the desired property, then $s - \varepsilon$ would be a supremum, contradicting minimality of s. □

This proposition becomes useful later in many arguments.

The infimum $\inf A$ is also something we need and classically we define it as being the greatest lower bound, but for the purposes of keeping things simple, we just let the infimum of a set A, $\inf A$, be defined as

$$\inf A = -\sup(-A)$$

Theorem 4.1 (Monotone convergence). *Let (x_n) be a monotone increasing sequence (a sequence with which $x_n \leq x_{n+1}$ for every $n \in \mathbb{N}$) which is bounded above.*
Then (x_n) converges, and

$$\lim x_n = \sup\{x_n : n \in \mathbb{N}\}$$

Proof. Let

$$A = \{x_n : n \in \mathbb{N}\}$$

Since A is bounded above, A has some supremum s. We claim $x_n \to s$.

Let $\varepsilon > 0$.

By the supremum property, there exists N such that

$$s - \varepsilon < x_N <\leq s$$

Since the sequence is increasing, for all $n \geq N$,

$$s - \varepsilon < x_n \leq s$$

Thus

$$|x_n - s| < \varepsilon$$

□

Lemma 4.1. *Every cauchy sequence in $\mathbb{R}$ is bounded.*

Proof. Let (x_n) be cauchy.

Choose $\varepsilon = 1$. Then there exists N such that

$$|x_n - x_m| < 1, \ \forall n, m \geq N$$

Fix $m = N$. Then for all $n \geq N$:

$$|x_n| \leq |x_N| + |x_n - x_N| < |x_N| + 1$$

Thus all terms are bounded. □

Theorem 4.2. *Every cauchy sequence in $\mathbb{R}$ converges.*

Proof idea. Let (x_n) be cauchy.

(x_n) is bounded. Use the cauchy property to construct nested intervals. Show the intersection of these to be nonempty via completeness of $\mathbb{R}$. (x_n) will converge to a point in this intersection.

A bit more formally, we choose indices $n_1, n_2, \ldots$ such that

$$|x_n - x_m| < 2^{-k},\ \forall n, m \geq n_k$$

and define the intervals

$$I_k = [x_{n_k} - 2^{-k}, x_{n_k} + 2^{-k}]$$

Then each I_k is closed and bounded with $I_{k+1} \subset I_k$. Completeness wil give us

$$\bigcap_{k \in \mathbb{N}} I_k \neq \emptyset$$

There will be an x in this set which is the limit point. $\square$

5 Open and Closed Sets in Metric Spaces

Let X be a set and $A \subseteq X$. Intuitively, a point x is either **inside of** A, **outside of** A, or **on the boundary** of A. We know what it means to not be in A; $a \notin A \Leftrightarrow a \in A^c$. We also know what $a \in A$ means, but set-theoretically we don't have a notion of being on the "boundary" of A. We can make sense of this notion in a metric space.

Recall that the open ball is the set $B(x;r) = \{z \in X : d(x,z) < r\}$. Likewise we can define the closed ball as follows.

Definition 5.1 (Closed Ball). *The closed ball centered at* $x \in X$ *with radius* $r \geq 0$ *in* $B(x;r) = \{z \in X : d(x,z) \leq r\}$.

Let (X,d) be a metric space with $A \subseteq X$.

- $x \in X$ is an interior point of A if $B(x;r) \subseteq A$ for some $r > 0$.
- $y \in X$ is an interior point of A if $B(x;r) \subseteq A^c$ for some $r > 0$.
- $z \in X$ is a boundary point of A if it is neither of the above, i.e. $B(z;r) \cap A \neq \emptyset$ and $B(z;r) \cap A^c \neq \emptyset$ for every $r > 0$.

Notably, every point in X is one of these three.

- $A^0 = \{$all interior points of $A\}$
- $\partial A = \{$all boundary points of $A\}$
- $\overline{A} = A \cup \partial A = ((A^c)^0)^c$

Proposition 5.1. *For any* $A \subset X$*, we have* $\partial A = \partial(A^c)$.

Theorem 5.1. *Useful characterizations of openness:*
$A \subseteq X$ *is open if* $A = A^0$
$A \subseteq X$ *is open if* A *contains none of its boundary points.*
$A \subseteq X$ *is open if* $A \cap \partial A = \emptyset$
$A \subseteq X$ *is open if* $\forall x \in A$ *there is some* $r > 0$ *s.t.* $B(x;r) \subseteq A$

Theorem 5.2. *Useful characterizations of openness:*
$B \subseteq X$ *is closed if* $B = \overline{B}$
$B \subseteq X$ *is closed if* A *contains all of its boundary points.*
$A \subseteq X$ *is closed if* $\partial B \subseteq B$

Problem 5.1. *Consider* $X = \mathbb{R}$ *with the canonical metric* $|\cdot|$*. Show that* (a,b) *is open and* $[a,b]$ *is open for* $a \leq b \in \mathbb{R}$*, and* $(a,b]$ *is neither open nor closed for* $a < b$*.*

Proof. Let $x \in (a,b)$, i.e. $a < x < b$. Take

$$r := \mathtt{min}\{x-a, x-b\} > 0$$

Take any $y \in (x-r, x+y)$. Then

$$y > x-r \geq a,\ y < x+r \leq b$$

Therefore

$$(x-r, x+r) \subset (a,b)$$

In other words, every $x \in (a,b)$ admits and open ball contained entirely in the interval, so (a,b) is open.

Notice that $\partial(a,b) = \{a,b\}$ and that $[a,b] = \partial(a,b) \cup (a,b)$ hence $[a,b]$ is the closure of (a,b) and thus is closed.

It is clear that $(a,b]$ is neither open nor closed as $\partial(a,b] = \{a,b\} \not\subseteq (a,b]$ and $(a,b] \cap \partial(a,b] \neq \emptyset$. $\square$

6 Week 4 problem set

Proposition 6.1 (Proposition 3.1.4 in Spaces). *If (X, d) is a metric space with $x, y, z \in X$, then*

$$|d(x, y) - d(y, z)| \leq d(x, z)$$

Problem 6.1. *Prove Proposition 6.1.*

Proof. Let (X, d) be a metric space and recall the standard triangle inequality:

$$d(x, z) \leq d(x, y) + d(y, z)$$

Now observe the following rearangements:

$$d(x, y) \leq d(y, z) + d(x, z)$$
$$d(x, y) - d(y, z) \leq d(x, z)$$

and

$$d(x, y) \leq d(y, z) + d(x, z)$$
$$d(y, z) \leq d(x, y) + d(x, z)$$
$$d(y, z) - d(x, z) \leq d(x, y)$$
$$-d(x, z) \leq d(x, y) - d(y, z)$$

Together:

$$|d(x, y) - d(y, z)| \leq d(x, z)$$

□

Problem 6.2 (3.1.6 Spaces). *Let $(V, ||\cdot||)$ be a normed space. Show that it induces a norm, i.e.*

$$d(x, y) := ||x - y||$$

is a norm.

Proof. Let $(V, ||\cdot||)$ be a normed space.

Non-negativity and Identity of Indiscernibles:

$$d(x, y) = ||x - y|| \geq 0$$

since $x - y \in V$ and $||\cdot||$ is non-negative.

$$d(x, x) = ||x - x|| = ||0|| = 0$$

Assume $d(x, y) = 0$.

$$d(x, y) = ||x - y|| = 0$$

Then $x - y = 0 \Rightarrow x = y$.

Symmetry:

Recall for a norm we have $||\alpha x|| = |\alpha| ||x||$, hence

$$d(x, y) = |1| ||x - y|| = || - 1(x - y)|| = ||y - x|| = d(y, x)$$

Triangle Inequality:

$$\begin{aligned} d(x, z) = ||x - z|| &= ||x - (y + y) - z|| \\ &\leq ||x - y|| + ||y - z|| \\ &= ||x - y|| + ||z - y|| \\ &= d(x, y) + d(y, z) \end{aligned}$$

$\square$

Problem 6.3 (3.1.7 Spaces). *Show that if $x_1, x_2, \ldots, x_n$ are points in a metric space, then*

$$d(x_1, x_n) \leq \sum_{i=1}^{n-1} d(x_i, x_{i+1})$$

Proof. We proceed by induction on $n \in \mathbb{N}$.

Base Case ($n = 1$).

Clearly,

$$d(x_1, x_1) \leq d(x_1, x_1)$$

Hypothesis. Now assume, for any $k \in \mathbb{N}$,

$$d(x_1, x_k) \leq \sum_{i=1}^{k-1} d(x_i, x_{i+1})$$

Induction. Let $n = k + 1$.

$$\begin{aligned} d(x_1, x_n) = d(x_1, x_{k+1}) &\leq d(x_1, x_k) + d(x_k, x_{k+1}) \\ &\leq \left(\sum_{i=1}^{k-1} d(x_i, x_{i+1}) \right) + d(x_k, x_{k+1}) \\ &= \sum_{i=1}^{n-1} d(x_i, x_{i+1}) \end{aligned}$$

as desired. $\square$

Problem 6.4 (3.2.1 Spaces). *Assume that (X, d) is a discrete metric space. Show that the sequence $\{x_n\}$ converges to a if and only if there is an $N \in \mathbb{N}$ such that $x_n = a$ for all $n \geq N$.*

Proof. Let (X, d) be a discrete metric space, i.e. a metric space where

$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{otherwise} \end{cases}$$

$\Leftarrow$.

Assume there is some $N \in \mathbb{N}$ such that $\forall n \geq N$,

$$x_n = a$$

Then it is clear that for any $\varepsilon > 0$,

$$x_n = 0 \Rightarrow d(x_n, a) = 0 < \varepsilon$$

so $\{x_n\} \to a$.

$\Rightarrow$.

Assume that $\{x_n\} \to a$. Then for any $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for $n \geq N$, then

$$d(x_n, a) < \varepsilon$$

In particular, this must hold for $\varepsilon = 0.5$. Notice that this requires $x_n = a$ since otherwise we would have $1 < 0.5$, a contradiction. □

Problem 6.5 (3.2.5 Spaces). *Let (X, d) be a metric space and fix some $a \in X$. Show that the function $f : X \to \mathbb{R}$ defined as*

$$f(x) = d(x, a)$$

is continuous. Note: $\mathbb{R}$ is finite dimensional so we are free to choose any metric. $d\mathbb{R}(x, y) = |x - y|$ is fine.

Proof. Let (X, d) be a metric space and fix $a \in X$. Let f be defined as above.

Consider a family of sequences in X such that every sequence $\{x_n\}$ converges to $a \in X$. Let $\{x_n\}$ be some arbitrary sequence in this family.

Then, $\forall \varepsilon > 0$ we have some natural number $\mathbb{N}$, s.t. for any $n \geq N$,

$$d(x_n, a) < \varepsilon$$

Notice that

$$d(x_n, a) = |d(x_n, a) - 0| < \varepsilon$$

where $\varepsilon > 0$ was arbitrary. Hence $f(\{x_n\})$ converges to $0 = d(a,a) = f(a)$.

Thus f is continuous. □

Problem 6.6. *Let (X,d) be a metric space. Prove that every finite subset of X is closed.*

Proof. Recall that proposition $3.3.13$ from Spaces tells us that for any finite collection $F_1, \dots, F_n$ of closed sets, their union

$$\bigcup_{i \in [n]} F_i$$

is closed.

Consider any singleton $\{x\} \subseteq X$. It is clear that $\{x\}$ is closed since $\{x\}^c$ is open. Observe that the open ball $B(y;\varepsilon) \subseteq \{x\}^c$ for any $y \in X \setminus \{x\}$ with $\varepsilon = d(x,y)$.

Let $\{x_1, \dots, x_n\} \subseteq X$ be any finite subset. As shown above, $\{x_i\}$ is closed, and by proposition 3.3.13

$$\{x_1, \dots, x_n\} = \bigcup_{i \in [n]} \{x_i\}$$

is closed. □

Problem 6.7 (3.3.13 Spaces). *A metric space (X,d) is disconnected if it is the union of two non-empty, disjoint and open subsets. If it is not disconnected it is connected.*
a) Let $X = (-1,1) \setminus \{0\}$ and let d be the usual metric on X. Show that (X,d) is disconnected.
b) Let $X = \mathbb{Q}$ and let d be the usual metric again. Show that (X,d) is disconnected.
c) Assume that (X,d) is a connected metric space and that $f : X \to Y$ is continuous and surjective. Show that Y is connected.

Proof of (a). It is clear that $(-1,0)$ and $(0,1)$ are open with respect to $d(x,y) = |x-y|$ as, we can find an open ball around any point in either. Since

$$X = (-1,0) \cup (0,1)$$

we conclude that X is disconnected. □

Proof of (b). Recall that $\sqrt{2} \notin \mathbb{Q}$ so the open subsets

$$(-\infty, \sqrt{2}) \cap \mathbb{Q} \text{ and } (\sqrt{2}, \infty) \cap \mathbb{Q}$$

Both of these sets are open in $\mathbb{R}$ and thus they are also open in $\mathbb{Q}$. □

Proof of (c). Suppose, for a contradiction that we have some continuous surjection $f : X \to Y$ where X is connected and Y is not.

From surjection we can gleam that for any $y \in Y$ there is some $x \in X$ such that $f(x) = y$. From continuity at every point we have that there is some $\delta > 0$ for any $\varepsilon > 0$ such that

$$f\left[B_X(x;\delta)\right] \subseteq B_Y(f(x);\varepsilon)$$

By assumption

$$Y = O_1 \cup O_2$$

for some nonempty, disjoint, open O_1 and O_2. So for any point in O_1 or O_2 we have an open neighbourhood surrounding it. Consider some family open sets in O_1, $\{\omega_n\}_{n\in\mathbb{N}}$ such that

$$\bigcup_{n\in\mathbb{N}} \omega_n = O_1$$

It is clear that $\omega_i \notin O_2$ for any ω_i. Suppose ω_i is centered around $y_i \in Y$. By surjectivity we have that there are some $x_i \in X$ such that the image open ball around x_i goes to the open ball around y_i. It is not hard to see that we are characterizing an open set in X which is mapped to O_1. Repeat this process for O_2 and we will see that since there is no open ball which falls in both sets, and correspondingly via continuity there then is no ball in X which falls in the pre-images of both O_1 and O_2, hence we have partitioned X into two disjoint non-empty subsets. So X is disconnected, a catastrophic contradiction. $\square$

Another attempt at a proof of (c). Suppose, for a contradiction that we have a continuous surjection $f : X \to Y$ where X is connected and Y is not.

Let $A \subseteq X$ be a subset such that $f|_A : A \to Y$ is a bijection. Since it is the restriction of f to A is still continuous f constitutes a homeomorphism from $A \to Y$. Then, the inverse $g : Y \to A$ exists, which is still continuous of course.

Recall that a function g from a metric space $Y \to A$ is continuous if and only if for any $\varepsilon > 0$ we have some $\delta > 0$ such that

$$g\left[B_Y(y;\delta)\right] \subseteq B_A(g(y);\varepsilon)$$

for any $y \in Y$. By assumption, there exist O_1 and O_2 such that

$$Y = O_1 \cup O_2$$

with $O_1, O_2 \neq \emptyset$ and disjoint. Furthermore, O_1 and O_2 are open. We now claim that A is partitioned into two disjoint, nonempty, open sets

$$g[O_1] \text{ and } g[O_2]$$

It is not hard to see that they are both open as the open ball around any point in O_1 or O_2 is sent to some open ball around a point in their image via continuity. Furthermore it is clear that they are nonempty since g is a bijection and O_1, O_2 are nonempty.

Lastly, assume (for contradiction) that there is some $y \in Y$ such that $g(y) \in g[O_1]$ and $g(y) \in g[O_2]$, i.e. they are not disjoint. Then we would have $f|_A(g(y)) \in O_1$ and $f|_A(g(y)) \in O_2$. Recalling that $f|_A \circ g = id_Y$ we have that $y \in O_1$ and $y \in O_2$, but it is established that no such point exists, contradicting the assumption that $g[O_1] \cap g[O_2] \neq \emptyset$.

Thus we have shown that A is disconnected, but then X is also disconnected, a catastrophic contradiction to our original assumption. □

Problem 6.8 (3.3.13 Spaces (again)). *A space X is path connected if there is a continuous $r : [0,1] \to X$ for every pair $x, y \in X$ such that $r(0) = x$ and $r(1) = y$.*
d) Let d be the usual metric on $\mathbb{R}^n$:

$$d(x,y) = ||x-y|| = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$

Show that $(\mathbb{R}^n, d)$ is path-connected.
e) Show that every path-connected metric space is connected.

Proof of (d). Consider two distinct points $x, y \in \mathbb{R}^n$. Then

$$x = (x_1, \ldots, x_n), \;\; y = (y_1, \ldots, y_n)$$

Consider the vector

$$(y-x) \cdot z = (y_1 - x_1, \ldots, y_n - x_n) \cdot z, \;\; z \in [0,1]$$

It is clear that

$$x + (y-x) \cdot 0 = x \text{ and } x + (y-x) \cdot 1 = y$$

So define $r : [0,1] \to X$ by

$$r(z) := x + (y-x) \cdot z$$

Let $z \in [0,1]$ and fix any $z_0 \in \mathbb{Z}$, let $\varepsilon > 0$ and choose $\delta := \frac{\varepsilon}{||x-y||} > 0$. Assume

$$|z - z_0| < \delta$$

Then,

$$
\begin{aligned}
||r(z) - r(z_0)|| &= ||(x + (y-x)\cdot z) - (x + (y-x)\cdot z_0)|| \\
&= \left(\sum_{i=1}^{n} \left((x_i - zx_i + zy_i) - (x_i - z_0x_i + z_0y_i) \right)^2 \right)^{1/2} \\
&= \left(\sum_{i=1}^{n} \left((z_0 - z)x_i + (z - z_0)y_i \right)^2 \right)^{1/2} \\
&= \left(\sum_{i=1}^{n} \left((z - z_0)(y_i - x_i) \right)^2 \right)^{1/2} \\
&= |z - z_0| \left(\sum_{i=1}^{n} (y_i - x_i)^2 \right)^{1/2} \\
&= |z - z_0| \cdot ||x - y|| < \delta \cdot ||x - y|| = \frac{\varepsilon}{||x - y||} ||x - y|| \\
&= \varepsilon
\end{aligned}
$$

$\square$

7 Week 6 problem set

Problem 7.1 (Problem relating to Section 3.3). *Let (X, d_X) and (Y, d_Y) be metric spaces, let $f : X \to Y$ be a continuous function, let $y \in Y$ be some given point, and consider the problem of finding an $x \in X$ such that $f(x) = y$ (that is, we wish to solve the above equation). Prove that the set of solutions of this equation is a closed subset of X.*
Hint: *Phrase the question in terms of finding f^{-1} of a closed set.*

Proof. Apply the assumptions of the problem description.

Let $S_y \subseteq X$ denote the set of solutions to $f(x) = y \in Y$. Notice that S_y is the set $f^{-1}[\{y\}]$. Since $\{y\}$ is a singleton it is a closed set. In the topological sense, if f is continuous then the preimage of every closed set in Y is closed in X. This would complete the proof, but let us give an analytical way of doing it instead.

Once again let $S_y \subset X$ denote the set of $x \in X$ such that $f(x) = y \in Y$. Let $\{x_n\}$ be some sequence in S_y such that $x_n \to x$ in X. Now we need to show that $x \in S_n$, i.e. $f(x) = y$. Since $x_n \in S_y$ we have that (for every $n \in \mathbb{N}$)

$$f(x_n) = y$$

f is continuous so via sequential continuity we get

$$f(x_n) = f(x)$$

i.e.

$$f(x) = y$$

Thus $x \in S_y$. Since $\{x_n\}$ was an arbitrarily chosen convergent sequence in S_y and we found that $x_n \to x \in S_y$ we conclude that S_y contains all it's limit points. In other words S_y is closed. □

Problem 7.2. *Prove that $(\mathbb{Z}, d)$, where $d(x, y) = |x - y|$, is complete.*
Hint: *What does it mean for a sequence in $(\mathbb{Z}, d)$ to converge or to be Cauchy?*

Proof. Let $\{x_n\}$ be a cauchy sequence in $\mathbb{Z}$. Then, for every $\varepsilon > 0$ there is some $N \in \mathbb{N}$ s.t.

$$|x_n - x_m| < \varepsilon$$

for every $n, m \geq N$. Notice that we then require $x_n = x_m$ for all $n, m \geq N$ since if that weren't the case we would have $0 < \varepsilon := 0.5 < 1 \leq |x_n - x_m|$. Thus every cauchy sequence in $\mathbb{Z}$ will necessarily settle at some $x \in \mathbb{Z}$ and be constant. In other words, $\{x_n\} \to x \in \mathbb{Z}$, as desired. □

Problem 7.3 (Problem 3.3.3 Spaces). *Assume that F is a nonempty, closed and bounded subset of $\mathbb{R}$ with the usual metric. Show that* $\inf F \in F$ *and* $\sup F \in F$. *Give an example of a bounded, but not closed set such that it contains it's supremum and infinmum.*

Proof. Suppose $\emptyset \neq F \subseteq \mathbb{R}$ with F closed and bounded.

Recall that since F is bounded there is some finite $r > 0$, $x \in F$ such that

$$F \subseteq (x - r, x + r)$$

Recall that for any nonempty subset of $\mathbb{R}$ there exists a least upper bound and greates lower bound, i.e. $\inf F \in \mathbb{R}$ and $\sup F \in \mathbb{R}$.

Let $\{x_n\}$ be a sequence in F such that $\{x_n\} \to \sup F$. By closedness $\sup F \in F$. Apply this same reasoning to the infimum to get the desired result. Notice that we required boundedness as, if F wasn't bounded above and below we might have had $\inf F = -\infty$ or $\sup F = \infty$. In that case we couldn't have constructed sequences converging to these points.

Let $F = [0, 1] \setminus \{\frac{1}{2}\}$. Clearly not closed since any sequence converging to $1/2$ contradicts it being closed. It is bounded above and below and $\inf F = 0 \in F$ and $\sup F = 1 \in F$. $\square$

Problem 7.4 (Problem 3.3.11 Spaces). *Prove Proposition 3.3.12. Find an infinite collection of open sets whose intersection is closed.*

Proposition 7.1 (Proposition 3.3.12 from Spaces). *Let (X, d) be a metric space.*
a) If $\mathcal{G}$ is a finite or infinite collection of open sets, then

$$\bigcup_{G \in \mathcal{G}} G$$

is open.
b) If $G_1, \dots, G_n$ is a finite collection of open sets then

$$\bigcap_{i=1}^{n} G_i$$

is open.

Proof of (a). Consider a (potentially infinite) family $\mathcal{G} = \{G_1, G_2, \dots\}$ of open sets. Define

$$\Gamma := \bigcup_{G \in \mathcal{G}} G$$

Take any arbitrary $x \in \Gamma$. Then there is some G_i such that $x \in G_i$. As G_i is open there is some $r \in \mathbb{R}^+$ with

$$B(x; r) \subseteq G_i \subseteq \Gamma$$

Hence Γ is open. □

Proof of (b). Let $\mathcal{G} = \{G_i\}_{i \in [n]}$ be a finite family of open sets. Define

$$\Gamma := \bigcap_{i \in [n]} G_i$$

Suppose $x \in \Gamma$. Then $\bigwedge_{i=1}^{n} x \in G_i$ and for each G_i there is some $r_i > 0$ such that

$$B(x; r_i) \subseteq G_i$$

Then

$$B(x; \min\{r_i : 1 \leq i \leq n\}) = \bigcap_{i=1}^{n} B(x; r_i) \subseteq \bigcap_{i \in [n]} G_i = \Gamma$$

so Γ is open. □

Example 7.1 (Counter example for infinite intersection.). *Consider the metric space* $(\mathbb{R}, |\cdot|)$. *Consider the following infinite family of open sets*

$$\{(-1/n, 1/n)\}_{n \in \mathbb{N}}$$

Clearly,

$$\bigcap_{n \in \mathbb{N}} (-1/n, 1/n) = \{0\}$$

which is closed in $\mathbb{R}$.

Problem 7.5. *prob Consider* $X = \mathbb{R}$ *with the canonical metric* $d(x, y) = |x - y|$. *The support of a function* $f : \mathbb{R} \to \mathbb{R}$ *is the set of points* x *where* $f(x) \neq 0$. *Prove that the support of a continuous function is always open.* Note: *In the literature, the "support" of* f *is usually defined to be the closure of the above set, which is of course closed, not open.*

Proof. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous and let

$$\Omega := \{x \in \mathbb{R} : f(x) \neq 0\}.$$

Suppose, for a contradiction, that Ω is not open. Then there exists $x \in \Omega$ such that for every $r > 0$ there is some $y \in (x - r, x + r)$ with $f(y) = 0$.

Since f is continuous at x, for $\varepsilon := |f(x)| > 0$ there exists $\delta > 0$ such that

$$|f(y) - f(x)| < \varepsilon \quad \text{whenever} \quad |y - x| < \delta.$$

By assumption, there exists $y \in (x-\delta, x+\delta)$ with $f(y) = 0$. Then

$$|f(x)| = |f(x) - f(y)| < |f(x)|,$$

a contradiction. Hence Ω is open. $\square$

Problem 7.6. *Show that the equation* $\cos t = 2t$ *has a unique solution.* Hint: *Formulate the problem as finding the fixed point of a function* f.

Proof. Banach's fixed-point theorem tells us that for some complete metric space X, a contraction $F : X \to X$ will have a unique $t \in X$ such that

$$t = F(t)$$

Let $F : [0, \pi/2] \to [0, \pi/2]$ be defined by

$$F(t) := \frac{1}{2}\cos t$$

Clearly, $F(t) = t$ if and only if $\cos t = 2t$. Furthermore, as a closed bounded subset of a compelte emtric space, $([0, \pi/2], |\cdot|_{[0,\pi/2]})$ is complete. Now all we need is to show that F constitutes a contraction on $\mathbb{R}$. Notice that

$$\frac{1}{2}\cos(t) \in [0, 1/2],\ t \in [0, \pi/2]$$

so F ceirtanly constitutes a contraction. Thus by Banach's fixed point theorem there is a unique point $t \in [0, \pi/2]$ such that $t = F(t)$. t constitutes the unique solution to

$$\cos(t) = 2t$$

$\square$

8 3.4 & 3.5

Proposition 8.1. *Let (X, d) be a metric space. Let $f : X \to X$ be a contraction. Then f is continuous.*

Proof. Let $f : X \to X$ be a contraction. In other words, for some $\omega \in (0, 1)$, let f satisfy

$$d(f(x), f(y)) \leq \omega d(x, y), \ \forall x, y \in X$$

Let $\varepsilon > 0$. Define $\delta = \frac{\varepsilon}{\omega}$. Suppose $d(x, y) < \delta$. Then

$$d(f(x), f(y)) \leq \omega d(x, y) < \omega \cdot \delta = \varepsilon$$

$\square$

Recall that completeness is a very usefull property as if a sequence seems to converge then it does converge, and to a point in the space. However, sometimes it is hard to show that a sequence is cauchy. It would be nice to substitute this requirement to get some other way of showing convergence of sequence in a space. We could require that every sequence converges, but that is a ridiculously strict requirement and false most of the time. So instead, we require all sequences to have convergent subseqneces.

Recall the definition of a subsequence:

Definition 8.1 (Subsequence). *Let $\{x_n\}_{n\in\mathbb{N}}$ be a sequence. A subsequence is a sequence of the form $\{x_{n(k)}\}_{k\in\mathbb{N}}$, where $n(k) \in \mathbb{N}$ and*

$$n(1) < n(2) < n(3) < \cdots$$

Useful note: $n(k) \geq k$.

Proposition 8.2. *Let (X, d) be a metric space. If $\{x_n\}_n$ converges, then all possible subsequences converge, and the limit is the same.*

Proof. Let $\varepsilon > 0$, let $N \in \mathbb{N}$ be such that $d(x_n, x) < \varepsilon$. Then $n(k) \geq n \geq N$ so also

$$d(x_{n(k)}, x) < \varepsilon$$

$\square$

Definition 8.2 (Compactness). *A metric space (X, d) is said to be* compact *if every sequence $\{x_n\}_n$ has a convergent subsequence $\{x_n(k)\}_k$.*

Definition 8.3. *A subset $K \subseteq X$ of a metric space (X, d) is compact if (K, d) is compact.*

Theorem 8.1. *Let (X, d) be a metric space. If $K \subseteq X$ is finite then (K, d) is compact.*

Proof. Let $\{x_n\}_{n\in\mathbb{N}}$ be a sequence in $K = \{k_1, k_2, \dots, k_m\}$. As $\{x_n\}_n$ is infinite, there must be at least one element k_i where $x_j = k_i$ for infinitely many $j \in \mathbb{N}$. Then we can pick out indices such that $\{x_n(p)\}_{p\in\mathbb{N}}$ is a constant sequence for some $k_i \in K$ and thus convergent. □

Theorem 8.2. *Let (X, d) is compact. Then (X, d) is complete.*

Proof. Let $\{x_n\}_n$ be cauchy. As (X, d) is compact there exists some convergent $\{x_{n(k)}\}_k$ which converges to $x \in X$. For $\varepsilon > 0$, let $N_1 \in \mathbb{N}$ be s.t.

$$d(x_{n(k)}, x) < \varepsilon,\ n \geq N_1$$

and let $N_2 \in \mathbb{N}$ s.t.

$$d(x_n, x_m) < \varepsilon,\ n, m \geq N_2$$

Let $N = \max\{n(N_1), N_2\}$. If $m \geq N$ then

$$d(x_m, x) \leq \underbrace{d(x_m, x_{n(m)})}_{<\varepsilon} + \underbrace{d(x_{n(m)}, x)}_{<\varepsilon} < 2\varepsilon$$

Hence $\{x_n\}_n$ converges to $x \in X$. □

Theorem 8.3. *If $K \subseteq X$ is compact then it is closed and bounded.*

Proof. Let $K \subseteq X$ be compact.

Closed.

Let $\{x_n\}_n$ be a sequence in K converging to $x \in X$. By compactness we can find a subsequence $\{x_{n(k)}\}_k$ converging to some $y \in K$. Then $x = y \in K$, as the subsequence converges to the same value.

Bounded.

Assume K is not bounded. Fix $\overline{x} \in X$. Then for every $n \in \mathbb{N}$, there is some $x_n \in K$ with

$$d(x_n, \overline{x}) \geq n$$

Let $\{x_{n(k)}\}_k$ be a convergent subsequence of $\{x_n\}_n$. Then

$$n(k) \leq d(x_{n(k)}, \overline{x}) \underset{k\to\infty}{\to} d(x, \overline{x})$$

In other words, a value which tends to infinity is less than a fixed value. □

Proposition 8.3. *Let $X = \mathbb{R}^n$, $d(x,y) = ||x - y||$ (all norms are equivalent in $\mathbb{R}^n$). Let $K \subseteq \mathbb{R}^n$. Then*

$$K \text{ is compact} \Leftrightarrow K \text{ is closed and bounded}$$

Proof. The l2r direction is already covered so we tackle the other way.

Let K be closed and bounded. Let $\{x_n\}_n$ be a sequence in K. Then $\{x_n\}_n$ is bounded, so by Bolzano-Weierstrass, there is a convergent subsequence $\{x_{n(k)}\}_k$ converging to some $x \in \mathbb{R}^n$. Since K is closed $x \in K$. We conclude that K is compact. □

Theorem 8.4. *Let (X, d_X) and (Y, d_Y) be metric spaces with $f : X \to Y$ continuous. If $K \subseteq X$ is compact then $f[K]$ is, also.*

Theorem 8.5. *If (X, d) is compact and $f : X \to \mathbb{R}$ continuous. Then f attains a minimum and maximum.*

Definition 8.4. *Let (X, d) be a metric space. A set $K \subseteq X$ is* totally bounded *if $\forall \varepsilon > 0$, there exists finitely many points $x_1, \dots, x_n \in K$ such that*

$$K \subseteq \bigcup_{i=1}^{n} B(x_i; \varepsilon)$$

In other words, we say K is totally bounded if we can find finitely many open balls which cover it for any radius ε. This sounds a lot like the topological definition of compactness (Every open cover has a finite subcover). What we have here is a stronger property than meere boundedness.

Lemma 8.1. *Every compact set is totally bounded.*

Proof. Let $K \subseteq X$ be a compact subspace of a metric space (X, d), and assume it is not totally bounded. So K compact with

$$K \nsubseteq \bigcup_{i=1}^{n} B(x_i; \varepsilon)$$

for any choice of $x_1, \dots, x_n$ for some ε. Pick any $x_1 \in K$. Then $K \nsubseteq B(x_1; \varepsilon)$, so there is some $x_2 \in K \setminus B(x_1; \varepsilon)$. Iteratively pick

$x_1, \ldots, x_n$, pick any $x_{n+1} \in K \setminus \bigcup_{i=1}^{n} B(x_i; \varepsilon)$. Note: $d(x_n, x_m) \geq \varepsilon$ for any $x \neq m$. Then $\{x_n\}_n$ is a sequence in K, so has convergent subsequence with limit $x \in K$. But then $\varepsilon \leq d(x_n, x) + d(x, x_m) \underset{x,m\to\infty}{\to} 0$. □

Definition 8.5. *Let (X, d) be a metric space. A set $B \subseteq X$ is separable if there exists a dense, countable set $D \subseteq B$.*

Example 8.1.

- *$\mathbb{R}$ is separable ($\mathbb{Q} \subseteq \mathbb{R}$)*
- *Any subset of $\mathbb{R}$ is separable*
- *$B_1, B_2, \ldots$ separable, then*

$$\bigcup_{i\in\mathbb{N}} B_i$$

  is, also
- *$\ell^p(\mathbb{R})$ is separable for all $p < \infty$, but not $p = \infty$*

Theorem 8.6. *Let (X, d) be a complete metric space, $K \subseteq X$.*

$$K \text{ compact} \Leftrightarrow K \text{ totally bounded}$$

Theorem 8.7. *Every compact set is separable.*

9 3.6 - Compactness IV

Definition 9.1. *Let (X, d) be a metric space and let $K \subseteq X$. An* open covering *of K is a family of open sets $\mathcal{F}$ satisfying*

$$K \subseteq \bigcup_{\mathcal{O} \in \mathcal{F}} \mathcal{O}$$

Definition 9.2. *A set $K \subseteq X$ has the open covering property if for any open covering $\mathcal{F}$ of K, there exists $\mathcal{O}_1, \ldots, \mathcal{O}_n \in \mathcal{F}$ such that $\{\mathcal{O}_1, \ldots, \mathcal{O}_n\}$ is an open covering of K.*

Theorem 9.1. *A set $K \subseteq X$ is compact if and only if it has the open covering property.*

This is a very powerful classification of compactness and is the one you typically first encounter in a topology course. Notice that we don't need to assume completeness of X, closedness of K (as that comes for free), etc.

10 4.1 - Models of Continuity

Definition 10.1. *A function* $f : X \to Y$ *is*

- *continuous at* x *if* $\forall \varepsilon > 0$, $\exists \delta > 0$ *such that*
$$d_Y(f(x), f(y)) < \varepsilon$$
for every $y \in X$ *satisfying* $d_X(x, y) < \delta$.
- *continuous if it is continuous at all* $x \in X$.
- *uniformly continuous if* $\forall \varepsilon > 0$, $\exists \delta > 0$ *s.t.*
$$d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y))$$
$\forall x, y \in X$.
- *Lipschitz continuous if* $\exists C > 0$ *s.t.*
$$d_Y(f(x), f(y)) < C d_x(x, y), \quad \forall x, y \in X$$

with each subsequent property implying the one before it.

Definition 10.2. *Denote*
$$C(X, Y) = \{\text{all continuous } f : X \to Y\}$$
$$C(X) = C(X, \mathbb{R})$$

Lemma 10.1. *If* $f : A \to \mathbb{R}$ *(*$A \subseteq \mathbb{R}$*) satisfies* $|f'(x)| \leq M$, $\forall x \in A$, *then* f *is Lipschitz with constant* M.

Lemma 10.2. *If* $f : [a, b] \to \mathbb{R}$ *is continuously differentiable, then it is Lipschitz.*

Theorem 10.1. *Let* $f : X \to Y$ *be continuous and* (X, d_X) *compact. Then* f *is uniformly continuous.*

11 Problems from week of 9/2 to 15/2

Problem 11.1 (Problem 5, Spaces p.67). *Let (X, d) be a metric space. For a subset $A \subseteq X$, let ∂A denote the set of all boundary points of A. Recall that the closure of A is $\overline{A} = A \cup \partial A$.*

a) A subset A of X is called precompact *if $\overline{A}$ is compact. Show that A is precompact if and only if all sequences in A have a convergent subsequence.*

b) Show that a subset of $\mathbb{R}^m$ is precompact if and only if it is bounded.

Proof of (a). Notice that what we're really saying here is that A is precompact if and only if A is compact.

By an earlier theorem we have that if $A \subseteq X$ is compact then it is closed and bounded. Namely A is closed so $A = \overline{A}$. Thus it is clear to see that A is precompact if and only if it is compact. □

Proof of (b). By an earlier proposition we have that a $A \subseteq \mathbb{R}^m$ is compact if and only if it is closed and bounded. As we saw in (a) we know that A is precompact if and only if it is compact. □

Problem 11.2 (Problem 6, Spaces p.67). *Assume that (X, d) is a metric space and that $f : X \to [0, \infty)$ is continuous. Assume that for each $\varepsilon > 0$, there is a compact set $K_\varepsilon \subseteq X$ such taht $f(x) < \varepsilon$ when $x \notin K_\varepsilon$, Show that f has a maximum point.*

Proof. If $f \equiv 0$, then f trivially attains its maximum. Assume therefore that there exists $x_0 \in X$ such that $f(x_0) > 0$.

Let

$$\varepsilon := \frac{1}{2} f(x_0) > 0.$$

By assumption, there exists a compact set $K_\varepsilon \subseteq X$ such that

$$f(x) < \varepsilon \text{ for all } x \notin K_\varepsilon.$$

Since f is continuous and K_ε is compact, the restriction

$$f\,\big|_{K_\varepsilon} : K_\varepsilon \to [0, \infty)$$

attains a maximum. Let $x^* \in K_\varepsilon$ be such that

$$f(x^*) = \max_{x \in K_\varepsilon} f(x) =: M.$$

Note that
$$M \geq f(x_0) > \varepsilon.$$

Now let $x \in X$ be arbitrary. If $x \in K_\varepsilon$, then $f(x) \leq M$. If $x \notin K_\varepsilon$, then
$$f(x) < \varepsilon < M.$$

Hence in all cases,
$$f(x) \leq M = f(x^*).$$

Therefore f attains its maximum at x^*. □

Problem 11.3 (Problem 7 Spaces p.67). *Let (X, d) be a compact metric space, and assume that $f : X \to \mathbb{R}$ is continuous when $\mathbb{R}$ is given the usual metric. Show that if $f(x) > 0$ for every $x \in X$, then there is a positive, real number a such that $f(x) > a$ for every $x \in X$.*

Proof. Notice that what the claim gives us is that $f[X]$ is bounded below by some *positive* real number a. By assumption $f[X]$ is already bounded below by 0. We give a constructive argument for the existence of $a > 0$ as follows:

As f is continuous, X compact, we conclude that $f[X] \subseteq \mathbb{R}$ is compact. By the Heine-Borel theorem $f[X]$ is closed and bounded. Thus $\inf f[X]$ exists and furthermore $0 < \inf f[X] \in f[X]$. (We conclude that $\inf f[X]$ is strictly greater than 0 as $f(x) > 0$ by assumption). Let $a = \frac{1}{2} \inf f[X]$.

Clearly,
$$0 < a < \inf f[X]$$

so
$$0 < a < f(x), \forall x \in X$$

□

Problem 11.4 (Problem 8 Spaces p.68). *Assume that $f : X \to Y$ is a continuous function between metric spaces, and let K be a compact subset of Y. Show that $f^{-1}(K)$ is closed.*

Proof. Let $K \subseteq Y$ compact, $f : X \to Y$ continuous, and denote
$$\Omega := f^{-1}[K] = \{x \in X \mid f(x) \in K\}$$

Suppose Ω is open. Then there is some sequence $\{x_n\}_{n\in\mathbb{N}}$ in Ω which converges to a limit point $x \notin \Omega$. By sequential continuity we have that
$$f(x_n) \to f(x)$$

$f(x_n) \in K$. Since K is compact (and thus closed), we conclude that

$$f(x) \in K$$

so

$$x \in f^{-1}[K] = \Omega$$

In other words, $x \in \Omega$ and $x \notin \Omega$.

(No this did not need to be a proof by contradiction) □

Problem 11.5 (Problem 9 Spaces p.68). *Show that a totally bounded subset of a metric space is always bounded. Find an example of a bounded set in a metric space that is not totally bounded.*

Proof. Let (X, d) be a metric space with $K \subseteq X$ totally bounded. Recall that what we mean by totally bounded is that for every $\varepsilon > 0$, there exists finitely many $x_1, \dots, x_n \in K$ such that

$$K \subseteq \bigcup_{i=1}^{n} B(x_i; \varepsilon)$$

K is bounded if

$$K \subseteq B(x; r)$$

for some $x \in K$ and $r > 0$.

Pick $x := x_i$ for any $i \in [n]$, and let

$$r := \sup_{1 \le j \le n} d_X(x, x_j) + \varepsilon$$

Consider any $y \in B(x_j; \varepsilon)$. Then

$$d_X(y, x_j) < \varepsilon$$

and, by triangle inequality

$$d(y, x) \le d(y, x_j) + d(x_j, x)$$

Hence,

$$d(y, x) < \varepsilon + d(x_j, x)$$

Since $d(x_j, x) \le \sup_{1 \le j \le n} d(x, x_j)$, we get

$$d(y, x) < \varepsilon + \sup_{1 \le j \le n} d(x, x_j)$$

As r is defined,

$$\varepsilon + \sup_j d(x, x_j) = r$$

Thus

$$d(y, x) < r.$$

As $y \in B(x_j; \varepsilon)$ was arbitrarily chosen we may conclude that

$$K \subseteq B(x; r)$$

□

Problem 11.6 (Problem 11 Spaces p.68). *A metric space (X, d) is locally compact if there for each $a \in X$ is some $r > 0$ such that the closed ball $\overline{B}(a; r)$ is compact.*
a) Show that $\mathbb{R}^n$ is locally compact.
b) Show that if $X = \mathbb{R} \setminus \{0\}$, and $d : X \to \mathbb{R}$ is the metric defined by $d(x, y) = |x - y|$, then (X, d) is locally compact, but not complete.

Proof of (a). Let $a \in \mathbb{R}^n$ and take *any* $r > 0$. Clearly

$$\overline{B}(a; r)$$

is closed and furthermore

$$\overline{B}(a; r) \subseteq \overline{B}(a; r)$$

so bounded.

By the Heine-Borel theorem $\overline{B}(a; r)$ is compact. As $a \in \mathbb{R}^n$ was chosen arbitrarily we conclude that $\mathbb{R}^n$ is locally compact. $\square$

Proof of (b). Pick any $a \in X$. If $a > 0$ we define the closed ball

$$a \pm \frac{|a|}{2}$$

which is clearly closed and bounded (thus compact).

The space is not complete because

$$\left\{\frac{1}{n}\right\}_{n \in \mathbb{N}} \underset{n \to \infty}{\to} 0 \notin X$$

i.e. we have a cauchy sequence which does not converge to a point in the space. $\square$

Problem 11.7 (Problem 15 Spaces p.68). *Assume that C and K are disjoint, compact subsets of a metric space (X, d), and define*

$$a = \text{inf}\left\{d(x, y) \mid x \in C, y \in K\right\}$$

Show that a is strictly positive and that there are point $x_0 \in C$, $y_0 \in K$ such that $d(x_0, y_0) = a$. Show by an example that the result does not hold if we only assume that one of the sets is compact and the other is closed.

Proof. As the metric is non-negative we can clearly see that $a \geq 0$. Assume, in search of contradiction, that $a = 0$. Then, for every $n \in \mathbb{N}$ there exists $x_n \in C$, $y_n \in K$ such that

$$d(x_n, y_n) < \frac{1}{n}$$

C is compact so you can take a convergent subsequence

$$\{x_n(k)\}_k \to x_0 \in C$$

Then, as

$$d(x_{n(k)}, y_{n(k)}) \to 0$$

we see

$$\{y_{n(k)}\}_k \to x_0 \in K$$

and thus conclude

$$x_0 \in C \cap K \neq \emptyset$$

contradiction our assumption. Hence $a > 0$.

Now consider two sequences $\{x_n\}_n \subseteq C$ and $\{y_n\}_n \subseteq K$ such that

$$d(x_n, y_n) \underset{n\to\infty}{\to} a$$

By sequential continuity (metric is continuous) we get that

$$d(x_n, y_n) \to d(x, y)$$

and by uniqueness of limit point we conclude that

$$d(x, y) = a$$

Now, as C and K are compact we can find convergent subsequences which must then again converge to x and y in their respective sets. In other words,

$$x \in C \text{ and } y \in K$$

Therefore you can find points in C and K such that the minimum distance is attained.

(Perhaps this could have been shown by arguing that as d is continuous and thus the set of distances between C and K are compact in $\mathbb{R}$ ($\Rightarrow$ closed and bounded), the extreme-value theorem applies?) $\square$

Problem 11.8 (Problem 2 Spaces p.80). *Prove that $f : (0, 1) \to \mathbb{R}$ given by $f(x) = \frac{1}{x}$ is not uniformly continuous.*

Proof. Recall that a function $f : X \to Y$ is uniformly continuous if

$$\forall \varepsilon > 0,\ \exists \delta > 0 : d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)),\ \forall x, y \in X$$

Crucially there must be a δ for every ε which works at every point similtaneously. Informally, the problem with $\frac{1}{x}$ is that it grows without bound when $x \to 0$.

Consider what happens as $x \to 0$. Let $x_n = \frac{1}{n}$ and $y = \frac{1}{n+1}$. Then

$$d_X(x_n, y_n) \underset{n\to\infty}{\to} 0$$

but

$$d_Y(f(x_n), f(y_n)) = |n - (n+1)| = 1$$

So we can fix $\varepsilon_0 = 1/2$ and for any $\delta > 0$ we can pick n large enough so that

$$d_X(x_n, y_n) < \delta$$

but

$$d_Y(f(x_n), f(y_n)) = 1 > \varepsilon_0$$

□

Problem 11.9 (Problem 4 Spaces p.80). *Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function and assume that the derivative f' is bounded. Show that f is uniformly continuous.*

Proof. We know that f is a real-valued function such that there exists $M \in \mathbb{N}$ for which

$$\left|\frac{d}{dx} f(x)\right| = \left|\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}\right| \leq M,\ \forall x \in \mathbb{R}$$

Suppose f is not uniformly continuous. Then it is ceirtainly not Lipschitz continuous. I.e. for every constant C there are points x_C, y_C such that

$$|f(x_C) - f(y_C)| > C|x_C - y_C|$$

Specifically:

$$\begin{aligned} |f(x_M) - f(y_M)| &> M|x_M - y_M| \\ \left|\frac{f(x_M) - f(y_M)}{x_M - y_M}\right| &> M \\ M \geq \left|\frac{f(x_M) - f(y_M)}{x_M - y_M}\right| &> M \end{aligned}$$

Our assumption that f is not Lipschitz continuous has led to a contradiction, so f is Lipschitz and hence uniformly continuous. □

Problem 11.10 (A reach problem from section 4.6). *Assume that $X \subset \mathbb{R}^n$ is not compact. Show that there is an unbounded continuous function $f : X \to \mathbb{R}$.*

Proof. Recall that from the Heine-Borel theorem a subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded. Therefore we may conclude that X is either open or unbounded. In the case where X is unbounded we construct the function $f : X \to \mathbb{R}$ with

$$f(x) = ||x||$$

which very clearly is continuous and not bounded.

Now suppose X is open. Fix some $y \in \partial X \setminus X$. Define $f : X \to \mathbb{R}$ as

$$f(x) = \frac{1}{||x - y||} = \frac{1}{d_X(x, y)}$$

□

12 Spaces 4.2 - Models of convergence

Definition 12.1. *Let $(X, d_X), (Y, d_Y)$ be metric spaces. Let $\{f_n\}_{n\in\mathbb{N}}$ be a sequence of functions with domain X and co-domain Y. Let $f : X \to Y$ be another function. We say that $\{f_n\}_{n\in\mathbb{N}}$ converges to f* pointwise *if $\forall \varepsilon > 0$ and $x \in X$ there exists $n \in \mathbb{N}$ such that*

$$d_Y(f_n(x), f(x)) < \varepsilon, \; \forall n \geq N$$

and will often write $f_n \underset{n\to\infty}{\to} f$ pointwise. We say that $\{f_n\}_{n\in\mathbb{N}}$ converges to f uniformly *if $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that*

$$d_Y(f_(x), f(x)) < \varepsilon, \; \forall n \geq N \text{ and } \forall x \in X$$

and we often write $f_n \underset{n\to\infty}{\to} f$ uniformly. The difference being that for uniform convergence, the choice of N doens't depend on $x \in X$.

Proposition 12.1. *Let $\{f_n\}_n$ be a sequence of $f_n : X \to Y$, and let $f : X \to Y$. Then the following are equivalent.*

1. *$f_n \to f$ pointwise.*
2. *$f_n(x) \to f(x)$ for every $x \in X$.*

Definition 12.2. *For $f, g : X \to Y$, define the* supremum metric

$$\rho(f, g) = \sup_{x\in X} d_Y(f(x), g(x))$$

also denoted

$$d_\infty(f, g), \;\; ||f - g||_\infty \text{ or } ||f - g||_{L^\infty}$$

Proposition 12.2. *Let $\{f_n\}_n$ be a sequence of functions from $X \to Y$, and let $f : X \to Y$. Then the following are equivalent.*

1. *$f_n \to f$ uniformly.*
2. *$\rho(f_n, f) \underset{n\to\infty}{\to} 0$.*

Example 12.1. *Let $f_n, f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, $f_n(x) = \min(n, x^2)$. Show that f_n converges to f pointwise, but not uniformly.*

Proof. For any $x \in \mathbb{R}$, we have $f_n(x) = f(x)$ for every $n \geq x^2$, so $f_n(x) \to f(x)$. Next $\rho(f_n, f) = \sup_{x\in\mathbb{R}} |f_n(x) - f(x)| = \sup_{x\in\mathbb{R}} |\min(n, x^2) - x^2| = \infty$. $\square$

Theorem 12.1. *Let $\{f_n\}_n$ be a sequence of $f_n : X \to Y$ and suppose $f_n \to f$ uniformly for some $f : X \to Y$.*
Then, f is continuous.

13 Some problems for week of Feb 16th

Problem 13.1. *Let $f_n : \mathbb{R} \to \mathbb{R}$ and assume that the sequence $\{f_n\}_n$ of continuous functions converges uniformly to $f : \mathbb{R} \to \mathbb{R}$ on all intervals $[-k, k]$, $k \in \mathbb{N}$. Show that f is continuous.*

Proof. Let $x_0 \in \mathbb{R}$ be arbitrary. We prove that f is continuous at x_0.

Choose $k \in \mathbb{N}$ such that $k > |x_0| + 1$. Then a neighbourhood of x_0 is contained in $[-k, k]$. By assumption, $f_n \to f$ uniformly on $[-k, k]$.

Let $\varepsilon > 0$ be given. Since the convergence is uniform on $[-k, k]$, there exists $N \in \mathbb{N}$ such that for all $x \in [-k, k]$ and all $n \geq N$,

$$|f_n(x) - f(x)| < \frac{\varepsilon}{3}.$$

Fix such an N. Because f_N is continuous at x_0, there exists $\delta > 0$ such that

$$|x - x_0| < \delta \quad \Rightarrow \quad |f_N(x) - f_N(x_0)| < \frac{\varepsilon}{3}.$$

We may assume $\delta \leq 1$, so that $|x - x_0| < \delta$ implies $x \in [-k, k]$.

For such x we estimate:

$$\begin{aligned} |f(x) - f(x_0)| &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Thus,

$$|x - x_0| < \delta \quad \Rightarrow \quad |f(x) - f(x_0)| < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, f is continuous at x_0. Because x_0 was arbitrary, f is continuous on $\mathbb{R}$. $\square$

Problem 13.2 (Problem 9, Spaces p.85). *Let (X, d) be a metric space and assume that the sequence f_n of continuous functions on X converges uniformly to f. Show that if $\{x_n\}_n$ is a sequence in X converging to x, then $f_n(x_n) \to f(x)$. Find an example which shows that this is not necessarily the case if the sequence of functions converges pointwise.*

Proof. Recall that, as $f_n \to f$ uniformly, we can for every ε, find $N \in \mathbb{N}$ such that

$$d(f_n(x), f(x)) < \varepsilon, \ \forall n \geq N \text{ and } \forall x \in X$$

As f_n is continuous we can use the property of sequential continuity. I.e.

$$f_n(x_{n'}) \underset{n' \to \infty}{\to} f_n(x)$$

Thus

$$d(f_n(x_{n'}), f_n(x)) \to 0,\ n' \to \infty$$

Let $\varepsilon > 0$. Pick $N \in \mathbb{N}$ such that

$$d(f_n(x_n), f_n(x)) < \varepsilon/2$$

and

$$d(f_n(x), f(x)) < \varepsilon/2,\ \forall x \in X$$

Then

$$d(f_n(x_n), f(x)) \leq d(f_n(x_n), f_n(x)) + d(f_n(x), f(x)) < \varepsilon$$

$\square$

Problem 13.3 (Problem 4.5.1, Spaces p.100). *Let $f, g : [0,1] \to \mathbb{R}$ be given by $f(x) = x$, $g(x) = x^2$. Find $\rho(f,g)$.*

Solution.

$$\rho(f,g) = \sup_{0 \leq x \leq 1} |g(x) - f(x)|$$

As f, g continuous we know that there exists extremal values in $[0,1]$. $|f(0) - g(0)| = 0$ and $|f(1) - f(1)| = 0$ so the endpoints are not it. We know that there must be some point in $[0,1]$ where a maxima or minima of $g(x) - f(x)$ is attained.

$$\begin{aligned} g(x) - f(x) &= x^2 - x \\ &= x(x-1) \\ \frac{d}{dx}(g-f)(x) &= 2x - 1 \\ 2x - 1 &= 0 \\ x &= \frac{1}{2} \end{aligned}$$

$(g-f)(1/2) = -1/4$ so

$$\rho(f,g) = \frac{1}{4}$$

$\square$

Problem 13.4 (Problem 4.5.7, Spaces p.101). *For $f \in B(\mathbb{R}, \mathbb{R})$ and $r \in \mathbb{R}$, we define a function f_r by $f_r(x) = f(x+r)$.*
a) Show that if f is uniformly continuous, then $\lim_{r \to 0} \rho(f_r, f) = 0$.
b) Show that the function g defined by $g(x) = \cos(\pi x^2)$ is not uniformly continuous on $\mathbb{R}$.

c) Is it true that $\lim_{r\to 0}\rho(f_r, f) \to 0$ *for all* $f \in B(\mathbb{R}, \mathbb{R})$*?*

Proof of (a). Suppose $f \in B(\mathbb{R}, \mathbb{R})$ is uniformly continuous. Namely, $\forall \varepsilon > 0$, $\exists \delta > 0$ for which

$$|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon,\ \forall x, y \in \mathbb{R}$$

We are asked to show that

$$\lim_{r\to 0} \rho(f_r, f) = 0$$

In other words

$$\sup_{x\in\mathbb{R}} |f(x + r) - f(x)| \underset{r\to 0}{\to} 0$$

Notice then that $f(x + r)$ converges to $f(x)$ for every x. Let $\varepsilon > 0$ and choose δ from uniform continuity. If $|r| < \varepsilon$, then for every $x \in \mathbb{R}$,

$$|(x + r) - r| = |r| < \delta$$

Therefore

$$|f(x + r) - f(x)| < \varepsilon,\ \forall x \in \mathbb{R}$$

Hence

$$\rho(f_r, f) \leq \varepsilon \text{ whenever } |r| < \delta$$

so

$$\lim_{r\to 0} \rho(f_r, f) = 0$$

$\square$

Problem 13.5. *The space* $C_b(X, Y)$ *is always "larger" than* Y*, in the sense that* Y *can be embedded in* $C_b(X, Y)$*:*
Indeed, show that the map $i : Y \to C_b(X, Y)$ *which maps* $y \in Y$ *to the constant function* $f(x) \equiv y$*, is an embedding (cf. Definition 3.1.3).*
Show that $i(Y)$ *is precisely the subset of constant functions, and that this set is a closed subset of* $C_b(X, Y)$*.*
Conclude that $C_b(X, Y)$ *is complete if and only if* Y *is complete.*

Proof sketch. Define $i : Y \to C_b(X, Y)$ by

$$y \mapsto f(x) \equiv y$$

We begin by showing the necessary condition of an embedding before arguing that it is injective (as the latter will follow clearly from the former).

Let $y_1, y_2 \in Y$ and denote constant functions which map to them by f_{y_1}, f_{y_2} respectively.

$$\begin{aligned}\rho(f_{y_1}, f_{y_2}) &= \sup_{x \in X} d_Y(f_{y_1}(x), f_{y_2}(x)) \\ &= \sup\{d_Y(y_1, y_2), d_Y(y_1, y_2) \dots\} \\ &= d_Y(y_1, y_2)\end{aligned}$$

It is clear that for distinct y_1, y_2 their images are distinct since otherwise the supremum distance between f_{y_1}, f_{y_2} would be 0, but then it would follow that $d_Y(y_1, y_2) = 0$. Therefore we conclude that i is an injection.

From a similar argument we can also see that there necessarily exists one and only one constant function f_y for every $y \in Y$, so $i[Y]$ must be exactly the set of all of these, and therefore we could create a one-to-one mapping from y's and constant functions. □

Problem 13.6 (Problem 4.6.1, Spaces p.102). *Let* $X, Y = \mathbb{R}$. *Find functions* $f, g \in C(X, Y)$ *such that*

$$\sup_x\{d_Y(f(x), g(x))\} = \infty$$

Solution. Consider $f, g : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = x, \quad g(x) = -x$$

As $\mathbb{R}$ is finite-dimensional, every norm-induced metric is equivalent so we work with $d_Y(x, y) = |x - y|$. Thus

$$\rho(f, g) = \sup_x |f(x) - g(x)| = 2 \sup_x |x|$$

As $\mathbb{R}$ is unbounded $\sup_x |x| = \infty$. □

Problem 13.7 (Problem 4.6.2, Spaces p.102). *Assume that* $X \subseteq \mathbb{R}^n$ *is not compact. Show that there is an unbounded, continuous function* $f : X \to \mathbb{R}$.

Proof. Recall that, from the Heine-Borel theorem, X is either unbounded or open. If X is unbounded we can trivially find the unbounded continuous function

$$f : X \to \mathbb{R}, f(x) = x$$

Constant functions are (Lipschitz) continuous and this function f is unbounded as X is unbounded. (For every $M > 0$ there exists $x \in X$ such that $||x|| > M$).

If X is open we can define $f : X \to \mathbb{R}$ as follows. Fix some $y \in \partial X$ ($y \notin X$ by definition) and let

$$f(x) = \frac{1}{||x - y||}$$

can take on arbitrarily large values. (Consider some sequence in X converging to y will always be well-defined for x_n, but grow undbounded). $\square$

Problem 13.8 (Problem 4.6.3, Spaces p.102). *Assume that $f : \mathbb{R} \to \mathbb{R}$ is a bounded continuous function. If $u \in C([0,1], \mathbb{R})$, we define $L(u) : [0,1] \to \mathbb{R}$ to be the function*

$$L(u)(t) \int_0^1 \frac{1}{1+t+s} f(u(s)) ds.$$

a) Show that L is a function from $C([0,1], \mathbb{R})$ to $C([0,1], \mathbb{R})$.
b) Show that ...

Proof of (a). proof $\square$

14 Integrating sequences of functions

Theorem 14.1. *Let $f_n : [a,b] \to \mathbb{R}$ be continuous $\forall n \in \mathbb{N}$ and assume $f_n \to f$ uniformly. Let $F_n(x) = \int_a^x f_n(t)\,dt$, $F(x) = \int_a^x f(t)\,dt$. Then $F_n \to F$ uniformly.*

Proof. Let $\varepsilon > 0$ and let $N \in \mathbb{N}$ be such that

$$\rho(f_n, f) < \varepsilon$$

when $n \geq N$. Then

$$\begin{aligned} |F_n(x) - F(x)| &= \left| \int_a^x f_n(t) - f(t)\,dt \right| \\ &\leq \int_a^x |f_n(t) - f(t)|\,dt \\ &\leq \int_a^x \rho(f_n, f)\,dt \\ &= (x-a)\rho(f_n, f) < (b-a)\varepsilon \end{aligned}$$

□

Next we look at the following. Given $v_1, v_2, \ldots : [a,b] \to \mathbb{R}$, what do we mean by

$$\sum_{n=1}^{\infty} v_n(x) \text{ ?}$$

Recall that if $a_1, a_2, \ldots$ are numbers we say that

$$\sum_{n=1}^{\infty} a_n \textbf{ converges}$$

if the partial sums $S_n = \sum_{n=1}^{N} a_n$ converge as $N \to \infty$.

Definition 14.1. *Let (X,d) be a metric space. Let $A \subseteq X$ and let $\{v_n\}_{n\in\mathbb{N}}$ be a sequence of functions $v_n : A \to \mathbb{R}$. We say that the series $\sum_{n=1}^{\infty} v_n$*

- ***converges pointwise** if for every $x \in A$, the partial sums $S_N(x) = \sum_{n=1}^{N} v_n(x)$ converge as $N \to \infty$*
- ***converges uniformly** if the partial sums S_N converge uniformly as $N \to \infty$.*

Lemma 14.1 (Weierstrass M-test)**.** *Assume that there are numbers $M_n \geq 0$ such that*

- $|v_n(x)| \leq M_n,\ \forall x \in A$
- $\sum_{n=1}^{\infty} M_n < \infty$

Then $\sum_{n=1}^{\infty} v_n$ converges uniformly.

Theorem 14.2. *Let $v_n : [a,b] \to \mathbb{R}$ be continuous $\forall n \in \mathbb{N}$ and assume $\sum_{n=1}^{\infty} v_n$ converges uniformly. Let $V_n(x) := \displaystyle\int_a^x v_n(t)\, dt$.*

Then the series $\sum_{n=1}^{\infty} V_n$ converges uniformly, and

$$\sum_{n=1}^{\infty} V_n(x) = \int_a^x \sum_{n=1}^{\infty} v_n(t)\, dt,\ \ \forall x \in [a,b]$$

Theorem 14.3. *Let $f_n : [a,b] \to \mathbb{R}$ be continuously differentiable $\forall n \in \mathbb{N}$. Assume*

- *$\{f_n'\}_{n\in\mathbb{N}}$ converges uniformly to some $g : [a,b] \to \mathbb{R}$*
- *$\{f_n(x_0)\}_{n\in\mathbb{N}}$ converges to some $\alpha \in \mathbb{R}$, for some $x_0 \in \mathbb{R}$*

Then $f_n \to f$ uniformly for some $f : [a,b] \to \mathbb{R}$ with $f' = g$ and $f(x_0) = \alpha$.

Theorem 14.4. *If $\{u_n\}_n$ is a sequence of continuously differentiable functions $u_n : [a,b] \to \mathbb{R}$ such that $\sum_{n=1}^{\infty} u_n'$ converges uniformly and $\sum_{n=1}^{\infty} u_n(x_0)$ converges for some $x_0 \in [a,b]$, then $\sum_{n=1}^{\infty} u_n$ converges uniformly, and*

$$\frac{d}{dx}\left(\sum_{n=1}^{\infty} u_n\right) = \sum_{n=1}^{\infty}\left(\frac{d}{dx}u_n\right)$$

Problem 14.1 (Problem 1, Spaces p.91). *Show that* $\sum_{n=1}^{\infty} \frac{\cos(nx)}{n^2+1}$ *converges uniformly on* $\mathbb{R}$*.*

Proof. We prove the claim by applying the Weierstrass M-test.

It is clear that $\cos(nx)/(n^2+1)$ is bounded by $\frac{1}{n^2+1}$ as $|\cos(nx)| \leq 1$. Furthermore

$$\sum_{n=1}^{\infty} \frac{1}{n^2+1} \leq \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} < \infty$$

By the Weierstrass M-test,

$$\sum_{n=1}^{\infty} \frac{\cos(nx)}{n^2+1}$$

converges uniformly on $\mathbb{R}$. $\square$

14.1 Power Series

Definition 14.2. *A* ***power series*** *is an expression of the form*

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

where $a \in \mathbb{R}$ and $c_0, c_1, \ldots \in \mathbb{R}$ are fixed, and x takes values in $\mathbb{R}$.

Example 14.1.

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \ln x = -\sum_{n=1}^{\infty} \frac{(1-x)^n}{n}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$$

Definition 14.3. *A function $f : B \to \mathbb{R}$ (for open $B \subseteq \mathbb{R}$) is* ***real analytic*** *if there are $c_0, c_1, \ldots \in \mathbb{R}$ and $a \in B$ such that*

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n, \ \ \forall x \in B$$

Notably, the series must converge at all $x \in B$.

Definition 14.4. *The* ***radius of convergence*** *of a power series $\sum_{n=0}^{\infty} c_n (x-a)^n$ is*

$$R = \frac{1}{\limsup_{n\to\infty} |c_n|^{1/n}}$$

where we use the convention that $1/0 = \infty$ and $1/\infty = 0$.

Proposition 14.1. *The power series converges if $|x-a| < R$, and diverges if $|x-a| > R$. It converges uniformly if on every interval $[a-r, a+r]$ for $r < R$.*

Proof. Recall that

$$R = \frac{1}{\limsup_{n\to\infty} |c_n|^{1/n}}$$

If $|x-a| > R$ then $\frac{1}{|x-a|} < \frac{1}{R}$, so for any $N \in \mathbb{N}$ there is an $n \geq N$ such that

$$\frac{1}{|x-a|} < |c_n|^{1/n}$$

but then $|c_n(x-a)^n| > 1$. Hence the sum cannot converge.

If $R = 0$ we are done. Otherwise let $r < R$. Then $\frac{1}{r} > \frac{1}{R} = \limsup_{n\to\infty} |c_n|^{1/n}$, so for some $N \in \mathbb{N}$ we have $|c_n|^{1/n} \leq \eta$, $\forall n \geq N$, where $\eta = \frac{1}{2}(\frac{1}{r} + \frac{1}{R}$. Note that $\eta^r < \frac{1}{r}r = 1$. We now have $|c_n(x-a)^n| = |c_n||x-a|^n \leq 2\eta^n r^n = (\eta r)^n$, $\forall n \geq N$, so $\sum_{n=0}^{\infty} |c_n(x-a)^n| \leq \sum_{n=0}^{N-1} |c_n| r^n + \sum_{n=N}^{\infty} (\eta r)^n < \infty$. Thus by Weierstrass M-test, the series converges uniformly on $[a-r, a+r]$. □

Theorem 14.5. *Let R be the radius of convergence of the power series*

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n,$$

and assume $R > 0$. Then f is continuous and differentiable on $(a-R, a+R)$, and

$$\frac{d}{dx} f(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1} = \sum_{n=0}^{\infty} (n+1) c_{n+1} (x-a)^n$$

Theorem 14.6. *Let R be the radius of convergence of the power series $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$, and assume $R > 0$. Then all derivatives of f exist, and $f^{(k)}(a) = k! c_k$, so*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

I.e. a power seris can be written as it's Taylor-series.

Problem 14.2 (Problem 2, Spaces p.91). *Show that $\sum_{n=1}^{\infty} \frac{x}{n^2}$ converges uniformly on $[-1, 1]$.*

Proof. We proceed by Weierstrass M-test.

Notice that for x in the closed unit-interval, $|\frac{x}{n^2}| \leq \frac{1}{n^2} =: M_n$.

Then it is well known from Euler's classic result that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} < \infty$$

We conclude that

$$\sum_{n=1}^{\infty} \frac{x}{n^2}$$

converges uniformly on $[-1, 1]$. ☐

Problem 14.3 (Problem 3, Spaces p.91). *Let $f_n : [0,1] \to \mathbb{R}$ be defined by $f_n(x) = nx(1-x^2)^n$. Show that $f_n(x) \to 0$ for all $x \in [0,1]$, but that $\int_0^1 f_n(x)\, dx \to 1/2$.*

Proof. Let $f : [0,1] \to \mathbb{R}$ be defined by

$$f(x) = \sum_{n=1}^{\infty} f_n(x) = \sum_{n=1}^{\infty} nx(1-x^2)^n$$

Notice that for x in the closed unit interval,

$$|nx(1-x^2)^n| \leq \frac{n}{2}\frac{1}{4^n} = \frac{n}{2 \cdot 4^n} \leq \frac{n}{4^{n+1}} \leq \frac{1}{4^{n+1}} =: M_n$$

and

$$\sum_{n=1}^{\infty} \frac{1}{4^{n+1}} < \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} < \infty$$

So by Weierstrass M-test, $\sum_{n=1}^{\infty} f_n(x)$ converges (uniformly) so $f_n \to 0$.

As f_n are all continuous for $n \in \mathbb{N}$ and we showed that $\sum_{n=1}^{\infty} f_n$ converges uniformly by M-test so we can interchange limits when taking the integral of f_n's. In other words

$$\lim_{n\to\infty} \int_0^1 f_n(x)\, dx = \int_0^1 \lim_{n\to\infty} f_n(x)\, dx$$

Computing the integral:

$$\begin{aligned}
\int_0^1 f_n(x)\, dx &= \int_0^1 nx(1-x^2)^n\, dx \\
&= \left[\frac{n}{2(n+1)}(1-x^2)^{n+1}\right]_{x=0}^{1} \\
&= \frac{n}{2(n+1)}
\end{aligned}$$

Taking the limit:

$$\begin{aligned}
\lim_{n\to\infty} \frac{n}{2(n+1)} &= \frac{1}{2} \lim_{n\to\infty} \frac{n}{n+1} \\
&= \frac{1}{2}
\end{aligned}$$

as desired. ☐

Problem 14.4 (Exam problem 1, 2021-kont). *Consider $\mathbb{R}^2$ equipped with the Euclidian norm $||x|| = \sqrt{x_1^2 + x_2^2}$ and the corresponding metric $d(x, y) = ||x - y||$. Define the unit circle $S = \{x \in \mathbb{R}^2 \mid ||x|| = 1\}$. Explain why*

- *(S, d) is a metric space,*
- *but $(S, ||\cdot||)$ is not a normed vector space.*

Solution. It is clear that (S, d) is a metric space as $||\cdot||$ is a well defined norm on $S \subseteq \mathbb{R}^2$ and $d(x, y) := ||x - y||$ always constitutes a metric. $(S, ||\cdot||)$ is not a normed space because it is not even a vector space. $(0, 1), (1, 0) \in S$, but $||(0, 1) + (1, 0)|| > 1$ so S is not closed. □

Problem 14.5 (Exam problem 2, 2021-kont). *Prove that $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = e^{-x^2}$ is uniformly continuous.*

Proof. Let $f(x) = e^{-x^2}$. Then

$$f'(x) = -2xe^{-x^2}.$$

Hence

$$|f'(x)| = 2|x|e^{-x^2}.$$

Consider $g(x) = 2xe^{-x^2}$ for $x \geq 0$. Then

$$g'(x) = 2e^{-x^2}(1 - 2x^2).$$

Thus $g'(x) = 0$ when $x = \frac{1}{\sqrt{2}}$, and

$$|f'(x)| \leq g\left(\frac{1}{\sqrt{2}}\right) = \sqrt{2}e^{-1/2}.$$

Therefore f' is bounded on $\mathbb{R}$. By the Mean Value Theorem,

$$|f(x) - f(y)| \leq M|x - y|$$

for some constant M, so f is Lipschitz continuous and hence uniformly continuous on $\mathbb{R}$. □

15 Problems week of March 2

Problem 15.1 (Problem from Spaces sec. 4.7). *Let f be the function $f(y,t) = y$. Choose initial data, say, $\bar{y} = 1$. Perform a fixed point iteration of the equation $y(t) = \bar{y} + \int_0^t f(y(s), s)\,ds$ that is, for some continuous function y^0 (here, the superscripts are indices, not powers) let $y^{n+1}(t) = \bar{y} + \int_0^t f(y^n(s), s)\,ds$ for $n = 0, 1, 2, \ldots$. (It's easiest to start with $y^0 \equiv 0$.) Compute y^1, y^2, y^3, y^4. Give an expression for y^n for any n, and prove that $y^n \to y$, where $y(t) = e^t$.*

Solution and proof.

$$
\begin{aligned}
y^1(t) &= 1 + \int_0^t 0\ ds \\
&= 1 + t \\
y^2(t) &= 1 + \int_0^t 1 + s\ ds \\
&= 1 + \left[s + \frac{s^2}{2} \right]_{s=0}^t \\
&= 1 + t + \frac{t^2}{2} \\
y^3(t) &= \cdots \\
&= 1 + t + \frac{t^2}{2} + \frac{t^3}{3!}
\end{aligned}
$$

and so on to arrive at

$$y^n(t) = \sum_{n=0}^{n-1} \frac{t^n}{n!}$$

It is well known that the taylor expansion of e^t is precisely

$$e^t = \sum_{n=0}^{\infty} \frac{t^n}{n!}$$

Let $\varepsilon > 0$ for $t \in \mathbb{R}$ choose $N \in \mathbb{N}$ such that

$$\left| \frac{t^n}{n!} \right| + \left| \frac{t^{n+1}}{(n+1)!} \right| + \cdots < \varepsilon,\ n \geq N$$

which we can do since $n!$ grows faster than any polynomial so the terms go to 0. Then

$$\left|\sum_{k=0}^{n-1} \frac{t^k}{k!} - e^t\right| = \left|\sum_{k=n}^{\infty} \frac{t^k}{k!}\right| \leq \sum_{k=n}^{\infty} \left|\frac{t^k}{k!}\right| < \varepsilon$$

□

Problem 15.2 (Problem 4.8.3 Spaces p.111). *Assume that (X, d) is a metric space, and that A is a dense subset of X. We let A have the subset metric d_A.*
a) Assume that $f : A \to \mathbb{R}$ is uniformly continuous. Explain that if $\{a_n\}$ is a sequence from A converging to a point $x \in X$, then $\{f(a_n)\}$ converges. Show that the limit is the same for all such sequences $\{a_n\}$ converging to the same point x.
b) Define $f : X \to \mathbb{R}$ by putting $\overline{f}(x) = \lim_{n\to\infty} f(a_n)$ where $\{a_n\}$ is a sequence from A converging to x. We call $\overline{f}$ the continuous extension *of f to X. Show that f is uniformly continuous.*
c) Let $f : \mathbb{Q} \to \mathbb{R}$ be defined by

$$f(q) = \begin{cases} 0 & \text{if } q < \sqrt{2} \\ 1 & \text{if } q > \sqrt{2} \end{cases}$$

Show that f is continuous on $\mathbb{Q}$ (we are using the usual metric $d_{\mathbb{Q}}(q, r) = |q - r|$). Is f uniformly continuous?
d) Show that f does not have a continuous extension to $\mathbb{R}$.

Proof of (a). Suppose $\{a_n\}$ is some sequence in A converging to a point $x \in X$. Let $f : A \to \mathbb{R}$ be some uniformly continuous function.

For any $\varepsilon > 0$ we can find $\delta > 0$ such that

$$d_A(a_n, a_m) < \delta \Rightarrow |f(a_n) - f(a_m)| < \varepsilon$$

so since $\{a_n\}$ is cauchy we conclude that $\{f(a_n)\}$ is cauchy and thus convergent since $\mathbb{R}$ is complete.

Suppose $\{a_n\}, \{b_n\}$ are sequences which both converge to $x \in \mathbb{X}$. Then

$$d(a_n, b_n) \leq d(a_n, x) + d(x, b_n) \to 0$$

so for large enough n we have

$$|f(a_n) - f(b_n)| < \varepsilon$$

in particular

$$\lim_{n\to\infty} f(a_n) = \lim_{n\to\infty} f(b_n)$$

□

Proof of (b). We are given that $f : A \to \mathbb{R}$ is uniformly continuous. For some $x \in A$ we have constant sequences $\{x\}_n$ for which $\lim_{n\to\infty} f(x) = f(x)$ so $\overline{f}$ is well-defined on A. For a dense subset $A \subseteq X$ we can, for every $x \in X$ and $\varepsilon > 0$, find $a \in A$ such that

$$d(x, a) < \varepsilon$$

So we can construct, for $x \in X$, sequences $\{a_n\}$ which minimize the distance

$$d(x, a_n)$$

with $a_n \to x$ when $n \to \infty$, so $\overline{f}$ is well-defined on $X \setminus A$. Now to check uniform continuity we ignore checking on A as this follows trivially. For $x, y \in X \setminus A$ we can find sequences $\{x_n\}, \{y_n\}$ in A which minimize distance to x, y and converge to those values. Consider then

$$d(x_n, y_n) \leq d(x_n, x) + d(y, y_n) \leq d(x_n, x) + d(x, y) + d(y, y_n) \to d(x, y)$$

Let $\varepsilon > 0$ and choose $\delta > 0$ such that

$$d(x, y) < \delta$$

then $d(x_n, y_n) < \delta$ so by uniform continuity of f we have

$$|f(x_n) - f(y_n)| < \varepsilon$$

Taking limits we get

$$\left|\overline{f}(x) - \overline{f}(y)\right| = \left|\lim_{n\to\infty} f(x_n) - \lim_{n\to\infty} f(y_n)\right| = \lim_{n\to\infty} |f(x_n) - f(y_n)| \leq \varepsilon$$

□

Problem 15.3 (Exam problem 1 2018). *Suppose that $u : \mathbb{R} \to \mathbb{R}$ is a continuous map, which has finite limits in $\pm\infty$, i.e. there are $a, b \in \mathbb{R}$ such that*

$$\lim_{x\to-\infty} u(x) = a \text{ and } \lim_{x\to\infty} u(x) = b$$

a)
Show that there exists $B \in \mathbb{R}_+$ and $C \in \mathbb{R}$ such that for all $x \in \mathbb{R}$, if $x \geq C$ then $|u(x)| \leq B$.

Proof of (a). Since

$$\lim_{x\to\infty} u(x) = b,$$

by definition of the limit there exists $C \in \mathbb{R}$ such that for every $\varepsilon > 0$ there exists C with

$$|u(x) - b| < \varepsilon \quad \text{whenever } x \geq C.$$

Choose $\varepsilon = 1$. Then there exists $C \in \mathbb{R}$ such that

$$|u(x) - b| < 1 \quad \text{for all } x \geq C.$$

Using the triangle inequality we obtain

$$|u(x)| = |u(x) - b + b| \leq |u(x) - b| + |b| < 1 + |b|.$$

Thus if we set

$$B := 1 + |b|,$$

then for all $x \geq C$ we have

$$|u(x)| \leq B.$$

This proves the claim. □

Problem 15.4 (Exam problem 2 2018). *Let X be a normed vector space. We suppose that $(u_n)_{n\in\mathbb{N}}$ is a sequence in X that converges to a limit $\ell \in X$, that is:*

$$\lim_{n\to\infty} u_n = \ell.$$

We define a new sequence $(v_n)_{n\in\mathbb{N}}$ by putting, for any $n \in \mathbb{N}$:

$$v_n = \frac{1}{n+1} \sum_{k=n}^{2n} u_k.$$

Show that $(v_n)_{n\in\mathbb{N}}$ converges to ℓ, that is:

$$\lim_{n\to\infty} v_n = \ell.$$

Proof. Recall that a sequence $(u_n)_n$ converges to ℓ in a normed space $(X, ||\cdot||)$ if $\forall\varepsilon > 0,\ \exists N \in \mathbb{N}$ s.t.

$$||u_n - \ell|| < \varepsilon$$

for $n \geq N$. Assuming this condition holds, observe the following.

$$\begin{aligned}
||v_n - \ell|| &= \frac{1}{n+1} \left\| \sum_{k=n}^{2n} (u_k - \ell) \right\| \\
&= \frac{1}{n+1} ||u_n - \ell + u_{n+1} - \ell + \cdots + u_{2n} - \ell|| \\
&\leq \frac{1}{n+1} \left(\sum_{k=n}^{2n} ||u_k - \ell|| \right) \\
&< \frac{1}{n+1} \left(\sum_{k=n}^{2n} \varepsilon \right) \\
&= \frac{n \cdot \varepsilon}{n+1} \\
&< \varepsilon
\end{aligned}$$

Therefore v_n converges to ℓ. □

Problem 15.5 (Problem 4.8.7 Spaces p.112). *A metric space* (X, d) *is called* locally compact *if for each point* $a \in X$*, there is a* closed *ball* $\overline{B}(a; r)$ *centered at* a *that is compact. Show that* $\mathbb{R}^m$ *is locally compact, but that* $C([0, 1], \mathbb{R})$ *is not.*

Proof. By the Heiner-Borel theorem, subsets of $\mathbb{R}^m$ are compact if and only if they are closed and bounded. Take some $x \in \mathbb{R}^m$. Clearly $\overline{B}(x; r)$ is compact and as $y \in \overline{B}(x; r)$ satisfies

$$||y - x|| \leq r$$

we can conclude that

$$||y|| \leq r + ||x||$$

thus $\overline{B}(a; r)$ is bounded, hence compact. As $x \in \mathbb{R}^m$ was arbitrarily chosen we conclude that $\mathbb{R}^m$ is locally compact.

Next consider $C([0, 1], \mathbb{R})$ with the supremum norm

$$\rho(f, g) := \sup_{x \in [0,1]} |f(x) - g(x)|.$$

We show that the closed unit ball around $f(x) \equiv 0$

$$\overline{B}(0; 1) = \{g \in C([0, 1], \mathbb{R}) \mid \|g\|_\infty \leq 1\}$$

is not compact.

For $n \in \mathbb{N}$, define

$$f_n(x) := x^n.$$

Then $f_n \in C([0, 1], \mathbb{R})$ and

$$\|f_n\|_\infty = 1,$$

so $f_n \in \overline{B}(0; 1)$ for all n.

The sequence (f_n) converges pointwise to the function

$$f(x) = \begin{cases} 0 & x \in [0, 1) \\ 1 & x = 1. \end{cases}$$

This limit is not continuous. Hence (f_n) does not converge uniformly on $[0, 1]$, because the uniform limit of continuous functions must be continuous.

Furthermore, any uniformly convergent subsequence would also have to converge to this same pointwise limit, which is not continuous. Therefore (f_n) has no uniformly convergent subsequence.

Thus $\overline{B}(0; 1)$ is not sequentially compact, and hence not compact. Therefore $C([0, 1], \mathbb{R})$ is not locally compact. $\square$

16 Weierstrass Approximation & some Normed Space stuff

Theorem 16.1. *Let*

$$f \in C([a,b])$$

Then for every $\varepsilon > 0$ there exists a polynomial p such that

$$\sup_{x\in[a,b]} |f(x) - p(x)| < \varepsilon$$

meaning every continuous function can be uniformly approximated by polynomials.

The intuition behind this is that we use the fact that the polynomials are dense in $C([0,1])$. So the infinite-dimensional space can be approximated arbitrarily well by using a simple finite basis.

The proof is brilliant and uses ideas from probability where

$$B_n(f)(x) := \sum_{k=0}^{n} f\left(\frac{k}{n}\right)\binom{n}{k} x^k (1-x)^{n-k}$$

These polynomials converge uniformly to f.

Theorem 16.2. *Let X be a normed space with $\{x_n\}_{n\in\mathbb{N}} \subseteq X$. Then*

$$\sum_{n\in\mathbb{N}} x_n$$

converges if partial sums converge.

Problem 16.1 (Problem 4.10.1 Spaces p.122). *Show that there is no sequence of polynomials that converges uniformly to the continuous function $f(x) = \frac{1}{x}$ on $(0,1)$.*

Proof. Suppose, for a contradiction, that there exists a family of polynomials $\{p_n\}_{n\to\mathbb{N}}$ such that

$$p_n \underset{n\to\infty}{\to} f$$

Then f is bounded since

$$\sup_{x\in(0,1)} |p_n(x) - f(x)| < \varepsilon$$

for large enough n and $p_n(x)$ is bounded. But $\lim_{x\to 0} f(x) = \infty$. Pick $\varepsilon = 1$ to get

$$\frac{1}{x} \leq |p_n(x)| + 1$$

which would give

$$\infty \leq |p_n(x)| + 1$$

a finite value. $\square$

Problem 16.2 (Problem 4.10.2 Spaces p.122). *Show that there is no sequence of polynomials that converges uniformly to the function $f(x) = e^x$ on $\mathbb{R}$.*

Proof. We know that e^x is

$$\sum_{k=0}^{\infty} \frac{x^k}{k!}$$

so one would think that we get uniform convergence, but on $\mathbb{R}$, e^x grows exponentially fast while any polynomial does not.

Suppose there is some family of polynomials, $\{p_n\}_n$, which converges uniformly to e^x. Then, for any $\varepsilon > 0$

$$\rho(p_n, e^x) = \sup_{x \in \mathbb{R}} |p_n(x) - e^x| < \varepsilon$$

for sufficiently large n. Fix $x \in \mathbb{R}$ to maximize $|p_n(x) - e^x|$. By assumption

$$|p_n(x) - e^x| = |e^x - p_n(x)| < \varepsilon$$

so pick $\varepsilon = 1$ to get

$$|e^x| \leq |e^x - p_n(x)| + |p_n(x)| < 1 + |p_n(x)|$$

i.e. e^x is bounded above by $1 + |p_n(x)|$. Observe that

$$\lim_{x \to \infty} \frac{e^x}{p_n(x)} = \infty$$

so e^x grows too fast to be bounded by a polynomial. $\square$

Problem 16.3 (Problem 5.2.1 Spaces p.141). *Prove that the set $\{e_k\}_{k \in \mathbb{N}}$ in Example 1 really is a basis for c_0.*

Proof. Let e_k be the sequence of real numbers which is 1 at element number k and 0 everywhere else. We prove that this is a basis for the space of real-valued sequences which converge to 0, with the norm

$$||x|| = \sup\{|x_k| : k \in \mathbb{N}\}$$

Let $\{x_k\}_k$ be a sequence of real numbers. It is easy to see that $x_k = x_k \cdot e_k$ and that

$$x = \sum_{k=1}^{\infty} x_k e_k$$

It is also easy to see that

$$\sum_{k=1}^{\infty} \alpha_k e_k = 0$$

if and only if $\alpha_k = 0$ for all k. $\square$

Problem 16.4 (Problem 5.2.2 Spaces p.141). *Show that if a normed vector space V has a basis, then it is separable.*

Proof. Let $\{v_1, v_2, \ldots\}$ be a countable basis of X. Every $x \in X$ can be written uniquely as a finite linear combination

$$x = \sum_{j=1}^{k} a_j v_{i_j}$$

for some $k \in \mathbb{N}$, coefficients $a_j \in \mathbb{R}$ (or $\mathbb{C}$), and indices $i_j \in \mathbb{N}$.

Define

$$D = \left\{ \sum_{j=1}^{k} q_j v_{i_j} \;:\; k \in \mathbb{N},\ i_j \in \mathbb{N},\ q_j \in \mathbb{Q} \right\}.$$

We first show that D is countable. For each fixed k, the set

$$\mathbb{Q}^k \times \mathbb{N}^k$$

is countable, hence the set of all expressions $\sum_{j=1}^{k} q_j v_{i_j}$ is countable. Since

$$D = \bigcup_{k=1}^{\infty} \left\{ \sum_{j=1}^{k} q_j v_{i_j} \right\},$$

it follows that D is a countable union of countable sets and therefore countable.

Next we show that D is dense in X. Let $x \in X$ and $\varepsilon > 0$. Write

$$x = \sum_{j=1}^{k} a_j v_{i_j}.$$

Since $\mathbb{Q}$ is dense in $\mathbb{R}$, choose $q_j \in \mathbb{Q}$ such that

$$|a_j - q_j| < \delta$$

for some $\delta > 0$ to be chosen later. Define

$$y = \sum_{j=1}^{k} q_j v_{i_j} \in D.$$

Then

$$x - y = \sum_{j=1}^{k} (a_j - q_j) v_{i_j}.$$

By the triangle inequality,

$$\|x - y\| \le \sum_{j=1}^{k} |a_j - q_j| \, \|v_{i_j}\| < \delta \sum_{j=1}^{k} \|v_{i_j}\|.$$

Choose

$$\delta < \frac{\varepsilon}{\sum_{j=1}^{k} \|v_{i_j}\|}.$$

Then $\|x - y\| < \varepsilon$. Hence for every $x \in X$ and $\varepsilon > 0$ there exists $y \in D$ with $\|x - y\| < \varepsilon$.

Thus D is a countable dense subset of X, so X is separable. □

> **Problem 16.5** (Problem 5.2.3 Spaces p.141). *ℓ_1 is the set of all sequences $x = \{e_n\}_n$ of real numbers such that $\sum_{n=1}^{\infty} |x_n|$ converges.*
> *c) Show that the space is complete.*

Proof of (c). We consider the norm

$$||x|| = \sum_{i=n}^{\infty} |x_n|$$

Take some cauchy sequence $\{\chi_k\}_k$, where $\chi_k = \{x_n\}_n$, i.e. a sequence of sequences. By assumption, for every $\varepsilon > 0$, there exists N s.t.

$$||\chi_n - \chi_m|| < \varepsilon$$

whenever $n, m \ge N$. Recalling that the sequences are convergent and the sum (and difference) of sequences converge we can pick N large enough so that their difference converges. The candidate limit is ... etc. □

> **Problem 16.6** (Problem 5.3.10 Spaces p.149). *Let $\{e_1, e_2, \dots, e_n\}$ be an orthonormal set in some inner product space V. Show that the projection $P = P_{e_1, e_2, \dots, e_n}$ is linear.*

Proof. Recall that the projection of x is the element in the span of $\{e_1, \dots, e_n\}$ which is ”closest” to x.

Let $u, v \in V$ with $P(u) = \sum_{i=1}^{n} u_i e_i$ and $P(v) = \sum_{i=1}^{n} v_i e_i$. With each of these being so that

$$||u - P(u)|| = \min_{x \in \text{Sp}(\{e_n\})} ||u - x||$$

and

$$||v - P(v)|| = \min_{x \in \text{Sp}(\{v_n\})} ||u - x||$$

It is clear that for scaled αu, $\alpha P(u)$ is in the span and $||\alpha u - \alpha P(u)|| = \alpha ||u - P(u)||$, ... etc. □

Problem 16.7. *Show that the functions $e_k(t) = e^{ikt}$ (for $k \in \mathbb{Z}$) constitute a basis for the space $C(\mathbb{T}, \mathbb{C})$ of complex-valued, continuous functions on the torus, equipped with the L^2 norm $||f|| = \frac{1}{2\pi} \int_0^{2\pi} |f(t)|^2 \, dt$.*

Proof sketch. We can note that functions in this space $f \in C(\mathbb{T}, \mathbb{C})$ converge to their fourier series. I.e. functions of the form

$$f = \sum_{k \in \mathbb{Z}} \langle f, e_k \rangle e_k$$

where

$$\langle f, e_k \rangle = \int_0^{2\pi} f(t) e_k(t) \, dt$$

and apply a theorem about how this converges is indicative of $\{e_k\}_k$ being a basis for the space. □

Problem 16.8 (Problem 5.3.12 Spaces p.150). *If S is a nonempty subset of an inner product space V, let*

$$S^\perp = \{u \in V : \langle u, s \rangle = 0 \textit{ for all } s \in S\}$$

a) Show that $S^\perp$ is a closed subspace of V.
b) Show that if $S \subseteq T$, then $S^\perp \supseteq T^\perp$.

Proof of (a). Since I am unsure of whether the problem refers to closed under multiplication or being a closed set we will prove both.

Let $a, b \in S^\perp$. Then, for any $s \in S$ we have

$$\langle a, s \rangle = 0$$

and

$$\langle b, s \rangle = 0$$

Then

$$\begin{aligned}\langle a+b, s\rangle &= \langle a, s\rangle + \langle b, s\rangle \\ &= 0+0 \\ &= 0\end{aligned}$$

so $a+b \in S^{\perp}$. Hence S is closed under multiplication.

Now let $\{x_n\}_n \to x$ be a sequence in $S^{\perp}$. For every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ so that

$$||x_n - x|| = \sqrt{\langle x_n - x, x_n - x\rangle} < \varepsilon$$

whenever $n \geq N$. Crucially

$$0 = \langle x_n, s\rangle = \langle x, s\rangle$$

□

Problem 16.9 (Problem 2, Exam 2018). *Let X be the space from Problem 1 ($C([0,1], \mathbb{R})$), and define a map $T : X \to X$ by*

$$T(f)(x) = \int_0^x \cos\left(\frac{f(s)}{2}\right)\, ds.$$

(a) Show that the function $g(x) = \cos(\frac{x}{2})$ is Lipschitz continuous on $\mathbb{R}$.
(b) Show that T is a continuous mapping from X to X (We will take it for granted that $T(f) \in X$ for each $f \in X$).
(c) Show that $T : X \to X$ has a unique fixed point.

Proof of (a). Let $g(x) = \cos(\frac{x}{2})$. For $y \in \mathbb{R}$ we know that $|\cos(y)| \leq 1$. In particular, $x \in \mathbb{R}$ then $y = x/2 \in \mathbb{R}$, so $|\cos(y)| = |\cos(x/2)| \leq 1$. Recall that g on $\mathbb{R}$ is Lipschitz if there exists $C > 0$ so that for every $x, y \in \mathbb{R}$

$$|g(x) - g(y)| < C|x-y|$$

Notice that $|g'(x)| = \frac{1}{2}\sin(x/2)$, implying

$$|g(x) - g(y)| \leq (1/2)|x-y|$$

for all $x, y \in \mathbb{R}$. □

Proof of (b). For $u, v \in X$, we have that

$$\begin{aligned}|T(u)(x) - T(v)(x)| &= \left|\int_0^x \cos\left(\frac{u(s)}{2}\right) - \cos\left(\frac{v(s)}{2}\right)\, ds\right| \\ &\leq \int_0^x \left|\cos\left(\frac{u(s)}{2}\right) - \cos\left(\frac{v(s)}{2}\right)\right|\, ds \\ &\leq \int_0^x (1/2)|u(s) - v(s)|\, ds \\ &\leq (1/2)||u - v||\end{aligned}$$

□

Proof of (c). We see that T is a contraction mapping, so it follows from Banach's fixed point theorem that T has a unique fixed point, namely $f \equiv 0$. □

Problem 16.10 (Problem 3, Exam 2018). *Let Y be a nonempty set, and let d_1, d_2 be metrics on Y. We let $d : Y \times Y \to \mathbb{R}$ be*

$$d(x, y) := \sup\{d_1(x, y), d_2(x, y)\},$$

for all $x, y \in Y$.
(a) Show that Y is a metric.
(b) Let $E = \{y_j\}_{j\in\mathbb{N}} \subset Y$ be a sequence of points which is cauchy w.r.t d_1 and d_2. Show that E is cauchy w.r.t d.

Proof of (a). Since d_1, d_2 are metrics it is clear that d is non-negative. Furthermore, as

$$d_i(x, y) = 0 \text{ iff } x = y$$

for $i = 1, 2$ we can see that as

$$d(x, y) = \sup\{d_1(x, y), d_2(x, y)\} = 0$$

iff $d_1(x, y) = d_2(x, y) = 0$, $d(x, y) = 0$ iff $x = y$. Symmetry follows similarily.

Let $x, y, z \in Y$. We know that the triangle-inequality holds for d_1, d_2.

$$d(x, z) = \sup\{d_1(x, z), d_2(x, z)\}$$

and

$$d(x, y) + d(y, z) = \sup\{d_1(x, y), d_2(x, y)\} + \sup\{d_1(y, z), d_1(y, z)\}$$

Suppose wlog $d_1(x, z) \leq d_2(x, z)$. Then

$$\begin{aligned} d(x, z) &= d_2(x, z) \\ &\leq d_2(x, y) + d_2(y, z) \\ &\leq \sup\{d_1(x, y), d_2(x, y)\} + \sup\{d_1(y, z), d_2(y, z)\} \\ &= d(x, y) + d(x, z) \end{aligned}$$

□

Proof of (b). Suppose $E = \{y_j\}_j$ is a sequence in Y which is cauchy w.r.t d_1, d_2, i.e. $\forall \varepsilon > 0$, $\exists N_1, N_2 \in \mathbb{N}$ so that

$$d_1(y_j, y_i) < \varepsilon$$

whenever $j, i \geq N_1$ and

$$d_2(y_j, y_i) < \varepsilon$$

whenever $j, i \geq N_2$. Let $N = \max\{N_1, N_2\}$. Then both hold for $j, i \geq N$. Then either

$$d_1(y_j, y_i) \leq d_2(y_j, y_i) < \varepsilon$$

or

$$d_2(y_j, y_i) \leq d_1(y_j, y_i) < \varepsilon$$

so

$$d(y_j, y_i) < \varepsilon$$

$\square$

17 Some problems from week of March 16

Problem 17.1 (Problem 5.4.5 Spaces p.154). *Define $F : C([0,1],\mathbb{R}) \to \mathbb{R}$ by $F(u) = u(0)$. Show that F is a linear functional. Is F continuous?*

Proof. F is well-defined since $u(0) \in \mathbb{R}$ for all $u \in C([0,1],\mathbb{R})$, and is linear since

$$\begin{aligned} F(u+v) &= (u+v)(0) \\ &= u(0) + v(0) \\ &= F(u) + F(v)F(\alpha u) & = (\alpha u)(0) \\ &= \alpha u(0) \\ &= \alpha F(u) \end{aligned}$$

for arbitrary $u, v \in C([0,1],\mathbb{R})$ and $\alpha \in \mathbb{R}$.

Let $\varepsilon > 0$ and pick $\delta = \varepsilon > 0$. Then if

$$\rho(u,v) = \sup_{x\in[0,1]} |u(x) - v(x)| < \delta$$

we know that

$$|F(u) - F(v)| = |u(0) - v(0)| \leq \rho(u,v) < \delta = \varepsilon$$

Therefore F is continuous. □

Problem 17.2 (Problem 5.4.7 Spaces p.155). *Check that $\mathcal{L}(V,W)$ is a linear space.*

Proof sketch. Suppose V, W are vector spaces, with

$$\mathcal{L}(V,W) := \{\text{linear operators } T : V \to W\}$$

Recognize that for $T, S \in \mathcal{L}(V,W)$ addition and multiplication is defined so that

$$\begin{aligned} (T+S)(v) &= Tv + Sv \\ (\alpha T)(v) &= \alpha Tv \end{aligned}$$

and that these operations are well-defined:

For β in the ground field of V, α in the ground field of W (and $\mathcal{L}(V,W)$), and $T, S \in \mathcal{L}(V,W)$ we see that

$$\begin{aligned} (T+S)(\beta v) &= T(\beta v) + S(\beta v) \\ &= \beta Tv + \beta Sv \\ &= \beta(Tv + Sv) \\ &= \beta(T+S)(v) \end{aligned}$$

and

$$\begin{aligned}(\alpha T)(\beta v) &= \alpha T(\beta v) \\ &= \alpha\beta T v \\ &= \beta\alpha T v \\ &= \beta(\alpha T)(v)\end{aligned}$$

The rest of the vector-space axioms follow nicely from here. $\square$

Problem 17.3 (Problem 1, Exam 2021). *Let (X, d) be a metric space $X = (0, 1]$, $d(x, y) = |x - y|$, and let $T : X \to X$ be given by $T(x) = x/2$.*
Show that T is a contraction. Does T have a fixed point?

Proof. Let $x, y \in X$, with $c = \frac{1}{2} \in (0, 1)$.

We see that

$$d(Tx, Ty) = \left|\frac{1}{2}x - \frac{1}{2}y\right| = \frac{1}{2}d(x, y) = cd(x, y)$$

so T is a contraction mapping.

If X was $[0, 1]$ a (unique) fixed point would follow from Banach's fixed point theorem, but notice that

$$Tx = \frac{1}{2}x = x$$

if and only if $x = 0 \notin X$ so X does not have a fixed point. $\square$

Problem 17.4 (Problem 2, Exam 2021). *Let (X, d) be a metric space and let $A \subseteq X$ be closed. We define the metric space $C_b(X, A) = \{$all continuous, bounded $f : X \to A\}$, equipped with the supremum metric*

$$\rho(f, g) = \sup_{x \in X} |f(x) - g(x)|$$

Show that $C_b(X, A)$ is a closed subspace of $C_b(X, \mathbb{R})$.

Proof. It is trivial that $C_b(X, A) \subseteq C_b(X, \mathbb{R})$ so it remains to show that this is closed w.r.t the supremum metric.

Let $\{f_n\}_{n \in \mathbb{N}}$ be some sequence of functions in $C_b(X, A)$ which converge to a point $f \in C_b(X, \mathbb{R})$.

I.e. we have

$$\rho(f_n, f) = \sup_{x \in X} |f_n(x) - f(x)| \to 0$$

Fix any $x \in X$. Then $f_n(x) \in A$ for all n. Consider the sequence of images of $f_n(x)$, $\{f_n(x)\}_{n\in\mathbb{N}}$. This sequence converges to $f(x)$ and since A is closed we know $f(x) \in A$.

As $\{f_n\}_n$ was arbitrarily chosen as well as $x \in X$ we conclude that $\{f_n\}$ converges to $f \in C_b(X, A)$, so $C_b(X, A)$ is closed. $\square$

Problem 17.5 (Problem 3, Exam 2021). *Let (X, d) be a metric space, and for every nonempty $E \subseteq X$ and $x \in X$, define*

$$\text{dist}(x, E) = \inf\{d(x, y) : y \in E\}.$$

(a) Show that if E is compact and nonempty, then there is some $z \in E$ such that $\text{dist}(x, E) = d(x, z)$.
(b) Give an example of a metric space (X, d), a point $x \in X$ and a nonempty subset $E \subseteq X$ for which there is no such point $z \in E$.

Proof of (a). We consider a nontrivial compact subset E.

Let $f : E \to \mathbb{R}$, $f(y) = d(x, y)$. Notice that f is continuous. Since f is continuous and E is compact, f attains its minimum, i.e. there is some $z \in E$ for which

$$f(z) = \inf_{y\in E} f(y) = \min_{y\in E} d(x, y)$$

$\square$

Solution of (b). Let $X = \mathbb{R}$ with the usual metric. Then fix $x = 0$ with $E = (0, 1)$. Then there is no point in E which is "as close to" x as possible.

Assume, for contracition, that there exists $y \in E$ for which

$$d(x, y) = \text{dist}(E)$$

Then $y > 0 = x$, but

$$x = 0 < y/2 < y$$

so

$$d(x, y/2) < d(x, y)$$

$\square$

Problem 17.6 (Problem 4, Exam 2021). *Let $(X, ||\cdot||)$ be a normed space and let $A : X \to X$ be an invertible, bounded linear operator. Define $||x||_A = ||Ax||$ for every $x \in X$.*
(a) Show that $||\cdot||_A$ is a norm on X.
(b) Show that a sequence in X converges w.r.t $||\cdot||$ if and only if it converges w.r.t $||\cdot||_A$.

Proof of (a). We let $(X, ||\cdot||)$ be a normed space with $||\cdot||_A$ defined as above.

A is invertible (bijective) and thus has $\ker A = \{0\}$ so $Ax = 0$ if and only if $x = 0$ so $||x||_A = 0$ iff $x = 0$ follows.

It is also clear that $||Ax|| \geq 0$ since $Ax \in X$ and $||\cdot||$ has non-negativity.

Now let α be in the ground field of X. Then

$$\begin{aligned} ||\alpha x||_A &= ||A(\alpha x)|| \\ &= ||\alpha Ax|| \\ &= \alpha||Ax|| \\ &= \alpha||x||_A \end{aligned}$$

Lastly we check the triangle inequality:

$$\begin{aligned} ||x+y||_A &= ||A(x+y)|| \\ &= ||Ax + Ay|| \\ &\leq ||Ax|| + ||Ay|| \\ &= ||x||_A + ||y||_A \end{aligned}$$

Thus $||\cdot||_A$ is a norm on X. □

Proof of (b). Let $\{x_n\}_{n\in\mathbb{N}}$ be some sequence in X.

First, suppose $\{x_n\}_n$ converges to x w.r.t $||\cdot||$, that is to say for every $\varepsilon > 0$ there is N large enough so that

$$||x_n - x|| < \varepsilon$$

whenever $n \geq N$. Then

$$\begin{aligned} ||x_n - x||_A &= ||A(x_n - x)|| \\ &= ||Ax_n - Ax|| \\ &= ||x_n - x||\frac{||Ax_n - Ax||}{||x_n - x||} \end{aligned}$$

Notice that

$$||x_n - x||_A = ||x_n - x||\frac{||Ax_n - Ax||}{||x_n - x||} \leq ||x_n - x||\cdot||A||_{\mathcal{L}} = ||x_n - x|| \sup_{y\in X,\ y\neq 0} \frac{||Ay||}{||y||}$$

So for $||\cdot||_A$ choose N large enough so that

$$||x_n - x|| < \frac{\varepsilon}{||A||_{\mathcal{L}}}$$

noticing that $||A||_{\mathcal{L}} > 0$ since A is invertible.

For the other direction proceed in a similar way, but choose N so that

$$||x_n - x||_A < \varepsilon||A||_{\mathcal{L}}$$

□

18 Repition Problems from Last Month

Problem 18.1. *Let*

$$f_n(x) = x^n \text{ on } [0,1]$$

1. *Find the pointwise limit.*
2. *Does* $f_n \to f$ *uniformly?*
3. *Is the convergence uniform on* $[0,a]$ *for* $a < 1$*?*

Proof. The pointwise limit is

$$f = \begin{cases} 0 & \text{for } x \in [0,1) \\ 1 & \text{for } x = 1 \end{cases}$$

as $0 \leq c^n < 1$ converges to 0 while $1^n = 1, \ \forall n \in \mathbb{N}$.

Then f_n does **not** converge unformly to f.

Recalling that

$$\rho(f_n, f) = \sup_{x\in[0,1]} |f_n(x) - f(x)|$$

Take $x = 1 - \varepsilon$ for some small $\varepsilon > 0$. Then

$$f(x) = 0$$

but

$$f_n(x) = (1-\varepsilon)^n$$

I.e.

$$|f_n(x) - f(x)| \leq \rho(f_n, f) = \sup_{x\in[0,1]} |1-\varepsilon|^n \to 1$$

So f_n does not converge pointwise to f.

Lastly we look at uniform convergence on $[0,a]$ for $a < 1$. Here $\lim_{n\to\infty} f_n(x) = 0$ for all $x \in [0,a]$, and

$$\begin{aligned} \rho(f_n, f) &= \sup_{x\in[0,a]} |x^n - 0| \\ &= \sup_{x\in[0,a]} |x^n| \end{aligned}$$

which converges to 0 for $n \to \infty$ on all $x \in [0,a]$. □

Problem 18.2. *Let*

$$f_n(x) = \frac{x}{1+nx}, \quad x \in [0,1]$$

1. *Show* $f_n \to 0$ *pointwise.*
2. *Show convergence is uniformy (or not).*

Proof. Fix any $x \in [0,1]$. Informally (for $x > 0$)

$$f_n(x) = \theta\left(\frac{1}{n}\right)$$

and for $x = 0$, $f_n(x) = 0$. And the asymptotic behaviour of $1/n$ is to approach 0.

Then,

$$\begin{aligned}\rho(f_n, 0) &= \sup_{0\le x\le 1} |f_n(x)| \\ &= \sup_{0\le x\le 1} \left|\frac{x}{1+nx}\right| \\ &= \sup_{0\le x\le 1} \left|\frac{x/n}{1/n + x}\right|\end{aligned}$$

It is easy to see that

$$\rho(f_n, 0) \le \left|\frac{1}{1+n}\right|.$$

which converges to 0 when $n \to \infty$. □

Problem 18.3. *Define*

$$f(x) = \frac{1}{x}$$

on $(0,1)$.

1. *Show that* f *is continuous.*
2. *Show that* f *is not uniformly continuous.*

Proof. Let $\varepsilon > 0$. We want to find $\delta > 0$ so that

$$|x - y| < \delta$$

then

$$|f(x) - f(y)| = \left|\frac{1}{x} - \frac{1}{y}\right| = \left|\frac{x-y}{xy}\right| = \delta \cdot \frac{1}{|xy|} < \varepsilon$$

So let $\delta := \frac{\varepsilon|xy|}{2} < \varepsilon|xy|$, but choice of δ is clearly dependent on $x, y \in (0,1)$.

In fact we find the following:

$$\sup_{x\in(0,1)} |f(x)| = \lim_{x\to 0} f(x) = \infty$$

since for any $n \in \mathbb{N}$ we can find $f(x) \ge n$ by setting $x = \frac{1}{n}$ since then

$$f(x) = f\left(\frac{1}{n}\right) = (n^{-1})^{-1} = n$$

so f grows without bound for $x \to 0$ and is therefore not uniformly continuous.

$\square$

Problem 18.4 (Problem 1, Exam 2021kont). *Consider $\mathbb{R}^2$ equipped with the Ruclidian norm, and the corresponding metric $d(x,y) = ||x-y||$. Define the unit circle $S = \{s \in \mathbb{R}^2 : ||x|| = 1\}$. Explain why*

- *(S,d) is a metric space*
- *$(S, ||\cdot||)$ is not a normed space*

Proof. (S,d) is a metric space since $\mathbb{R}$ with the Euclidian metric is a metric space and $S \subseteq \mathbb{R}$, so the necessary properties hold for $x,y,z \in S \Rightarrow x,y,z \in \mathbb{R}$.

$(S, ||\cdot||)$ is not a normed space because it is not even a vector space.

Notice that $(1,0),(0,1) \in S$, but

$$(1,0) + (0,1) = (1,1)$$

and $||(1,1)|| = \sqrt{2}$ so $(1,1) \notin S$. $\square$

Problem 18.5 (Problem 2, Exam 2021kont). *Prove that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = e^{-x^2}$ is uniformly continuous.*

Proof. Notice that $x^2 \in [0,\infty)$ and so

$$e^{-x^2} \in [0,1]$$

with

$$\max_{x\in\mathbb{R}} |e^{-x^2}| = 1$$

for $x = 0$.

So let $\varepsilon > 0$ and choose $\delta > 0$ so that

$$|x-y| < \delta$$

we assume $x \neq y$ so $|x-y| > 0$ (since otherwise $|f(x)-f(y)| = 0$). Then

$$\frac{|f(x)-f(y)|}{|x-y|} \leq \sup_{z\in\mathbb{R}} |f'(z)| = \frac{1}{\sqrt{2}} e^{1/\sqrt{2}}$$

so

$$|f(x)-f(y)| \leq \frac{1}{\sqrt{2}} e^{1\sqrt{2}} |x-y|$$

and so f is Lipschitz and hence uniformly continuous. $\square$

Problem 18.6 (Problem 3, Exam 2021kont). *Fix some* $p \in [1, \infty]$ *and let* $x \in \ell^p(\mathbb{R})$ *with components* $x = (x(1), x(2), \ldots)$. *Define* $y_n = (x(1), x(2), \ldots, x(n), 0, \ldots)$ *for* $n \in \mathbb{N}$.
(a) Explain why $y_n \in \ell^p(\mathbb{R})$ *for every* $n \in \mathbb{N}$. *Show that if* $p < \infty$ *then* $\{y_n\}_n$ *is Cauchy.*
(b) Is the same true for $p = \infty$*? If yes, prove it. Otherwise provide a counterexample.*

Proof of (a). $\ell^p(\mathbb{R}) = \{\{x_n\}_n : (\sum_{i=1}^{\infty} |x_i|^p)^{1/p} < \infty\}$ so if $x \in \ell^p(\mathbb{R})$ then

$$\left(\sum_{i=1}^{\infty} |y_i|^p\right)^{1/p} = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} \leq \left(\sum_{i=1}^{\infty} |x_i|^p\right)^{1/p} < \infty$$

so

$$y_n \in \ell^p(\mathbb{R})$$

Suppose $p < \infty$ then for y_n, y_m (supposing wlog $n \geq m$)

$$\begin{aligned} ||y_n - y_m|| &= \left(\sum_{i=1}^{\infty} |y_i - y_i|^p\right)^{1/p} \\ &= \left(\sum_{i=m}^{n} |x_i|^p\right)^{1/p} \end{aligned}$$

It is known that $x_n \to 0$ so we can pick N large enough so that the above sum is as small as we want. $\square$

Proof sketch of (b). It is not true that $\{y_n\}_n$ is cauchy for $p = \infty$ since

$$||x_n|| = \sup_{n \in \mathbb{N}} |x_n|$$

which cannot necessarily be made arbitrarily small. Blah blah blah something more rigorous... $\square$

19 Doing some cathing up on Normed- and Inner Product Spaces

The following section will not be very rigorous as it's just to re-iterate some things from the past month which I have not been very diligent in working on.

Given some series $\sum x_n$ in a normed space, the series converges if

$$\left\| \sum_{k=1}^{n} x_k - S \right\| \to 0$$

If

$$\sum ||x_n|| < \infty$$

then $\sum x_n$ converges.

This works in Banach spaces, but may fail in general.

We differentiate between algebraic bases, finite linear combinations so that every vector has a unique representation, and Schaudner bases which are infinite series

$$x = \sum_{n=1}^{\infty} a_n e_n$$

An example of this are elements of ℓ^2 which clearly have no finite representation.

Given some inner product space with i.p $\langle \cdot, \cdot \rangle$, the projection of x over v is

$$\mathtt{proj}_v(x) = \frac{\langle x, v \rangle}{\langle v, v \rangle} v$$

Given some linear operator T, T is bounded if and only if T is continuous.

The operator norm is

$$||T|| = \sup_{||x|| \leq 1} ||Tx||$$

This has many, many equivalent formulations.

The space of operators $\mathcal{L}(V, W)$ is a banach space if W is a banach space.

In infinite-dimensional spaces, T being bijective does not imply that T^{-1} is bijective, so we end up caring about the inverse being bounded.

Neumann series are the analogue of geometric series. If $||T|| < 1$ then

$$(I - T)^{-1} = \sum_{n=0}^{\infty} T^n$$

A nice application of this is that if T is invertible and S is ”small” then

$$T + S$$

is invertible if

$$||T^{-1} + S|| < 1$$

20 Problems from week of March 23

Problem 20.1. *Let* $A : \ell^\infty \to \ell^\infty$ *be the function*

$$A((a_1, a_2, a_3, ...)) = (0, a_1, a_2, a_3, ...)$$

Show that A *is a bounded linear operator.*
Find a bounded, linear operator $B : \ell^\infty \to \ell^\infty$ *such that* $BA = I$*.*
Show that $AB \neq I$*.*

Proof. Let $A : \ell^\infty \to \ell^\infty$ be as defined in the problem statement. If $(a_i)_{i\in\mathbb{N}} \in \ell^\infty$ then clearly $(0, a_1, a_2, ...) \in \ell^\infty$ so the map is well-defined.

Now let $a = (a_i)_{i\in\mathbb{N}}, b = (b_i)_{i\in\mathbb{N}} \in \ell^\infty$.

Then

$$\begin{aligned} A(a+b) &= A((a_1+b_1, a_2+b_2, ...)) \\ &= (0, a_1+b_1, a_2+b_2 ...) \\ &= (0, a_1, a_2, ...) + (0, b_1, b_2, ...) \\ &= A(a) + A(b) \end{aligned}$$

Now let α be some scalar from the ground-field. Then

$$\begin{aligned} A(\alpha a) &= (0, \alpha a_1, \alpha a_2, ...) \\ &= (\alpha \cdot 0, \alpha a_1, a_2, ...) \\ &= \alpha(0, a_1, a_2, ...) \\ &= \alpha A(a) \end{aligned}$$

Now it remains to show that A is bounded. Notice that

$$\begin{aligned} ||A(a)||_\infty &= \max\{0, |a_1|, |a_2|, ...\} \\ &= \max\{|a_1|, |a_2|, ...\} \\ &= ||a||_\infty \end{aligned}$$

for any $a \in \ell^\infty$, so A is bounded.

Let $B : \ell^\infty \to \ell^\infty$ be defined by

$$B((a_1, a_2, ...)) = (a_2, a_3, ...)$$

Then

$$BA((a_1, a_2, ...)) = B((0, a_1, a_2, ...)) = (a_1, a_2, ...)$$

but

$$AB((a_1, a_2, ...)) = (0, a_2, ...)$$

Now notice that

$$\begin{aligned} ||B(a)||_\infty &= ||(a_2, a_3, ...)||_\infty \\ &\leq ||(a_1, a_2, ...)||_\infty \\ &= ||a||_\infty \end{aligned}$$

so B is bounded (linearity would also be a quick check). □

Problem 20.2 (Problem 5.5.2 Spaces p.160). *Assume that U is a normed space and that $C : U \to U$ is a bounded, linear operator. Show that $||C^n|| \leq ||C||^n$.*

Proof sketch. Let C be some bounded, linear operator on a normed space U. Then

$$||C|| \leq m$$

for some $m \in \mathbb{R}_{\geq 0}$, recalling that

$$||C|| = \sup_{x \in U \setminus \{0\}} \frac{||Cx||}{||x||} = k$$

with the observation that

$$||C||^n = k^n \leq m^n$$

We also observe that

$$||C^n|| = \sup_{x \in U \setminus \{0\}} \frac{||C^n x||}{||x||}$$

Recall that for bounded, linear operators A, B the following holds

$$||AB|| \leq ||A|| \cdot ||B||$$

Observe then that

$$||C^n|| \leq ||C|| \cdot ||C^{n-1}|| \leq ||C|| \cdot ||C|| \cdot ||C^{n-2}||$$

and so on. Inductively

$$||C^n|| \leq \prod_{i \in [n]} ||C|| = ||C||^n$$

□

Problem 20.3 (Problem 5.5.3 Spaces p.160). *Assume that U, V are normed spaces and that $A : U \to V$ is a surjective, linear operator whose inverse A^{-1} is bounded. Show that $||A(u)||_V \geq \frac{1}{||A^{-1}||}||u||_U$ for all $u \in U$.*

Proof. Since A^{-1} exists, A is bijective. Because A^{-1} is bounded, we have

$$\|A^{-1}v\|_U \leq \|A^{-1}\| \, \|v\|_V \quad \text{for all } v \in V.$$

Now let $u \in U$. Since $A^{-1}(A(u)) = u$, we get

$$\|u\|_U = \|A^{-1}(A(u))\|_U \leq \|A^{-1}\| \, \|A(u)\|_V.$$

Dividing by $\|A^{-1}\|$ yields

$$\|A(u)\|_V \geq \frac{1}{\|A^{-1}\|} \|u\|_U,$$

for all $u \in U$. □

Problem 20.4 (Problem 4 Spaces p.160). *Assume that U, V are normed spaces and that $A : U \to V$ is a surjective, bounded, linear operator. Show that if there is a $c \in \mathbb{R}$ such that $\|A(u)\|_V \geq c\|u\|_U$ for all $u \in U$, then A is invertible.*

Proof. Assume there exists $c > 0$ such that

$$\|A(u)\|_V \geq c\|u\|_U \quad \text{for all } u \in U.$$

Since A is already assumed surjective, it remains only to show that A is injective.

Suppose $u \in U$ and $A(u) = 0$. Then

$$0 = \|A(u)\|_V \geq c\|u\|_U.$$

Because $c > 0$, this implies $\|u\|_U = 0$, and hence $u = 0$. Therefore $\ker(A) = \{0\}$, so A is injective.

Thus A is both injective and surjective, hence bijective. Therefore A is invertible. □

Proof. We show that the conclusion cannot be deduced if $c \leq 0$.

Assume $c \leq 0$, and let $A : U \to V$ be any map whatsoever. For every $u \in U$ we have

$$\|A(u)\|_V \geq 0,$$

since norms are always nonnegative. Also, because $c \leq 0$ and $\|u\|_U \geq 0$, we have

$$c\|u\|_U \leq 0.$$

Therefore

$$\|A(u)\|_V \geq 0 \geq c\|u\|_U$$

for every $u \in U$. Hence, whenever $c \leq 0$, the inequality

$$\|A(u)\|_V \geq c\|u\|_U \quad \forall u \in U$$

holds for every operator A.

So if the statement "if A is surjective, bounded, linear and $\|A(u)\|_V \geq c\|u\|_U$ $\forall u \in U$, then A is invertible" were true for some $c \leq 0$, it would follow that every surjective, bounded, linear operator is invertible. This is false.

Indeed, define

$$A : \mathbb{R}^2 \to \mathbb{R}, \qquad A(x, y) = x.$$

Then A is linear, bounded, and surjective. However, A is not injective, since

$$A(0, 0) = 0 = A(0, 1)$$

with $(0, 0) \neq (0, 1)$. Hence A is not invertible.

Thus the statement cannot hold for $c \leq 0$. □

Problem 20.5 (Problem 5.5.5 Spaces). *Assume U is a Banach space and define*

$$GL(U) := \{T \in \mathcal{L}(U) : T \text{ is invertible}\}$$

Show that $GL(U)$ is an open subset of $\mathcal{L}(U)$.

Proof. $GL(U)$ is open if we for any $A \in GL(U)$ can find $r > 0$ so that

$$\forall B \in \mathcal{L}(U), \ \ \|B - A\| < r \Rightarrow B \in GL(U).$$

By Banach's Lemma, if A is invertible (which we assume) then

$$\|A - B\| < \frac{1}{\|A^{-1}\|}$$

A is invertible so we can take $r = \frac{1}{\|A^{-1}\|}$. □

Problem 20.6 (Problem 3 Exam 2022). *Assume that (X, d) is a metric space and that $f : X \to \mathbb{R}$ is a continuous function. If $a \in \mathbb{R}$, we say that the equation $f(x) = a$ is almost solvable if there for each $\varepsilon > 0$ is an $x \in \mathbb{R}$ such that $|f(x) - a| < \varepsilon$. Show that if X is compact and $f(x) = a$ is almost solvable, then there is an $x \in X$ such that $f(x) = a$. Find an example which shows that this is not always true for noncompact X.*

Proof. We begin by giving an example where this does not work. Let $X = (0, \infty)$ with the usual norm and let

$$f(x) = x$$

Let $a = 0$ then $f(x) = a$ is almost solvable since for any $\varepsilon > 0$ you can simply choose $x = \varepsilon/2 \in X$ and the condition is satisfied, but there is no $x \in X$ so that $f(x) = a$.

Now we prove the claim. Consider a sequence $\{x_n\}_n$ in X so that $|f(x_n) - a| < \frac{1}{n}$. Since X is compact we can find a subsequence so that $x_{n_k} \to x \in X$. Then $f(x_{n_k}) \to f(x)$ and furthermore

$$|f(x_{n_k}) - a| < \frac{1}{n_k} \to 0$$

so $f(x_{n_k}) \to a$ and by uniqueness of the limit $f(x) = a$. □

21 Derivatives - Chapter 6 of Spaces

We recall that for a function $f : \mathbb{R} \to \mathbb{R}$ at $\overline{x} \in \mathbb{R}$, the derivative is

$$f'(x) = \lim_{h \to 0} \frac{f(\overline{x}+h) - f(\overline{x})}{h}$$

where the geometric intuition is that $f'(\overline{x})$ is the slope of the tangent of the graph of f at $\overline{x}$.

For our purposes, the following is the most important property of f'.

Theorem 21.1 (Taylor's Theorem). *Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function.*
Let $\overline{x} \in \mathbb{R}$. Then

$$f(\overline{x}+h) \approx f(\overline{x}) + f'(\overline{x})h$$

for small h. More precisely:

$$\frac{|f(\overline{x}+h) - (f(\overline{x}) + f'(\overline{x})h)|}{|h|} \underset{h \to 0}{\to} 0$$

With some manipulation of the expression in the above theorem, we get that

$$f'(\overline{x}) = \lim_{h \to 0} \frac{f(\overline{x}+h) - f(\overline{x})}{h}$$

as we already know.

For a function $f : \mathbb{R}^m \to \mathbb{R}^n$ we identify the derivative with its Jacobian ∇f,

$$\nabla f(\overline{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x^1}(\overline{x}) & \cdots & \frac{\partial f_1}{\partial x^m}(\overline{x}) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x^1}(\overline{x}) & \cdots & \frac{\partial f_n}{\partial x^m}(\overline{x}) \end{pmatrix} \in \mathbb{R}^{n \times m}$$

Then

$$\frac{\|f(\overline{x}+h) - f(\overline{x}) - \nabla f(\overline{x})h\|_2}{\|h\|_2} \underset{h \to 0}{\to} 0$$

Thinking of $\nabla f(\overline{x})h$ as some linear function of h we get the correct notion; The derivative of a function from some space to another shold be some linear function of h.

Definition 21.1 (Frechet Derivative). *Let X, Y be normed spaces, let $O \subseteq X$ be open and let $F : O \to Y$ be a function. Then F is (Frechet) differentiable at $\overline{x} \in O$ if there is some $A \in \mathcal{L}(X, Y)$ such that*

$$\lim_{h \in X,\ h \to 0} \frac{\|F(\overline{x}+h) - F(\overline{x}) - A(h)\|_Y}{\|h\|_X} = 0$$

A is the (Frechet) derivative of F.

Lemma 21.1. *If $F : O \to Y$ is differentiable at $\overline{x} \in O$, then the derivative is unique.*

Proof. Suppose $A, B \in \mathcal{L}(X, Y)$ are derivatives of F. Let

$$f(h) := \frac{\|F(\overline{x}+h) - F(\overline{x}) - A(h) - (F(\overline{x}+h) - F(\overline{x}) - B(h))\|_Y}{\|h\|_x}$$

Then

$$f(h) = \left\| (B - A) \left(\frac{h}{\|h\|_X} \right) \right\|_Y$$

By the triangle inequality

$$f(h) \leq \frac{\|F(\overline{x}+h) - F(\overline{x}) - A(h)\|_Y}{\|h\|_X} + \frac{\|F(\overline{x}+h) - F(\overline{x}) - B(h)\|_Y}{\|h\|_X} \to 0$$

when $h \to 0$. For any $x \in X$ and $\alpha > 0$, we get

$$\|(B - A)(x)\|_Y = \|x\|_X \|(B - A)(\alpha x / \|\alpha x\|)\|_Y = \|x\|_X f(\alpha x) \to 0$$

as $\alpha \to 0$. So $(B - A)(x) = 0$. Since x was arbitrary $A = B$. ◻

Problem 21.1 (Problem 6.1.1 Spaces). *Prove Proposition 6.1.6: Assume that X and Y are normed spaces, and that $F : X \to Y$ is constant. Then F is differentiable at all points $a \in X$, and*

$$F'(a) = 0$$

Proof. Let X, Y be normed spaces with $F : X \to Y$ constant.

Then F is

$$F(a) = b$$

for some $b \in Y$.

Fix some arbitrary $a \in X$ and observe that

$$\begin{aligned} \lim_{h \in X,\ h \to 0} \frac{\|F(a+h) - F(h) - 0\|}{\|h\|_X} &= \lim_{h \in X,\ h \to 0} \frac{\|b - b\|_Y}{\|h\|_X} \\ &= \lim_{h \in X,\ h \to 0} 0 \\ &= 0 \end{aligned}$$

so $F'(x) = 0$ (which is ceirtainly in $\mathcal{L}(X, Y)$) is the (Frechet) derivative of F. ◻

Problem 21.2 (Problem 6.1.2 Spaces). *Assume that X and Y are normed spaces. A function $F : X \to Y$ is called **affine** if there is a linear map $A : X \to Y$ and an element $c \in Y$ such that $F(x) = A(x) + c$ for all $x \in X$. Show that if A is bounded, then $F'(a) = A$ for all $a \in X$.*

Proof. Suppose $F : X \to Y$ is affine with some bounded linear $A \in \mathcal{L}(X, Y)$, meaning

$$F(x) = A(x) + c, \ \ c \in Y$$

for all $x \in X$.

Fix some arbitrary $a \in X$. Then

$$\begin{aligned}
\lim_{h \to 0} \frac{\|F(a+h) - F(a) - A(h)\|_Y}{\|h\|_X} &= \lim_{h \to 0} \frac{\|A(a+h) + c - (A(a) + c) - A(h)\|_Y}{\|h\|_X} \\
&= \lim_{h \to 0} \frac{\|A(a+h) - A(a) - A(h)\|}{\|h\|_X} \\
&= \lim_{h \to 0} \frac{\|A(a) + A(h) - A(a) - A(h)\|}{\|h\|_X} \\
&= \lim_{h \to 0} \frac{0}{\|h\|_X} \\
&= 0
\end{aligned}$$

Since $A \in \mathcal{L}(X, Y)$ and it satisfies the above and $a \in X$ was arbitrarily chosen, A is the derivative of F. $\square$

Problem 21.3 (Problem 6.1.3 Spaces). *Assume that $F, G : X \to Y$ are differentiable at $a \in X$. Show that for all constants $\alpha, \beta \in \mathbb{K}$, the function defined by $H(x) = \alpha F(x) + \beta G(x)$ is differentiable at a and $H'(a) = \alpha F'(a) + \beta G'(a)$.*

Proof. Assume that $F, G : X \to Y$ are differentiable at $a \in X$. Let

$$A = F'(a), \qquad B = G'(a).$$

Then $A, B \in \mathcal{L}(X, Y)$ are bounded linear maps, and

$$\lim_{h \to 0} \frac{\|F(a+h) - F(a) - A(h)\|_Y}{\|h\|_X} = 0,$$

$$\lim_{h \to 0} \frac{\|G(a+h) - G(a) - B(h)\|_Y}{\|h\|_X} = 0.$$

Let $\alpha, \beta \in \mathbb{K}$, and define

$$H(x) = \alpha F(x) + \beta G(x)$$

for all $x \in X$.

We claim that H is differentiable at a with

$$H'(a) = \alpha A + \beta B.$$

First note that since A and B are bounded linear maps, $\alpha A + \beta B$ is also a bounded linear map from X to Y.

Now for $h \in X$, $h \neq 0$, we have

$$\begin{aligned}
&\frac{\|H(a+h) - H(a) - (\alpha A + \beta B)(h)\|_Y}{\|h\|_X} \\
&= \frac{\|\alpha F(a+h) + \beta G(a+h) - \alpha F(a) - \beta G(a) - \alpha A(h) - \beta B(h)\|_Y}{\|h\|_X} \\
&= \frac{\|\alpha(F(a+h) - F(a) - A(h)) + \beta(G(a+h) - G(a) - B(h))\|_Y}{\|h\|_X} \\
&\le |\alpha|\frac{\|F(a+h) - F(a) - A(h)\|_Y}{\|h\|_X} + |\beta|\frac{\|G(a+h) - G(a) - B(h)\|_Y}{\|h\|_X}.
\end{aligned}$$

As $h \to 0$, the first term on the right tends to 0 because F is differentiable at a, and the second term tends to 0 because G is differentiable at a. Therefore

$$\lim_{h \to 0} \frac{\|H(a+h) - H(a) - (\alpha A + \beta B)(h)\|_Y}{\|h\|_X} = 0.$$

Hence H is differentiable at a, and

$$H'(a) = \alpha A + \beta B = \alpha F'(a) + \beta G'(a).$$

$\square$

Problem 21.4 (Problem 6.1.6 Spaces). *Assume that X, Y are normed spaces over $\mathbb{R}$ and that the norm in Y is generated by an inner product $\langle \cdot, \cdot \rangle$. Assume that the functions $F, G : X \to Y$ are differentiable at $a \in X$. Show that the function $\langle F(x), G(x) \rangle$ is differentiable at a, and that*

$$h'(a) = \langle F'(a), G(a) \rangle + \langle F(a), G'(a) \rangle$$

Proof. Define

$$h(x) = \langle F(x), G(x) \rangle, \qquad x \in X.$$

We will show that h is differentiable at a and that its derivative is the linear functional $L : X \to \mathbb{R}$ given by

$$L(u) = \langle F'(a)u, G(a) \rangle + \langle F(a), G'(a)u \rangle.$$

Since F and G are differentiable at a, we may write

$$F(a+u) = F(a) + F'(a)u + r_F(u),$$
$$G(a+u) = G(a) + G'(a)u + r_G(u),$$

where

$$\frac{\|r_F(u)\|_Y}{\|u\|_X} \to 0 \quad \text{and} \quad \frac{\|r_G(u)\|_Y}{\|u\|_X} \to 0 \qquad \text{as } u \to 0.$$

Then

$$\begin{aligned} h(a+u) &= \langle F(a+u), G(a+u) \rangle \\ &= \langle F(a) + F'(a)u + r_F(u),\, G(a) + G'(a)u + r_G(u) \rangle. \end{aligned}$$

Expanding this inner product gives

$$\begin{aligned} h(a+u) = \langle F(a), G(a) \rangle &+ \langle F'(a)u, G(a) \rangle + \langle F(a), G'(a)u \rangle \\ &+ \langle r_F(u), G(a) \rangle + \langle F(a), r_G(u) \rangle + \langle F'(a)u, G'(a)u \rangle \\ &+ \langle F'(a)u, r_G(u) \rangle + \langle r_F(u), G'(a)u \rangle + \langle r_F(u), r_G(u) \rangle. \end{aligned}$$

Hence

$$\begin{aligned} h(a+u) - h(a) - L(u) = \langle r_F(u), G(a) \rangle &+ \langle F(a), r_G(u) \rangle + \langle F'(a)u, G'(a)u \rangle \\ &+ \langle F'(a)u, r_G(u) \rangle + \langle r_F(u), G'(a)u \rangle + \langle r_F(u), r_G(u) \rangle. \end{aligned}$$

We now divide by $\|u\|_X$ and show that the quotient tends to 0. Using Cauchy-Schwarz and boundedness of the linear maps $F'(a)$ and $G'(a)$, we get

$$\begin{aligned} \frac{|\langle r_F(u), G(a) \rangle|}{\|u\|_X} &\le \|G(a)\|_Y \frac{\|r_F(u)\|_Y}{\|u\|_X} \to 0, \\ \frac{|\langle F(a), r_G(u) \rangle|}{\|u\|_X} &\le \|F(a)\|_Y \frac{\|r_G(u)\|_Y}{\|u\|_X} \to 0, \\ \frac{|\langle F'(a)u, G'(a)u \rangle|}{\|u\|_X} &\le \|F'(a)\|\,\|G'(a)\|\,\|u\|_X \to 0, \\ \frac{|\langle F'(a)u, r_G(u) \rangle|}{\|u\|_X} &\le \|F'(a)\|\,\|r_G(u)\|_Y \to 0, \\ \frac{|\langle r_F(u), G'(a)u \rangle|}{\|u\|_X} &\le \|G'(a)\|\,\|r_F(u)\|_Y \to 0, \\ \frac{|\langle r_F(u), r_G(u) \rangle|}{\|u\|_X} &= \left(\frac{\|r_F(u)\|_Y}{\|u\|_X} \right) \|r_G(u)\|_Y \to 0. \end{aligned}$$

Therefore

$$\frac{|h(a+u) - h(a) - L(u)|}{\|u\|_X} \to 0 \qquad \text{as } u \to 0.$$

So h is differentiable at a, with derivative

$$h'(a)(u) = \langle F'(a)u, G(a) \rangle + \langle F(a), G'(a)u \rangle.$$

In abbreviated notation, this is often written as

$$h'(a) = \langle F'(a), G(a) \rangle + \langle F(a), G'(a) \rangle.$$

$\square$

Problem 21.5 (Problem 6.1.10 Spaces). *Let X be a normed space and assume that the function $F : X \to \mathbb{R}$ has its maximal value at a point $a \in X$ where F is differentiable. Show that $F'(a) = 0$.*

Proof. Since F has a maximal value at a, we have

$$F(x) \leq F(a)$$

for all $x \in X$.

We want to show that $F'(a) = 0$. Since $F'(a)$ is a bounded linear functional on X, it suffices to show that

$$F'(a)(h) = 0$$

for every $h \in X$.

Let $h \in X$ be arbitrary, and define a function

$$\varphi : \mathbb{R} \to \mathbb{R}, \qquad \varphi(t) = F(a + th).$$

Since F has a maximum at a, the function φ has a maximum at $t = 0$, because for every $t \in \mathbb{R}$,

$$\varphi(t) = F(a + th) \leq F(a) = \varphi(0).$$

Since F is differentiable at a, φ is differentiable at 0, and

$$\varphi'(0) = F'(a)(h).$$

On the other hand, because φ has a maximum at 0, the usual one-variable result gives

$$\varphi'(0) = 0.$$

Hence

$$F'(a)(h) = 0.$$

Since $h \in X$ was arbitrary, it follows that

$$F'(a)(h) = 0 \qquad \text{for all } h \in X.$$

Therefore $F'(a)$ is the zero functional, i.e.

$$F'(a) = 0.$$

$\square$

21.1 Useful properties and rules

Proposition 21.1. *If F is Frechet differentiable at $\overline{x}$, then it is continuous at $\overline{x}$. In particular*

$$\limsup_{x\to\overline{x}} \frac{\|F(x)-F(\overline{x})\|_Y}{\|x-\overline{x}\|_X} \leq \|F'(\overline{x})\|_{\mathcal{L}}.$$

Proposition 21.2. *The following hold for normed spaces X, Y.*

1. *If $F \in \mathcal{L}(X,Y)$, then $F'(x) = F(x)$ for all $x \in X$,*
2. *If $F : X \to Y$ is constant then $F'(x) = 0$ at all $x \in X$.*

Proposition 21.3. *The following hold for $F, G : X \to Y$ where X, Y are normed spaces.*

1. $(F+G)'(x) = F'(x) + G'(x)$,
2. $(FG)'(x) = F'(x)G(x) + F(x)G'(x)$.

Proposition 21.4. *For normed spaces X, Y, Z and $x \in X$ let $F : X \to Y$ be differentiable at x and let $G : Y \to Z$ be differentiable at $F(x)$.*
Then $G \circ F$ is differentiable at x and

$$(G \circ F)'(x) = G(F(x)) \circ F'(x).$$

22 Gateaux Derivative

Definition 22.1. *Let $F : O \to Y$ ($O \subseteq X$ open) be a function and $x \in O$. The Gateaux Derivative (or directional derivative) of F at x in some direction $r \in X$ is*

$$F'(x;r) = \lim_{t\to 0, t\in\mathbb{R}} \frac{F(x+tr) - F(x)}{t}$$

Recall that partial derivatives of $F : \mathbb{R}^m \to \mathbb{R}^n$ are

$$\frac{\partial}{\partial x^j} F(x) = \lim_{t\to 0} \frac{F(x+te_j) - F(x)}{t} = F'(x;e_j)$$

where $e_j = (0, \dots, 1, \dots, 0)$ (j-th position is 1).

Proposition 22.1. *If F is Frechet differentiable at x, then*

$$F'(x;r) = F'(x)(r)$$

for all $r \in X$.

The opposite implication is not necessarily true.

Proposition 22.2. *Let $O \subseteq \mathbb{R}^m$ be open, let $x \in O$ and assume $F : O \to \mathbb{R}^n$ is such that all partial derivatives $\frac{\partial}{\partial x^j} F(x)$ ($j = 1, 2, \dots, m$) exists for all $y \in O$ and are continuous at $y = x$. Then F is frechet differentiable at x.*

22.1 Some problems from 6.2

Problem 22.1 (Problem 6.2.1 Spaces). *Let $F : \mathbb{R}^3 \to \mathbb{R}^2$ be given by*

$$F(x,y,z) = \begin{pmatrix} x^2y + z \\ xyz^2 \end{pmatrix}$$

Find the Jacobian matrix $JF(a)$ for $a = (1, -1, 2)$. Show that F is differentiable and compute $F'(a)(r)$ when $r = (2, 0, -2)$.

Solution & proof. Write $F_1 = x^2y + z$ and $F_2 = xyz^2$. Then the Jacobian matrix is the matrix of first partial derivatives:

$$JF(x,y,z) = \begin{pmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} & \frac{\partial F_1}{\partial z} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} & \frac{\partial F_2}{\partial z} \end{pmatrix}.$$

Computing the partials we get:

$$JF(x,y,z) = \begin{pmatrix} 2xy & x^2 & 1 \\ yz^2 & xz^2 & 2xyz \end{pmatrix}.$$

At $a = (1,-1,2)$ we get

$$\boxed{JF(a) = \begin{pmatrix} -2 & 1 & 1 \\ -4 & 4 & -4 \end{pmatrix}.}$$

All entries of $JF(x,y,z)$ are polynomials, therefore in C^∞, hence differentiable everywhere. Then

$$F'(a)(r) = \begin{pmatrix} -2 & 1 & 1 \\ -4 & 4 & -4 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} -6 \\ 0 \end{pmatrix}.$$

$\square$

Problem 22.2 (Problem 6.2.2 Spaces p.185). *Let $F : \mathbb{R}^2 \to \mathbb{R}^3$ be given by*

$$F(x,y) = \begin{pmatrix} xe^{x+y} \\ xy^2 \\ x^2 + y^2 \end{pmatrix}.$$

Find the Jacobian matrix $JF(a)$ for $a = (1,-1)$. Show that F is differentiable and compute $F'(a)(r)$ when $r = (2,1)$.

Solution & proof. Write $F_1 = xe^{x+y}$, $F_2 = xy^2$, $F_3 = x^2 + y^2$. Then the Jacobian is

$$JF(x,y) = \begin{pmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \\ \frac{\partial F_3}{\partial x} & \frac{\partial F_3}{\partial y} \end{pmatrix}$$

which comes out to

$$JF(x,y) = \begin{pmatrix} (x+1)e^{x+y} & xe^{x+y} \\ y^2 & 2xy \\ 2x & 2y \end{pmatrix}$$

Computing at $a = (1,-1)$ we get

$$JF(a) = \begin{pmatrix} 2 & 1 \\ 1 & -2 \\ 2 & -2 \end{pmatrix}$$

All entries of $JF(x,y,z)$ are infinitely differentiable everywhere so F is differentiable. Lastly

$$F'(a)(r) = \begin{pmatrix} 2 & 1 \\ 1 & -2 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4+1 \\ 2-2 \\ 4-2 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ 2 \end{pmatrix}$$

$\square$

Problem 22.3 (Problem 6.2.3 Spaces p.185). *Let* $f : \mathbb{R}^4 \to \mathbb{R}$ *be given by*

$$f(x, y, z, u) = xyu + xyz^2 \sin(xz).$$

Find the gradient $\nabla f(a)$ *for* $a = (2, 1, -1, 0)$. *Show that* f *is differentiable and compute* $f'(a)(r)$ *when* $r = (1, -1, 3, 1)$.

Solution & proof. Let

$$f(x, y, z, u) = xyu + xyz^2 \sin(xz).$$

We compute the partial derivatives.

Partial derivatives:

$$\begin{aligned}
\frac{\partial f}{\partial x} &= yu + yz^2(\sin(xz) + xz\cos(xz)),\\
\frac{\partial f}{\partial y} &= xu + xz^2 \sin(xz),\\
\frac{\partial f}{\partial z} &= xy(2z\sin(xz) + xz^2\cos(xz)),\\
\frac{\partial f}{\partial u} &= xy.
\end{aligned}$$

Hence the gradient is

$$\nabla f(x, y, z, u) = \begin{pmatrix} yu + yz^2(\sin(xz) + xz\cos(xz)) \\ xu + xz^2\sin(xz) \\ xy(2z\sin(xz) + xz^2\cos(xz)) \\ xy \end{pmatrix}.$$

Evaluate at $a = (2, 1, -1, 0)$**:**

Note that $xz = -2$, $z^2 = 1$, $yu = 0$, and $xu = 0$. Thus

$$\begin{aligned}
\nabla f(a) &= \begin{pmatrix} \sin(-2) - 2\cos(-2) \\ 2\sin(-2) \\ 2(2\sin 2 + 2\cos 2) \\ 2 \end{pmatrix}\\
&= \begin{pmatrix} -\sin 2 - 2\cos 2 \\ -2\sin 2 \\ 4\sin 2 + 4\cos 2 \\ 2 \end{pmatrix}.
\end{aligned}$$

Differentiability:

The function f is composed of polynomials and the sine function, both of which are infinitely differentiable. Hence f is differentiable on $\mathbb{R}^4$.

Compute $f'(a)(r)$ **for** $r = (1, -1, 3, 1)$:

Since $f'(a)(r) = \nabla f(a) \cdot r$, we obtain

$$\begin{aligned} f'(a)(r) &= (-\sin 2 - 2\cos 2)(1) + (-2\sin 2)(-1) \\ &\quad + (4\sin 2 + 4\cos 2)(3) + 2(1) \\ &= 13\sin 2 + 10\cos 2 + 2. \end{aligned}$$

□

Problem 22.4 (Problem 6.2.4 Spaces p.185). *In this problem,* $X = C([0,1], \mathbb{R})$ *with the usual supremum norm,* $\|y\| = \sup\{|y(t)| : t \in [0,1]\}$. *Define a function* $F : X \to \mathbb{R}$ *by*

$$F(y) = y(0)^2 + y(1)^3.$$

Show that F *is differentiable and find an expression for* F'.

Proof. Let $y, h \in X$. Then

$$F(y+h) = (y(0) + h(0))^2 + (y(1) + h(1))^3.$$

Expanding gives

$$\begin{aligned} F(y+h) &= y(0)^2 + 2y(0)h(0) + h(0)^2 \\ &\quad + y(1)^3 + 3y(1)^2h(1) + 3y(1)h(1)^2 + h(1)^3. \end{aligned}$$

Hence

$$F(y+h) - F(y) = 2y(0)h(0) + 3y(1)^2h(1) + h(0)^2 + 3y(1)h(1)^2 + h(1)^3.$$

Define $L_y : X \to \mathbb{R}$ by

$$L_y(h) = 2y(0)h(0) + 3y(1)^2h(1).$$

This map is linear, and

$$|L_y(h)| \le 2|y(0)||h(0)| + 3|y(1)|^2|h(1)| \le (2\|y\|_\infty + 3\|y\|_\infty^2)\|h\|_\infty,$$

so L_y is bounded.

The remainder is

$$R_y(h) := F(y+h) - F(y) - L_y(h) = h(0)^2 + 3y(1)h(1)^2 + h(1)^3.$$

Therefore

$$\begin{aligned} |R_y(h)| &\le |h(0)|^2 + 3|y(1)||h(1)|^2 + |h(1)|^3 \\ &\le \|h\|_\infty^2 + 3\|y\|_\infty\|h\|_\infty^2 + \|h\|_\infty^3. \end{aligned}$$

Dividing by $\|h\|_\infty$, we get

$$\frac{|R_y(h)|}{\|h\|_\infty} \leq (1 + 3\|y\|_\infty)\|h\|_\infty + \|h\|_\infty^2 \to 0$$

as $\|h\|_\infty \to 0$. Thus F is differentiable at y, and

$$F'(y)(h) = 2y(0)h(0) + 3y(1)^2 h(1).$$

Since $y \in X$ was arbitrary, F is differentiable on all of X. □

Problem 22.5 (Problem 6.2.7 Spaces p.186). *In this problem, $X = Y = C([0,1], \mathbb{R})$ with the usual supremum norm, $\|y\| = \sup_{t\in[0,1]} |y(t)|$. Assume that $f : \mathbb{R} \to \mathbb{R}$ is a function whose derivative f' is continuous and define $F : X \to Y$ by*

$$F(y)(x) = \int_0^x f(y(s))\, ds$$

a) Fine an expression for the directional derivative $F'(y; r)(x)$. You need not give a formal proof - a heuristic derivation suffices.
b) Prove that F is differentiable, and find an expression for the derivative. This time a formal proof is required.

Partial solution. **a)** We compute (with some "cheating"):

$$\begin{aligned}
F'(a; r)(x) &= \lim_{t\to 0} \int_0^x \frac{f(a(s) + tr(s)) - f(a(s))}{t}\, ds \\
&= \lim_{t\to 0} \int_0^x \frac{f(a(s) + tr(s)) - f(a(s))}{tr(s)}\, r(s)\, ds \\
&= \int_0^x \lim_{t\to 0} \frac{f(a(s) + tr(s)) - f(a(s))}{tr(s)}\, r(s)\, ds \\
&= \int_0^x f'(a(s))\, r(s)\, ds.
\end{aligned}$$

The "cheating" occurs when we move the limit inside the integral. This is not always justified. The interchange of limit and integral requires justification (e.g. dominated convergence).

b) Now, do the same thing as in step 2 in the previous exercise. □

Problem 22.6 (Problem 2 Exam 2022). *The map $F : C([0,1], \mathbb{R}) \to \mathbb{R}$ is given by $F(x) = x(0)x(1)$.*

(a) Find the directional derivative $F'(x;r)$.
(b) Show that F *is differentiable and find the derivative* $F'(x)(r)$.

Solution of (a). We compute the directional derivative from the definition:

$$F'(x;r) = \lim_{t\to 0} \frac{F(x+tr) - F(x)}{t}.$$

Since

$$F(x+tr) = (x(0) + tr(0))(x(1) + tr(1)),$$

we get

$$\begin{aligned}
\frac{F(x+tr) - F(x)}{t} &= \frac{(x(0) + tr(0))(x(1) + tr(1)) - x(0)x(1)}{t} \\
&= \frac{x(0)x(1) + tx(0)r(1) + tx(1)r(0) + t^2 r(0)r(1) - x(0)x(1)}{t} \\
&= x(0)r(1) + x(1)r(0) + tr(0)r(1).
\end{aligned}$$

Hence

$$F'(x;r) = x(0)r(1) + x(1)r(0).$$

□

Proof of (b). The derivative should be the linear map

$$F'(x)(r) = x(0)r(1) + x(1)r(0).$$

Then

$$F(x+r) - F(x) - F'(x)(r) = r(0)r(1).$$

Using the sup-norm we get

$$|r(0)r(1)| \leq \|r\|_\infty^2.$$

So the limit converges to

$$\|r\|_\infty \to 0.$$

So F is differentiable. □

Problem 22.7 (Problem 1 Exam 2022kont). *The map* $F : C([0,1],\mathbb{R}) \to \mathbb{R}$ *is given by*

$$F(x) = \int_0^1 e^{-s} x(s)^2 \, ds.$$

a) Find the directional derivative $F'(x;r)$.
b) Show that F *is differentiable and find* $F'(x)(r)$.

Solution of (a). Fix $x, r \in C([0,1],\mathbb{R})$. The directional derivative is

$$F'(x;r) = \lim_{t\to 0} \frac{F(x+tr)-F(x)}{t}.$$

We compute

$$\begin{aligned} F(x+tr) &= \int_0^1 e^{-s}(x(s)+tr(s))^2\,ds \\ &= \int_0^1 e^{-s}(x(s)^2+2tx(s)r(s)+t^2r(s)^2)\,ds. \end{aligned}$$

Hence

$$\begin{aligned} \frac{F(x+tr)-F(x)}{t} &= \frac{1}{t}\int_0^1 e^{-s}(2tx(s)r(s)+t^2r(s)^2)\,ds \\ &= \int_0^1 e^{-s}(2x(s)r(s)+tr(s)^2)\,ds. \end{aligned}$$

Letting $t \to 0$, we obtain

$$F'(x;r) = \int_0^1 2e^{-s}x(s)r(s)\,ds.$$

$\square$

Solution of (b). Define $A : C([0,1],\mathbb{R}) \to \mathbb{R}$ by

$$A(r) = \int_0^1 2e^{-s}x(s)r(s)\,ds.$$

This map is linear. Moreover,

$$\begin{aligned} |A(r)| &= \left|\int_0^1 2e^{-s}x(s)r(s)\,ds\right| \\ &\le \int_0^1 2e^{-s}|x(s)||r(s)|\,ds \\ &\le 2\|x\|\|r\| \int_0^1 e^{-s}\,ds, \end{aligned}$$

so A is bounded.

We now check differentiability. We have

$$\begin{aligned} F(x+r) &= \int_0^1 e^{-s}(x(s)+r(s))^2\,ds \\ &= \int_0^1 e^{-s}(x(s)^2+2x(s)r(s)+r(s)^2)\,ds. \end{aligned}$$

Hence

$$F(x+r) - F(x) - A(r) = \int_0^1 e^{-s} r(s)^2\, ds.$$

Therefore

$$\begin{aligned} |F(x+r) - F(x) - A(r)| &\leq \int_0^1 e^{-s} |r(s)|^2\, ds \\ &\leq \|r\|^2 \int_0^1 e^{-s}\, ds. \end{aligned}$$

Dividing by $\|r\|$, we get

$$\frac{|F(x+r) - F(x) - A(r)|}{\|r\|} \leq \|r\| \int_0^1 e^{-s}\, ds \to 0 \quad \text{as } \|r\| \to 0.$$

Thus F is differentiable and $F'(x) = A$. □

Problem 22.8 (Exam problem 2 Exam 2023kont). *Let X be a metric space equipped with a metric d. Let $\alpha \in \mathbb{R}$ be such that $0 < \alpha < 1$. Define a function $\rho : X \times X \to \mathbb{R}_+$ by, for all $x, y \in X$:*

$$\rho(x, y) = (d(x, y))^\alpha.$$

a)
Check that for any $a, b \in \mathbb{R}$ such that $a, b \geq 0$ and $a + b > 0$, we have

$$1 \leq \left(\frac{a}{a+b}\right)^\alpha + \left(\frac{b}{a+b}\right)^\alpha.$$

Show that ρ is a metric on X.
b)
Let $\mathrm{id}_X : X \to X$ be the identity map. Show that id_X is continuous from (X, d) to (X, ρ) and from (X, ρ) to (X, d).

Proof of (a).

$$\begin{aligned} \left(\frac{a}{a+b}\right)^\alpha + \left(\frac{b}{a+b}\right)^\alpha &= \left(\frac{a^\alpha}{(a+b)^\alpha}\right) + \left(\frac{b^\alpha}{(a+b)^\alpha}\right) \\ &= \frac{a^\alpha + b^\alpha}{(a+b)^\alpha} \end{aligned}$$

For $\alpha \in (0, 1)$ the quantity at the top is larger than or equal to the one of the bottom, with equality in the case where $a = 0$ or $b = 0$. Therefore the fraction evaluates to ≥ 1.

A similar argument is needed to show that the triangle inequality holds. The rest of the properties follow trivially from d. □

Proof of (b). Fix any $x, y \in X$. Let $\varepsilon > 0$ and suppose $d(x, y) < \delta := \varepsilon^{1/\alpha}$. Then

$$\rho(x, y) = (d(x, y))^\alpha < (\varepsilon^{1/\alpha})^\alpha = \varepsilon.$$

Now let $\varepsilon > 0$ and suppose $\rho(x, y) = (d(x, y))^\alpha < \delta := \varepsilon^\alpha$. Then

$$d(x, y) = (\rho(x, y))^{1/\alpha} < \varepsilon.$$

□

23 Exam Practice

Problem 23.1 (1 Exam 2021). *Let (X, d) be the metric space $X = (0, 1]$, $d(x, y) = |x - y|$, and let $T : X \to X$ be given by $Tx = x/2$. Show that T is a contraction. Does T have a fixed point? Justify your answer.*

Proof. Let $x, y \in X$. Then

$$d(Tx, Ty) = d(x/2, y/2) = \frac{1}{2}d(x, y)$$

so T is a contraction mapping with $0 < c = 1/2 < 1$.

T does not have a fixed point. Banach's fixed point theorem does not apply because $(0, 1]$ is not complete and we can see that the fixed point of T ought to be $0 \notin X$ since

$$Tx = x \Rightarrow \frac{1}{2}x = x$$

so $x = 0$. □

Problem 23.2 (2 Exam 2021). *Let (X, d) be a metric space and let $A \subseteq \mathbb{R}$ be closed. We define the metric space $C_b(X, A) = \{$all continuous, bounded $f : X \to A\}$, equipped with the supremum metric*

$$\rho(f, g) = \sup_{x \in X} |f(x) - g(x)|.$$

Show that $C_b(X, A)$ is a closed subspace of $C_b(X, \mathbb{R})$.

Proof. $C_b(X, A) \subseteq C_b(X, \mathbb{R})$ is trivial, so it remains to show that it is closed. Let $\{f_n\}_{n \in \mathbb{N}}$ be a convergent sequence of functions in $C_b(X, A)$. Fix $x \in X$. Then $\{f_n(x)\}_{n \in \mathbb{N}}$ defines a sequence in A. Then $f_n(x)$ converges to some $y \in A$, since A is closed, and f_n is continuous. So $f_n(x) \underset{n\to\infty}{\to} y \in A$, but by assumption $f_n \underset{n\to\infty}{\to} f \in C_b(X, \mathbb{R})$. Then by sequential continuity and uniqueness of the limit $f(x) = y \in A$. Since x was chosen arbitrarly, $f(x)$ is in A for any $x \in X$. Thus $f_n(x) - f(x) \to 0$ for any $x \in X$. Hence

$$\rho(f_n, f) = \sup_{x \in X} |f_n(x) - f(x)| \to 0,$$

so $f_n \to f$ uniformly. □

Problem 23.3 (4 Exam 2021). *Recall that a bounded linear operator A is invertible if it is bijective and A^{-1} is bounded.*
Let $(X, \|\cdot\|)$ be a normed vector space and let $A : X \to X$ be an invertible linear operator. Define $\|x\|_A = \|Ax\|$ for every $x \in A$.

a) *Show that $\|\cdot\|_A$ is a norm on X.*

b) *Show that a sequence $\{x_n\}_n$ in X converges w.r.t $\|\cdot\|$ iff it converges w.r.t $\|\cdot\|_A$.*

Proof of (a). $\|x\|_A$ is non-negative since $\|Ax\| \geq 0$. Suppose $\|x\|_A = 0$. Then $\|Ax\| = 0$ so $Ax = 0$. Then, since A is not the 0-operator, $x = 0$. The converse is trivial.

Now let α be in the ground field.

$$\|\alpha x\|_A = \|A(\alpha x)\| = \|\alpha Ax\| = \alpha\|Ax\| = \alpha\|x\|_A$$

Lastly we check the triangleinequality holds

$$\begin{aligned} \|x+y\|_A &= \|A(x+y)\| \\ &= \|Ax + Ay\| \\ &\leq \|Ax\| + \|Ay\| \\ &= \|x\|_A + \|y\|_A \end{aligned}$$

□

Proof of (b). Suppose $\{x_n\}_n$ is a sequence in X.

Assume first that $\{x_n\}_n \underset{n\to\infty}{\to} x$ w.r.t $\|\cdot\|$. Then $\forall \varepsilon > 0$ there is $N \in \mathbb{N}$ so that, for $n \geq N$, we have

$$\|x_n - x\| < \varepsilon$$

Recall that for bounded linear operator A we have

$$\|A\|_{\mathcal{L}} := \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|},$$

so $\|x\|_A = \|Ax\| \leq \|A\|_{\mathcal{L}}\|x\|$.

Pick N large enough so that

$$\|x_n - x\| < \varepsilon/\|A\|_{\mathcal{L}}.$$

Then

$$\|x_n - x\|_A \leq \|x_n - x\| \cdot \|A\|_{\mathcal{L}} < \varepsilon.$$

Now assume $\{x_n\}_n$ converges to x w.r.t $\|x\|_A$, i.e. $\forall \varepsilon > 0$ there exists N large enough so that

$$\|x_n - x\|_A = \|A(x_n - x)\| < \varepsilon.$$

We will now use the fact that $x = AA^{-1}x = A^{-1}Ax$, so

$$\|x\| = \|A^{-1}Ax\| \leq \|A^{-1}\|_{\mathcal{L}}\|x\|_A.$$

Pick N large enough so that

$$\|x_n - x\|_A < \varepsilon/\|A^{-1}\|_{\mathcal{L}}.$$

Then

$$\|x_n - x\| \leq \|A^{-1}\|_{\mathcal{L}}\|x_n - x\|_A < \varepsilon.$$

□

Problem 23.4 (Problem 1). *Let (X, d) be a metric space.*

a) Suppose (x_n) is a Cauchy sequence in X, and suppose it has a subsequence (x_{n_k}) converging to $a \in X$. Prove that $x_n \to a$.

b) Let $A \subseteq X$ be compact. Show that A is complete.

c) Let $A, B \subseteq X$ be compact. Show that $A \cup B$ is compact.

Proof of (a). Suppose (x_n) is a cauchy sequence in X. Then for $\varepsilon > 0$ we can find $N_1 \in \mathbb{N}$ such that

$$d(x_n, x_m) < \varepsilon/2$$

whenever $n, m \geq N_1$.

Furthermore there is a sub-sequence $(x_{n_k}) \to a$ then for every $\varepsilon > 0$ we can find $N \in \mathbb{N}$ such that when $k \geq N$ we have

$$d(x_{n_k}, a) < \varepsilon/2,$$

and $n_k \geq N_1$. Then

$$d(x_n, a) \leq d(x_n, x_{n_k}) + d(x_{n_k}, a) < \varepsilon.$$

□

Proof of (b). $A \subseteq X$ is compact if every sequence $(x_n) \subset A$ has a convergent subsequence with $a \in A$. Then by (a) every Cauchy sequence converges to $a \in A$. □

Proof of (c). Let (x_n) be a Cauchy sequence in $A \cup B$. Then denote the subsequences

$$(x_n)\,\big|_A = (x_n : x_n \in A),$$

and

$$(x_n)\,\big|_B = (x_n : x_n \in B).$$

One of these will contain infinitely many terms x_n, since if we assume for instance that A only contains finitely many then the pigeonhole principle yields infinitely many terms in B and vice-versa. Thus we can assume, without loss of generality, that A contains infinitely many terms. Then $(x_n)\big|_A$ is a Cauchy sequence in A and thus converges to $a \in A \subseteq A \cup B$. By (a) we get that

$$(x_n) \to a \in A \cup B.$$

As (x_n) was arbitrary we conclude that $A \cup B$ is complete. □

Problem 23.5 (Problem 2). *Let $X = \mathbb{R}^{n\times n}$ with the operator norm. Fix matrices $A, B \in X$, and define*

$$F : X \to X, \qquad F(M) = AMB.$$

a) Show that F is linear and bounded.

b) Now define

$$G : X \to X, \qquad G(M) = MAM.$$

Show that G is differentiable and compute $DG(M)H$.

c) Show that G is Lipschitz on every bounded subset of X.

Proof of (a). Let $M, N \in \mathbb{R}^{n\times n}$. Then

$$\begin{aligned} F(M+N) &= A(M+N)B \\ &= (AM + AN)B \\ &= AMB + ANB \\ &= F(M) + F(N). \end{aligned}$$

Now let $\lambda \in \mathbb{R}$, the ground-field. Then

$$\begin{aligned} F(\lambda M) &= A\lambda MB \\ &= \lambda AMB \\ &= \lambda F(M) \end{aligned}$$

It is trivial to see that

$$\|F(M)\| \leq \|A\|\|B\|\|M\|.$$

Hence bounded. □

Solution and proof of (b). We compute

$$\lim_{\|h\|\to 0} \frac{\|G(M+h) - G(M) - A(h)\|}{\|h\|}.$$

First

$$\begin{aligned} G(M+h)-G(M) &= (M+h)A(M+h)-(MAM) \\ &= (MA+hA)(M+h)-(MAM) \\ &= (MA(M+h)-hA(M+h))-(MAM) \\ &= MAM+MAh-hAM-hAh-MAM \\ &= MAh-hAM-hAh \end{aligned}$$

The linear part is $MAh-hAM =: A(h) = DG(M)h$.

The remainder is hAh so

$$\begin{aligned} \lim_{\|h\|\to 0} \frac{\|G(M+h)-G(M)-A(h)\|}{\|h\|} &= \lim_{\|h\|\to 0} \frac{\|hAh\|}{\|h\|} \\ &\leq \lim_{\|h\|\to 0} \frac{\|h\|^2\|A\|}{\|h\|} \\ &= \lim_{\|h\|\to 0} \|h\|\|A\| \\ &= 0 \end{aligned}$$

so F is differentiable. □

Proof of (c). Let $A \subseteq X$ be bounded. Then there is $\lambda > 0$ so $\|M\|, \|N\| \leq \lambda$ when $M, N \in A$.

Then

$$\begin{aligned} \|G(M)-G(N)\| &= \|MAM-NAN\| \\ &= \|MA(M-N)+(M-N)AN\| \\ &\leq \lambda\|A\|\|M-N\| + \|M-N\|\lambda\|A\| \\ &= 2\lambda\|A\|\|M-N\|. \end{aligned}$$

Thus G is Lipschitz with constant $2\lambda\|A\|$. □

Problem 23.6 (Problem 3). *Let*

$$X = C([0,1], \mathbb{R})$$

with the supremum norm, and let $k : [0,1]^2 \to \mathbb{R}$ *be continuous. Define*

$$(Lu)(x) = \int_0^1 k(x,y)u(y)\,dy.$$

a) *Show that* $Lu \in X$ *for every* $u \in X$.

b) *Show that* $L : X \to X$ *is bounded.*

c) Assume now that

$$k(x,y) = xy.$$

Show that $I - L : X \to X$ is bijective and has bounded inverse.

Proof of (a). Let $x_1, x_2 \in [0,1]$. If $|x_1 - x_2| < \delta$ then

$$|k(x_1,y) - k(x_2,y)| < \varepsilon/\|u\|, \quad \forall y,$$

assuming u is nonzero. Then

$$|(Lu)(x_1) - (Lu)(x_2)| \leq \int_0^1 |k(x_1,y) - k(x_2,y)| \cdot |u(y)| \, dy \leq \varepsilon.$$

□

Proof of (b). Since k is continuous on a closed and bounded (and thus compact) interval we have $|k(x,y)| \leq M$ for some $M > 0$.

Then by the same approximation as in (a) we have

$$|(Lu)(x)| \leq \int_0^1 |k(x,y)| \cdot |u(y)| \, dy \leq M\|u\|.$$

□

Proof of (c). If $k(x,y) = xy$ then

$$(Lu)(x) = \int_0^1 xy \cdot u(y) \, dy$$

and

$$((I - L)(u))(x) = u(x) - \int_0^1 xy \cdot u(y) \, dy$$

Now we observe that

$$((I - L)(u))(x) = u(x) - x \cdot c_u$$

where

$$c_u = \int_0^1 xy \cdot u(y) \, dy.$$

Solving $(I - L)(u) = f$ we get

$$u(x) = f(x) + c_u x.$$

Integrating c_u we get

$$c_u = \int_0^1 f(x) \, dx + (1/3)c_u.$$

Something something, unique solution... □

Problem 23.7 (Problem 4). *Let $(\Omega, \mathcal{A}, \mu)$ be a measure space.*

a) Let $A_1, A_2, \ldots \in \mathcal{A}$. Prove countable subadditivity:

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

b) Suppose $A_n \uparrow A$, meaning

$$A_1 \subseteq A_2 \subseteq \cdots, \qquad A = \bigcup_{n=1}^{\infty} A_n.$$

Show that

$$\mu(A) = \lim_{n\to\infty} \mu(A_n).$$

c) Let $f_n : \Omega \to [0, \infty]$ be measurable and suppose

$$f_n \uparrow f.$$

State the Monotone Convergence Theorem and explain what it says about

$$\int f_n \, d\mu \quad \text{and} \quad \int f \, d\mu.$$

Proof of (a). Let $B_1 = A_1$ and define $B_n = A_n \setminus \cup_{k=1}^{n-1} A_k$. Then $B_1, B_2, \ldots$ are disjoint and furthermore:

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n$$

so their meassure is the same and in particular

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(B_n).$$

Since $B_n \subseteq A_n$ for all $n \in \mathbb{N}$ we have

$$\sum_{n=1}^{\infty} \mu(B_n) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

Combining the latter two results gives the desired result. $\square$

Proof of (b). Similarily to (a) let $B_1 = A_1$ and define $B_n = A_n \setminus B_{n-1}$. Then as above B_n are all disjoint and

$$A = \bigcup_{n=1}^{\infty} B_n$$

so

$$\mu(A) = \sum_{n=1}^{\infty} \mu(B_n).$$

Furthermore

$$\mu(A_N) = \sum_{n=1}^{N} \mu(B_n)$$

so for $N \to \infty$ we get

$$\lim_{N\to\infty} \mu(A_N) = \mu(A).$$

□

Solution to (c). Let $(X, \mathcal{A}, \mu)$ be a meassure space and let $(f_n)_n \in M_+(X, \mathcal{A})$ be an increasing sequence, i.e. $0 \leq f_n \leq f_{n+1}$ for all $n \in \mathbb{N}$. Define

$$f(x) = \lim_{n\to\infty} f_n(x).$$

Then f is well-defined and meassurable, and

$$\int f \, d\mu = \int f_n \, d\mu.$$

The implications are clear. □

Problem 23.8 (Problem 5). *For $n \in \mathbb{N}$, define*

$$f_n : [0,1] \to \mathbb{R}, \qquad f_n(x) = \frac{nx}{1 + n^2x^2}.$$

a) Find the pointwise limit of f_n on $[0,1]$.

b) Decide whether $f_n \to f$ uniformly on $[0,1]$.

c) Decide whether $f_n \to f$ uniformly on $[\delta, 1]$ for every fixed $\delta > 0$.

Start of solution of (a). For $x = 0$, f_n is 0 for all n. For $x = 1$ we get

$$f_n(1) = \frac{n}{1+n^2} \leq \frac{1}{n}$$

which goes to 0 for increasing n.

We can differentiate to get the rate of change on $[0,1]$. □

23.1 Problem type set

Analyzing earlier relevant exam sets, ChatGPT came up with problem "types" and has comprised the following problem set with problem types which I personally struggle with. So the following is a no aid attempt at solving those questions.

The following list summarizes the main types of problems that seem relevant for MAT2400 practice. The goal is not to memorize solutions, but to recognize the structure of a problem and know which techniques are likely to apply.

23.2 Type 1: Basic metric-space structure

Type 1A: Subspaces as metric spaces / failure of vector-space structure

Typical task:

> Let $S \subseteq X$ inherit a metric from a metric space (X, d). Explain why (S, d) is a metric space. Then decide whether S is a normed vector space.

Typical trigger:

$$S \subseteq X \quad \Rightarrow \quad d|_{S\times S} \text{ is a metric.}$$

But for a normed vector space, S must be closed under addition and scalar multiplication.

Type 1B: Cauchy sequence with convergent subsequence

Typical task:

> Suppose (x_n) is Cauchy in a metric space and has a subsequence converging to a. Prove that $x_n \to a$.

Typical trigger:

$$d(x_n, a) \le d(x_n, x_{n_k}) + d(x_{n_k}, a).$$

Type 1C: Completeness of unions

Typical task:

> Let $A, B \subseteq X$ be complete. Prove that $A \cup B$ is complete.

Typical trigger: A Cauchy sequence in $A \cup B$ has infinitely many terms in A or infinitely many terms in B.

Type 1D: Closedness of function spaces

Typical task:

> Let $A \subseteq \mathbb{R}$ be closed. Show that $C_b(X, A)$ is closed in $C_b(X, \mathbb{R})$ with the supremum metric.

Typical trigger:

$$f_n \to f \text{ uniformly}, \quad f_n(x) \in A, \quad A \text{ closed} \quad \Rightarrow \quad f(x) \in A.$$

23.3 Type 2: Compactness and extreme-value arguments

Type 2A: Compact set gives minimizing point

Typical task:

> Let $E \subseteq X$ be compact and nonempty. Show that for each $x \in X$, there exists $z \in E$ such that
> $$d(x, z) = \text{dist}(x, E).$$

Typical trigger: The function

$$y \mapsto d(x, y)$$

is continuous on compact E, hence attains a minimum.

Type 2B: Counterexample when compactness fails

Typical task:

> Give an example where $\text{dist}(x, E)$ is not attained.

Typical trigger: Use an open set, for example

$$X = \mathbb{R}, \qquad E = (0, 1), \qquad x = 2.$$

Type 2C: Compactness via sequences

Typical task:

> If A and B are compact subsets of a metric space, prove that $A \cup B$ is compact.

Typical trigger: Every sequence in $A \cup B$ has infinitely many terms in either A or B.

Type 2D: Noncompactness by separated sequences

Typical task:

> Show that a set is not compact by constructing a sequence with no convergent subsequence.

Typical trigger:

$$\|x_n - x_m\| \geq c > 0 \quad \text{for all } n \neq m.$$

Then no subsequence is Cauchy.

23.4 Type 3: Modified metrics and equivalent convergence

Type 3A: Max metric from two metrics

Typical task:

> Given two metrics d_1, d_2 on X, define
> $$\rho(x, y) = \max\{d_1(x, y), d_2(x, y)\}.$$
> Show that ρ is a metric and characterize convergence in ρ.

Typical trigger:

$$\max\{a_1 + b_1, a_2 + b_2\} \leq \max\{a_1, a_2\} + \max\{b_1, b_2\}.$$

Convergence:

$$x_n \to x \text{ in } \rho \iff x_n \to x \text{ in both } d_1 \text{ and } d_2.$$

Type 3B: Snowflake metric

Typical task:

> Let (X, d) be a metric space and $0 < \alpha < 1$. Define
> $$\rho(x, y) = d(x, y)^\alpha.$$
> Show that ρ is a metric and compare continuity or Lipschitz properties of the identity maps.

Typical trigger:

$$(a + b)^\alpha \leq a^\alpha + b^\alpha \qquad (a, b \geq 0).$$

On $[0, 1]$:

$$|x - y| \leq |x - y|^\alpha,$$

but generally there is no constant C such that

$$|x - y|^\alpha \leq C|x - y|$$

near $x = y$.

23.5 Type 4: Uniform continuity and Lipschitz estimates

Type 4A: Uniform continuity on unbounded domains

Typical task:

> Prove that a function such as $f(x) = e^{-x^2}$ is uniformly continuous on $\mathbb{R}$.

Typical triggers:

$$f' \text{ bounded } \Rightarrow f \text{ Lipschitz } \Rightarrow f \text{ uniformly continuous.}$$

Or split the real line into a compact middle region and small tails.

Type 4B: Bounded derivative implies Lipschitz

Typical task:

> Show that a differentiable map is Lipschitz on a set because its derivative is bounded.

Typical trigger:

$$\|f(x) - f(y)\| \le \sup_z \|Df(z)\| \, \|x - y\|.$$

23.6 Type 5: Normed spaces and bounded linear maps

Type 5A: Show a map is linear and bounded

Typical task:

> Let $L : X \to Y$ be given by a formula. Show that L is linear and bounded.

Typical trigger:

$$\|Lf\|_Y \le C\|f\|_X.$$

Type 5B: Bounded bijection with unbounded inverse

Typical task:

> Show that a bounded linear bijection $L : X \to Y$ has an unbounded inverse.

Typical trigger: Construct a sequence (g_n) in Y such that

$$\|g_n\|_Y \le C, \qquad \|L^{-1} g_n\|_X \to \infty.$$

Type 5C: Equivalent norms from invertible operators

Typical task:

> Let $A : X \to X$ be an invertible bounded linear operator. Define
> $$\|x\|_A = \|Ax\|.$$
> Show that this is a norm and that convergence in $\|\cdot\|_A$ is equivalent to convergence in $\|\cdot\|$.

Typical trigger:

$$\|Ax\| \le \|A\|\|x\|,$$

and

$$\|x\| = \|A^{-1}Ax\| \le \|A^{-1}\|\|Ax\|.$$

Type 5D: Neumann-series inverse

Typical task:

> If $\|C\| < 1$, prove that
> $$(I - C)^{-1} = \sum_{k=0}^{\infty} C^k.$$

Typical trigger:

$$(I - C)\sum_{k=0}^{n} C^k = I - C^{n+1}.$$

If $\|C\| < 1$, then $C^{n+1} \to 0$ in operator norm.

Type 5E: Fixed-point iteration in Banach spaces

Typical task:

> Show that an iteration
> $$x_{n+1} = T(x_n)$$
> converges by proving that T is a contraction.

Typical trigger:

$$\|T(x) - T(y)\| \le q\|x - y\|, \qquad 0 \le q < 1.$$

Then Banach's fixed point theorem applies.

23.7 Type 6: Fréchet differentiability

Type 6A: Finite-dimensional Jacobian problems

Typical task:

> Compute $F'(x)$ for $F : \mathbb{R}^n \to \mathbb{R}^m$, and possibly use the inverse function theorem.

Typical triggers:

$$F'(x)h = DF(x)h.$$

If $DF(x_0)$ is invertible, then locally

$$(F^{-1})'(F(x_0)) = (F'(x_0))^{-1}.$$

Type 6B: Matrix product differentiability

Typical task:

> Let
> $$F(M) = MJM.$$
> Show that F is differentiable and compute $DF(M)H$.

Typical trigger:

$$(M + H)J(M + H) - MJM = MJH + HJM + HJH.$$

Therefore

$$DF(M)H = MJH + HJM.$$

Remainder estimate:

$$\frac{\|HJH\|}{\|H\|} \leq \|J\|\|H\| \to 0.$$

Type 6C: Functional differentiability on $C([0, 1])$

Typical task:

> Show that a functional such as
> $$F(x) = x(0)x(1)$$
> or
> $$F(x) = \int_0^1 e^{-s} x(s)^2 \, ds$$
> is Fréchet differentiable.

Typical triggers:
$$F'(x)r = x(0)r(1) + r(0)x(1)$$
for endpoint-product functionals, and
$$F'(x)r = \int_0^1 e^{-s} 2x(s)r(s)\, ds$$
for quadratic integral functionals.

Always check:

1. the candidate derivative is linear,
2. the candidate derivative is bounded,
3. the remainder is $o(\|r\|)$.

Type 6D: Nonlinear maps on C_b

Typical task:

> Show that a nonlinear map such as
> $$F(f)(t) = 2f(t)^2 - e^{f(t)-t^2}$$
> is Fréchet differentiable.

Typical trigger: Use Taylor expansion with uniform remainder estimates:
$$e^{f+r} - e^f - e^f r = e^f(e^r - 1 - r) = O(\|r\|^2).$$

Type 6E: Directional derivatives but no Fréchet derivative

Typical task:

> Study differentiability at the origin for a function with a formula involving
> $$\frac{xy}{x^2+y^2}.$$

Typical trigger: Check values along lines or compute directional derivatives. If the directional derivative is not linear in the direction, then the Fréchet derivative cannot exist.

23.8 Type 7: Integral operators

Type 7A: Show $Lu \in C([0,1])$

Typical task:

> Let
> $$(Lu)(x) = \int_0^1 k(x,y)u(y)\,dy$$
> with k continuous. Show that Lu is continuous.

Typical trigger: k is uniformly continuous on compact $[0,1]^2$, and

$$|(Lu)(x) - (Lu)(x')| \leq \int_0^1 |k(x,y) - k(x',y)|\,|u(y)|\,dy.$$

Type 7B: Show L is bounded

Typical task:

> Show that the same integral operator is bounded on $C([0,1])$.

Typical trigger:

$$\|Lu\|_\infty \leq \left(\sup_{x\in[0,1]} \int_0^1 |k(x,y)|\,dy \right) \|u\|_\infty.$$

Type 7C: Rank-one integral operator inverse

Typical task:

> For a separable kernel
> $$k(x,y) = a(x)b(y),$$
> show that $I \pm L$ is bijective and compute its inverse.

Typical trigger:

$$Lu = a(x)c(u), \qquad c(u) = \int b(y)u(y)\,dy.$$

Then solve
$$u \pm c(u)a = f$$
by applying the functional c to both sides.

Type 7D: Volterra/exponential integral operators

Typical task:

> Let
> $$F(u)(t) = e^t \int_t^\infty e^{-\tau} \sin(u(\tau))\, d\tau.$$
> Prove boundedness, differentiability in t, and compute the Fréchet derivative.

Typical triggers:

$$|F(u)(t)| \le \int_t^\infty e^{t-\tau}\, d\tau = 1.$$

Also:

$$F'(u)r(t) = e^t \int_t^\infty e^{-\tau} \cos(u(\tau)) r(\tau)\, d\tau.$$

23.9 Type 8: Series and sequences of functions

Type 8A: Power series radius and endpoints

Typical task:

> Determine the convergence set of a power series and of its derivative series.

Typical trigger:

$$R^{-1} = \limsup_{n\to\infty} |a_n|^{1/n}.$$

Then check endpoints separately.

Type 8B: Geometric series with parameter

Typical task:

> Study
> $$\sum_{n=0}^\infty e^{-nx}.$$
> Find where it converges, prove uniform convergence on suitable intervals, and justify termwise integration.

Typical trigger:

$$e^{-nx} \le e^{-na} \qquad x \in [a, \infty).$$

Then use the Weierstrass M-test.

Type 8C: Uniform convergence via M-test

Typical task:

> Prove uniform convergence of a series of functions using a summable numerical majorant.

Typical trigger:

$$|f_n(x)| \le M_n, \qquad \sum_{n=1}^{\infty} M_n < \infty.$$

Type 8D: Pointwise versus uniform convergence

Typical task:

> For a sequence such as
>
> $$f_n(x) = x^n$$
>
> or
>
> $$f_n(x) = \frac{nx}{1 + n^2 x^2},$$
>
> find the pointwise limit and decide whether convergence is uniform.

Typical trigger:

$$\sup_{x \in D} |f_n(x) - f(x)|.$$

Often the bad behavior occurs near one endpoint or one moving point.

23.10 Type 9: Sequence spaces and completeness

Type 9A: Truncations in ℓ^p

Typical task:

> For $x \in \ell^p$, define
>
> $$y_n = (x(1), \ldots, x(n), 0, 0, \ldots).$$
>
> Show that (y_n) is Cauchy for $p < \infty$, and decide what happens for $p = \infty$.

Typical trigger:

$$\|y_n - y_m\|_p^p = \sum_{i=m+1}^{n} |x(i)|^p,$$

which is controlled by tails of a convergent series.

For $p = \infty$, use

$$x = (1, 1, 1, ...)$$

as a counterexample.

Type 9B: Incomplete spaces via Cauchy sequences

Typical task:

> Construct a Cauchy sequence in a normed vector space that does not converge in the space.

Typical trigger: Use partial sums that converge to an object outside the space.

Example structure:

$$f_n = \sum_{m=0}^{n} 2^{-m} e_m,$$

where the limit has infinite support, while the space only contains finite-support functions.

23.11 Type 10: Hilbert-space and orthonormal-system problems

Type 10A: Uniqueness of coefficients

Typical task:

> Suppose
>
> $$u = \sum_{n=1}^{\infty} \alpha_n e_n = \sum_{n=1}^{\infty} \beta_n e_n$$
>
> for an orthonormal system (e_n). Prove that $\alpha_n = \beta_n$ for all n.

Typical trigger:

$$\alpha_i = \langle u, e_i \rangle = \beta_i.$$

Type 10B: Bessel inequality and convergence of orthogonal series

Typical task:

> Given $\alpha_n = \langle u, e_n \rangle$, show that
>
> $$\sum_{n=1}^{\infty} \alpha_n e_n$$
>
> converges.

Typical trigger:

$$\sum_{n=1}^{\infty} |\alpha_n|^2 \le \|u\|^2.$$

Then partial sums are Cauchy.

Type 10C: Failure of completeness of an orthonormal system

Typical task:

> Explain how an orthonormal sequence can fail to be a basis.

Typical trigger: There may exist u such that

$$u \neq \sum_{n=1}^{\infty} \langle u, e_n \rangle e_n.$$

23.12 Type 11: Measure theory

Type 11A: Empirical measures

Typical task:

> Given real numbers $x_1, \dots, x_N$, define
>
> $$\mu_N(A) = \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}_A(x_n).$$
>
> Show that μ_N is a measure and compute $\mu_N(\mathbb{R})$.

Typical trigger: For disjoint A_i,

$$\mathbf{1}_{\cup_i A_i}(x_n) = \sum_i \mathbf{1}_{A_i}(x_n).$$

Type 11B: Integrating simple functions with respect to empirical measures

Typical task:

> If
>
> $$g = \sum_{i=1}^{m} a_i \mathbf{1}_{A_i},$$
>
> compute
>
> $$\int g \, d\mu_N.$$

Typical trigger:

$$\int g\, d\mu_N = \frac{1}{N}\sum_{n=1}^{N} g(x_n).$$

Type 11C: Continuous functions under empirical measures

Typical task:

> Show that for continuous f,
>
> $$\int f\, d\mu_N = \frac{1}{N}\sum_{n=1}^{N} f(x_n).$$
>
> If $x_n \to \bar{x}$, compute
>
> $$\lim_{N\to\infty} \int f\, d\mu_N.$$

Typical trigger:

$$\frac{1}{N}\sum_{n=1}^{N} f(x_n) \to f(\bar{x})$$

when $f(x_n) \to f(\bar{x})$.

Type 11D: Markov and Chebyshev inequalities

Typical task:

> Let $f \geq 0$ be integrable. Show that
>
> $$\mu(\{x : f(x) \geq t\}) \leq \frac{1}{t}\int f\, d\mu.$$

Typical trigger:

$$\int f\, d\mu \geq \int_{\{f\geq t\}} f\, d\mu \geq t\mu(\{f \geq t\}).$$

For p-integrable functions:

$$\mu(\{x : f(x) \geq t\}) = \mu(\{x : f(x)^p \geq t^p\}) \leq \frac{1}{t^p}\int f^p\, d\mu.$$

Type 11E: Basic set identities for measures

Typical task:

> Prove that
> $$\mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B).$$

Typical trigger: Decompose into disjoint pieces:

$$A = (A \setminus B) \cup (A \cap B),$$
$$B = (B \setminus A) \cup (A \cap B),$$
$$A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A).$$

Type 11F: Countable subadditivity and continuity from below

Typical task:

> Prove
> $$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \le \sum_{n=1}^{\infty} \mu(A_n),$$
> or prove that if $A_n \uparrow A$, then
> $$\mu(A_n) \to \mu(A).$$

Typical trigger: Disjointify the sets:

$$B_1 = A_1, \qquad B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k.$$

For increasing sequences:

$$B_1 = A_1, \qquad B_n = A_n \setminus A_{n-1}.$$

Type 11G: Monotone convergence theorem

Typical task:

> State and apply the monotone convergence theorem.

Typical statement: If

$$0 \le f_1 \le f_2 \le \cdots$$

and

$$f_n(x) \to f(x)$$

pointwise, then

$$\int f \, d\mu = \lim_{n\to\infty} \int f_n \, d\mu.$$

23.13 Type 12: Compact operators and Arzelà-Ascoli

Type 12A: Show compactness using Arzelà-Ascoli

Typical task:

> Show that an operator L maps bounded sets into relatively compact sets.

Typical trigger:

1. Show $\{Lu : \|u\| \leq 1\}$ is uniformly bounded.
2. Show $\{Lu : \|u\| \leq 1\}$ is equicontinuous.
3. Apply Arzelà-Ascoli.

23.14 Type 13: Banach fixed point theorem

Type 13A: Contraction without fixed point because of incompleteness

Typical task:

> Let $X = (0, 1]$ and $T(x) = x/2$. Show T is a contraction. Does it have a fixed point?

Typical trigger:

$$d(Tx, Ty) = \frac{1}{2} d(x, y).$$

But the fixed point equation gives

$$x = 0,$$

which is not in X.

Type 13B: Contraction iteration in a complete space

Typical task:

> Show that an iteration converges by proving that its defining map is a contraction.

Typical trigger:

$$\|T(x) - T(y)\| \leq q\|x - y\|, \qquad q < 1.$$

Then apply Banach's fixed point theorem.

23.15 Type 14: Almost-solvable equations and compactness

Typical task:

> Let X be compact and $f : X \to \mathbb{R}$ continuous. Suppose that for every $\varepsilon > 0$ there exists $x \in X$ such that
> $$|f(x) - a| < \varepsilon.$$
> Show that there exists $x_0 \in X$ with $f(x_0) = a$.

Typical trigger: Let
$$g(x) = |f(x) - a|.$$
Then g is continuous on compact X, so it attains a minimum. The almost-solvability assumption forces that minimum to be zero.

23.16 Recommended practice order

For current exam preparation, the most important types to drill are:

1. Type 11: Measure theory.
2. Type 7: Integral operators.
3. Type 6: Fréchet differentiability.
4. Type 5: Normed spaces and bounded linear maps.
5. Type 8: Series and uniform convergence.
6. Type 1-2: Metric-space completeness and compactness.
7. Type 3: Modified metrics.
8. Type 13: Banach fixed point theorem.

Solution attempt of (11A). **Nonnegativity.**

We're taking the sum of $\mathbf{1}_A(x_n) \in \{0, 1\}$ and dividing by a positive number hence $\mu_N(A) \geq 0$.

Null Empty Set.

$$\mu_N(\emptyset) = \frac{1}{N} \sum_{n=1}^{N} 0 = 0$$

σ-additivity.

Let $A_1, A_2, \ldots$ be disjoint.

Then

$$\mu_N\left(\bigcup_{m=1}^{\infty} A_m\right) = \frac{1}{N}\sum_{n=1}^{N} \mathbf{1}_{\cup_m A_m}(x_n)$$
$$= \frac{1}{N}\sum_{n=1}^{N}\sum_{m=1}^{\infty} \mathbf{1}_{A_m}(x_n).$$

We can interchange the sums to get

$$\mu_N\left(\bigcup_{m=1}^{\infty} A_m\right) = \sum_{m=1}^{\infty}\frac{1}{N}\sum_{n=1}^{N} \mathbf{1}_{A_m}(x_n)$$
$$= \sum_{m=1}^{\infty} \mu_N(A_m).$$

Hence μ_N is a meassure. □

Solution attempt of (11B). We have

$$g := \sum_{i=1}^{m} a_i \mathbf{1} A_i,$$

and are asked to compute

$$\int g \, d\mu_N.$$

For a simple measurable function with standard form

$$g = \sum_{i=1}^{m} a_i \mathbf{1} A_i$$

the the integral with respect to the measure μ_N is

$$\int g \, d\mu_N = \sum_{n=1}^{m} a_n \mu_N(A_n).$$

So

$$\begin{aligned}\int g\, d\mu_N &= \sum_{n=1}^m a_n \frac{1}{N}\sum_{i=1}^N \mathbf{1}_{A_n}(x_i) \\ &= \sum_{n=1}^m a_n \frac{1}{N} |A_n \cap \{x_1, \dots, x_N\}| \\ &= \frac{1}{N}\sum_{n=1}^m a_n \cdot |A_n \cap \{x_1, \dots, x_N\}| \\ &\leq \frac{1}{N}\sum_{n=1}^m a_n \cdot N \\ &\leq \sum_{n=1}^m a_n \\ &< \infty.\end{aligned}$$

□

Solution attempt of (11E). We want to prove that

$$\mu(A \cup B) + \mu(A \cup B) = \mu(A) + \mu(B).$$

$A = (A \setminus B) \cup (A \cap B)$ and $B = (B \setminus A) \cup (A \cap B)$. Then since $B \setminus A$ and $A \setminus B$ are disjoint we have

$$\begin{aligned}\mu(A) + \mu(B) &= \mu((A \setminus B) \cup (A \cap B) \cup (B \setminus A)) + \mu(A \cap B) \\ &= \mu(A \cup B) + \mu(A \cap B).\end{aligned}$$

□

Solution attempt of (11F). We will prove that

$$\mu\left(\bigcup_{n=1}^\infty A_n\right) \leq \sum_{n=1}^\infty \mu(A_n)$$

and that if $A_1 \subseteq A_2 \subseteq \cdots$ so that

$$\bigcup_{n=1}^\infty A_n = A$$

then

$$\mu(A_n) \to \mu(A).$$

First.

Let $B_1 := A_1$ and define $B_n := A_n \setminus \cup_{k=1}^{n-1} A_k$. Then B_n are disjoin for all n so

$$\mu\left(\bigcup_{n=1}^\infty B_n\right) = \sum_{n=1}^\infty \mu(B_n),$$

and furthermore $\cup_{n=1}^{\infty} A_n = \cup_{n=1}^{\infty} B_n$ and since $|B_n| \leq |A_n|$ for all n we have

$$\sum_{n=1}^{\infty} \mu(B_n) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

Hence

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \mu\left(\bigcup_{n=1}^{\infty} B_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

Second.

Let $B_1 := A_1$ and define $B_n := A_n \setminus A_{n-1}$. Then all B_n are disjoint and

$$A = \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n$$

so their meassure is the same.

Also

$$\mu(A_N) = \sum_{n=1}^{N} \mu(B_n)$$

and

$$\mu(A) = \sum_{n=1}^{\infty} \mu(B_n).$$

Then

$$\begin{aligned} \lim_{N\to\infty} \mu(A_N) &= \lim_{N\to\infty} \sum_{n=1}^{N} \mu(B_n) \\ &= \sum_{n=1}^{\infty} \mu(B_n) \\ &= \mu(A). \end{aligned}$$

□

Solution attempt of (13A). If $X = [0,1]$ then Banach's fixed point theorem would apply, but $X = (0,1]$.

$$|x/2 - y/2| = \frac{1}{2}|x - y|$$

so T is a contraction with constant $\frac{1}{2}$.

If we try to find the fixed-point we get

$$Tx = \frac{x}{2} = 0 \Rightarrow x = 0 \notin X.$$

We can also recall that to find the fixed-point of a function T, if it exists, can be done by taking

$$\lim_{n\to\infty} T^n x$$

for any $x \in X$. Then we observe that

$$\lim_{n\to\infty} T^n x = \lim_{n\to\infty} \frac{x}{2^n} = 0 \notin X.$$

□

Solution attempt of (14A). Let $g := |f(x) - a|$. Then g is a continuous function on a compact set X. By the Extreme Value Theorem there exists $x \in X$ such that

$$g(x) = \inf_{y\in X} |f(x) - a| < \varepsilon$$

for every $\varepsilon > 0$. Since this has to hold for every positive ε we have that

$$g(x) = |f(x) - a| = 0$$

which, since $|\cdot|$ is a norm, implies that

$$f(x) - a = 0 \Rightarrow f(x) = a.$$

All of the above could (and probably should) be made more rigorous. □

(Short) Solution attempt of (7a). Since K is continuous we can find δ so that

$$|x_1 - x_2| < \delta \Rightarrow |k(x_1, y) - k(x_2, y)| < \varepsilon/\|u\|$$

for every y.

Then

$$\begin{aligned}|(Lu)(x_1) - (Lu)(x_2)| &\leq |k(x_1, y) - k(x - 2, y)| \cdot \|u\| \\ &< \varepsilon\end{aligned}$$

□

(Short) Solution attempt of (7b). Let $x \in [0, 1]$.

$$\begin{aligned}|(Lu)(x)| &= |\int_0^1 k(x, y)u(y)\ dy| \\ &\leq \|u\| \cdot \int_0^1 |k(x, y)|\ dy\end{aligned}$$

So taking the supremum over $x \in [0, 1]$ we get

$$\|Lu\| \leq \|u\| \cdot \sup_{x\in[0,1]} \int_0^1 |k(x, y)|\ dy.$$

□

Problem 23.9. *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f : X \to [0, \infty]$ be measurable.*
Assume
$$\int_X f\, d\mu < \infty.$$
For $t > 0$, define
$$E_t = \{x \in X : f(x) \geq t\}.$$

a) Prove that
$$\mu(E_t) \leq \frac{1}{t} \int_X f\, d\mu.$$

b) Use this to prove that
$$\mu(\{x : f(x) = \infty\}) = 0.$$

In other words, prove that f is finite μ-almost-everywhere.

Proof of (a). On E_t we have $f(x) \geq t$. So for every $x \in X$ we have
$$f(x) \geq t\mathbb{1}_{E_t}(x),$$
since if $x \in E_t$ then $t\mathbb{1}_{E_t}(x) = t \leq f(x)$ and otherwise $t\mathbb{1}_{E_t}(x) = 0 \leq f(x)$.

By the properties of the Lebesgue integral we have
$$\int_X f\, d\mu \geq \int_X t\mathbb{1}_{E_t}(x)\, d\mu.$$
Then observe that
$$\begin{aligned}\int_X t\mathbb{1}E_t(x)\, d\mu &= t\int_X \mathbb{1}_{E_t}(x)\, d\mu \\ &= t\mu(E_t).\end{aligned}$$
Then
$$\mu(E_t) \leq \frac{1}{t} \int_X f\, d\mu.$$
□

Proof of (b). The set $\{x : f(x) = \infty\}$ is the set of $x \in X$ such that $\forall N \in \mathbb{N}$, $f(x) \geq N$.

Then, for every $N \in \mathbb{N}$,
$$\mu(\{x : f(x) = \infty\}) \leq \frac{1}{N} \int_X f\, d\mu.$$
Since this holds for every $N \in \mathbb{N}$, and the meassure is non-negative, we have
$$0 \leq \mu(\{x : f(x) = \infty\}) \leq \frac{1}{N} \int_X f\, d\mu = 0.$$

$\square$

Problem 23.10. *Define* $F : \mathbb{R}^2 \to \mathbb{R}^2$ *by*

$$F(0,0) = 0,$$

and for $(x,y) \neq (0,0)$*,*

$$F(x,y) = \frac{x^3}{x^2+y^2}.$$

a) Compute the directional derivative of F *at* $(0,0)$ *in direction* $v = (a,b)$*.*

b) Do all directional derivatives exist?

c) Is F *frechet differentiable at* $(0,0)$*?*

Solution of (a/b/c). We compute

$$\begin{aligned} D_v f(0,0) &= \lim_{h\to 0} \frac{f((0,0)+h(a,b)) - f(0,0)}{h} \\ &= \lim_{h\to 0} \frac{f(ha,hb)}{h} \\ &= \lim_{h\to 0} \frac{\frac{(ha)^3}{(ha)^2+(hb)^2}}{h} \\ &= \lim_{h\to 0} \frac{h^3a^3}{h^3(a^2+b^2)} \\ &= \frac{a^3}{a^2+b^2} \\ &= f(a,b) \end{aligned}$$

Thus for any direction $v = (a,b) \in \mathbb{R}^2$ the directional derivative at $(0,0)$ is

$$D_v f(0,0) = f(a,b).$$

Since v was any arbitrary direction we conclude that all directional derivatives exist at $(0,0)$. It takes some work to show, but if one computes

$$D_v f(a,b)$$

for arbitrary $(a,b) \neq (0,0)$, then the one term will have factors of h at the bottom of the fraction meaning it doesn't resolve nicely. So directional derivatives do not exist everywhere, and since Frechet differentiability implies Gateux differentiability we conclude that f is indeed not Frechet differentiable. $\square$

Problem 23.11. *Let*

$$X = C([0,1],\mathbb{R})$$

with the supremum norm. Define

$$\Phi : X \to \mathbb{R}$$

by

$$\Phi(u) = \int_0^1 u(t)^2\, dt.$$

a) *Guess the derivative* $D\Phi(u)h$*.*

b) *Prove that* Φ *is Frechet differentiable.*

c) *Show that* $D\Phi(u) : X \to \mathbb{R}$ *is bounded and linear.*

Solution and proof to all. I refuse to guess.

We solve

$$\lim_{\|h\to 0\|} \frac{\|\Phi(u+h) - \Phi(u) - A(h)\|}{\|h\|}.$$

Compute

$$\begin{aligned}
\Phi(u+h) - \Phi(u) &= \int_0^1 (u+h)(t)^2\, dt - \int_0^1 u(t)^2\, dt \\
&= \int_0^1 (u(t)+h(t))^2 - u(t)^2\, dt \\
&= \int_0^1 u(t)^2 + h(t)^2 + 2u(t)h(t) - u(t)^2\, dt \\
&= \int_0^1 h(t)^2 + 2u(t)h(t)\, dt \\
&= \int_0^1 h(t)^2\, dt + \int_0^1 2u(t)h(t)\, dt
\end{aligned}$$

Since we are working with continuous functions with compact domains they are bounded so we can bound these integrals by

$$\int_0^1 h(t)^2\, dt + \int_0^1 2u(t)h(t)\, dt \le \|h\|^2 + \int_0^1 2u(t)h(t)\, dt$$

Let $D\Phi(u)h = A(h) := \int_0^1 2u(t)h(t)\, dt$. Then

$$\begin{aligned}
\lim_{\|h\|\to 0} \frac{\|\Phi(u+h) - \Phi(u) - A(h)\|}{\|h\|} &= \lim_{\|h\|\to 0} \frac{\|\int_0^1 h(t)^2\, dt\|}{\|h\|} \qquad \le \lim_{\|h\|\to 0} \frac{\|h\|^2}{\|h\|} \\
&= 0
\end{aligned}$$

We have $D\Phi(u)h = A(h)$ and A is a linear functional in $\mathcal{L}(X, \mathbb{R})$. $\square$

Problem 23.12. *Let X be a Banach space and $T \in \mathcal{L}(X)$ with*

$$\|T\| \leq \frac{1}{2}.$$

Show that $I - T$ is invertible and prove that

$$\|(I-T)^{-1}\| \leq 2.$$

Proof. For a linear operator T on a Banach space X we have $\|T\| < 1 \Rightarrow I - T$ invertible and

$$(I-T)^{-1} = \sum_{n=0}^{\infty} T^n.$$

Then

$$\begin{aligned}
\|(I-T)^{-1}\| &= \|\sum_{n=0}^{\infty} T^n\| \\
&\leq \sum_{n=0}^{\infty} \|T^n\| \\
&\leq \sum_{n=0}^{\infty} \|T\|^n \\
&\leq \sum_{n=0}^{\infty} \frac{1}{2^n} \\
&= 2.
\end{aligned}$$

$\square$